

Logic Manager Installation

LM20-500

**Implementation
Logic Manager**

Logic Manager Installation

**LM20-500
R500
3/97**

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Honeywell
Industrial Automation and Control
Automation College
2820 West Kelton Lane
Phoenix, AZ 85023-3028

1-800-852-3211

About This Publication

This manual provides instructions for the installation of the Logic Manager System (LM).

Site planning information is contained in *Logic Manager Site Planning* manual, which is in the *System Site Planning* binder. After installation of the LM, the checkout procedure can be found in the *Logic Manager Service* manual, which is in the *LM Service/Logic Manager* binder.

This publication supports **TotalPlant** Solution (TPS) system network Release 500. TPS is the evolution of TDC 3000^X.

Change bars are used to indicate paragraphs, tables, or illustrations containing changes that have been made by this update. Pages revised only to correct minor typographical errors contain no change bars.

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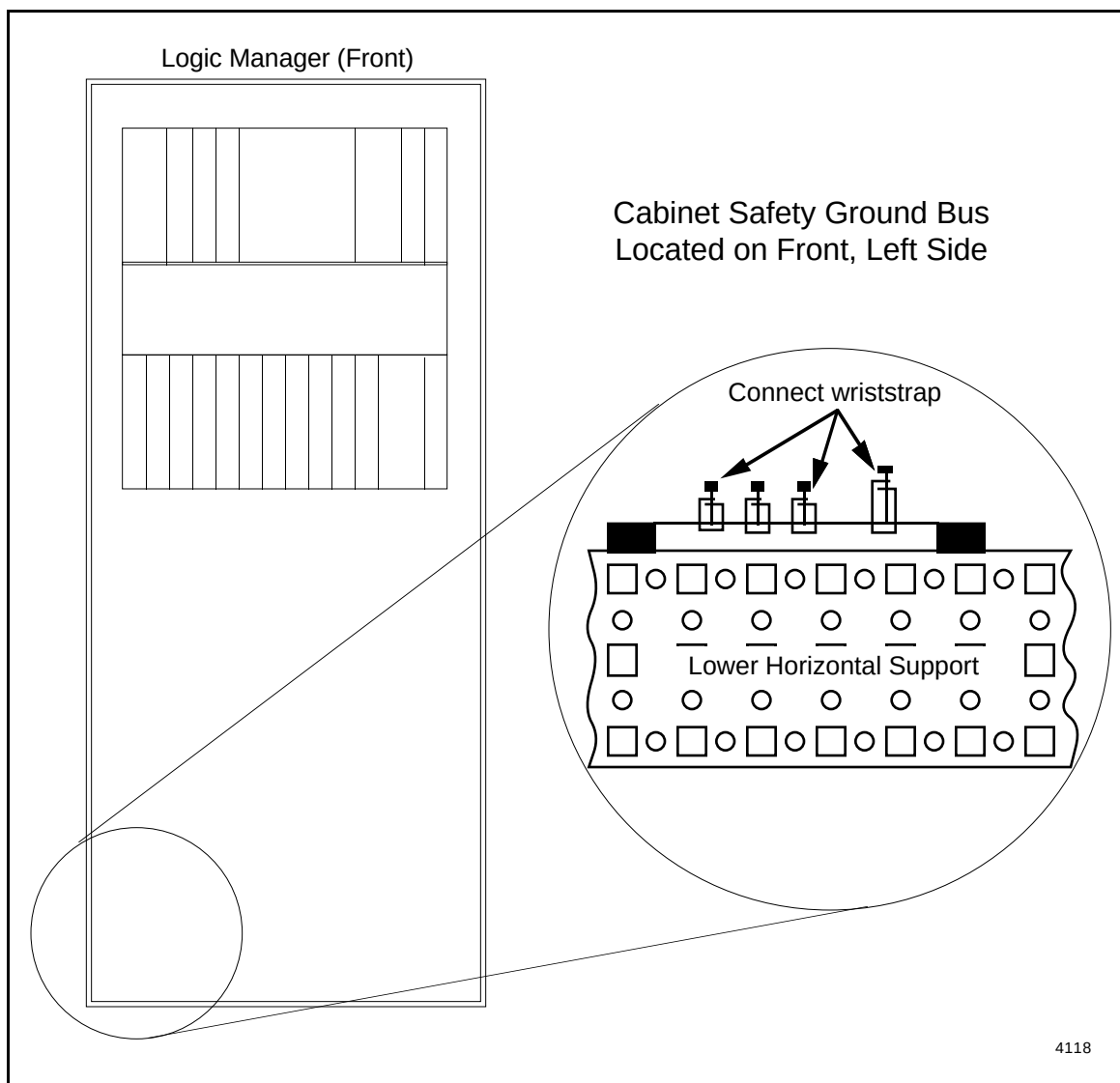
B FIBER OPTIC SERIAL I/O

B.1	Note
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ESD WARNING
Section 0**ESD WARNING**

This equipment is subject to damage by Electrostatic Discharge (ESD).

Wear your ESD wriststrap any time you make contact with the equipment.



INTRODUCTION

Section 1

This section introduces the contents of this manual.

This manual provides instructions for the installation of the Logic Manager System (LM). Installation is accomplished using the following procedures:

- Equipment Warehousing (Section 2)
- Move Preparation (Section 3)
- Unpacking and Placement (Section 4)
- System Component Installation (Section 5)
- Power and Ground Installation (Section 6)
- System Configuration (Section 7)
- Process I/O Wiring Installation (Section 8)
- UCN Connection (Section 9)

1.1 RELATED DOCUMENTATION

<u>Title</u>	<u>Binder</u>
Logic Manager Specification and Technical Data	Implementation/Logic Manager
Logic Manager Site Planning	System Site Planning
Universal Control Network Site Planning	System Site Planning
Universal Control Network Installation	Installation/Universal Control Network

1.2 TOOLS AND TEST EQUIPMENT

The following equipment is necessary for installation of the Logic Manager:

- Mobile lifter/lift truck—see Section 4.2 for requirements.
- Carrier Band MAP tester—see Section 5.1 for requirements.
- Hand tools and multimeter, as in a typical electrician's tool kit.
- Torque wrench to work in 0 to 70 cm/kg (0 to 60 inch-pound with one inch-pound resolution) range for assembly of UCN cable to their connectors.
- Set of SAE (inch) "crow's foot" wrenches for the above torque wrench.
- ESD kit, Honeywell part number 30185-H.

EQUIPMENT WAREHOUSING Section 2

This section describes the factory packing, transit and warehousing specifications.

2.1 FACTORY PACKING

For shipment, the Logic Manager is packed with desiccant packages, wrapped in a moisture barrier, placed on skids, and optionally enclosed for shipment as required by the user.

2.2 TRANSIT AND WAREHOUSING SPECIFICATIONS

The Logic Manager can be shipped by a variety of means, but any trucking should be done by enclosed air-ride electronic vans. The equipment is usually trucked on the final leg of the journey to the user warehouse or site. There it is stored indoors until needed for installation. During this whole process, the environment must be monitored and **corrected** if the following equipment ratings are exceeded:

- Transit/Storage Temperature Range: -55° to 85°C (-67° to 158°F)
- Transit/Storage Humidity Range: 5 to 95% RH noncondensing

Keep the factory wrapping intact to minimize humidity.

If it is necessary to unseal the equipment for customs or receiving, add more desiccant and reseal the package.

MOVE PREPARATION

Section 3

This section describes the site survey and preparations that are made before installing the Logic Manager.

3.1 SURVEY

A single Logic Manager cabinet is nominally 0.8 x 0.8 x 2 meters (32 x 32 x 79 inches). Lifting eye bolts, which are removable, add about five cm (2 inches) to the height. The weight varies with options and can go as high as 385 kg (850 pounds) per cabinet. A factory assembled complex uses up to four such cabinets bolted together. See Figures 3-1 and 4-5. Keep this in mind while planning the moving and unpacking of Logic Manager System.

Points to check for when planning the moving of cabinets are:

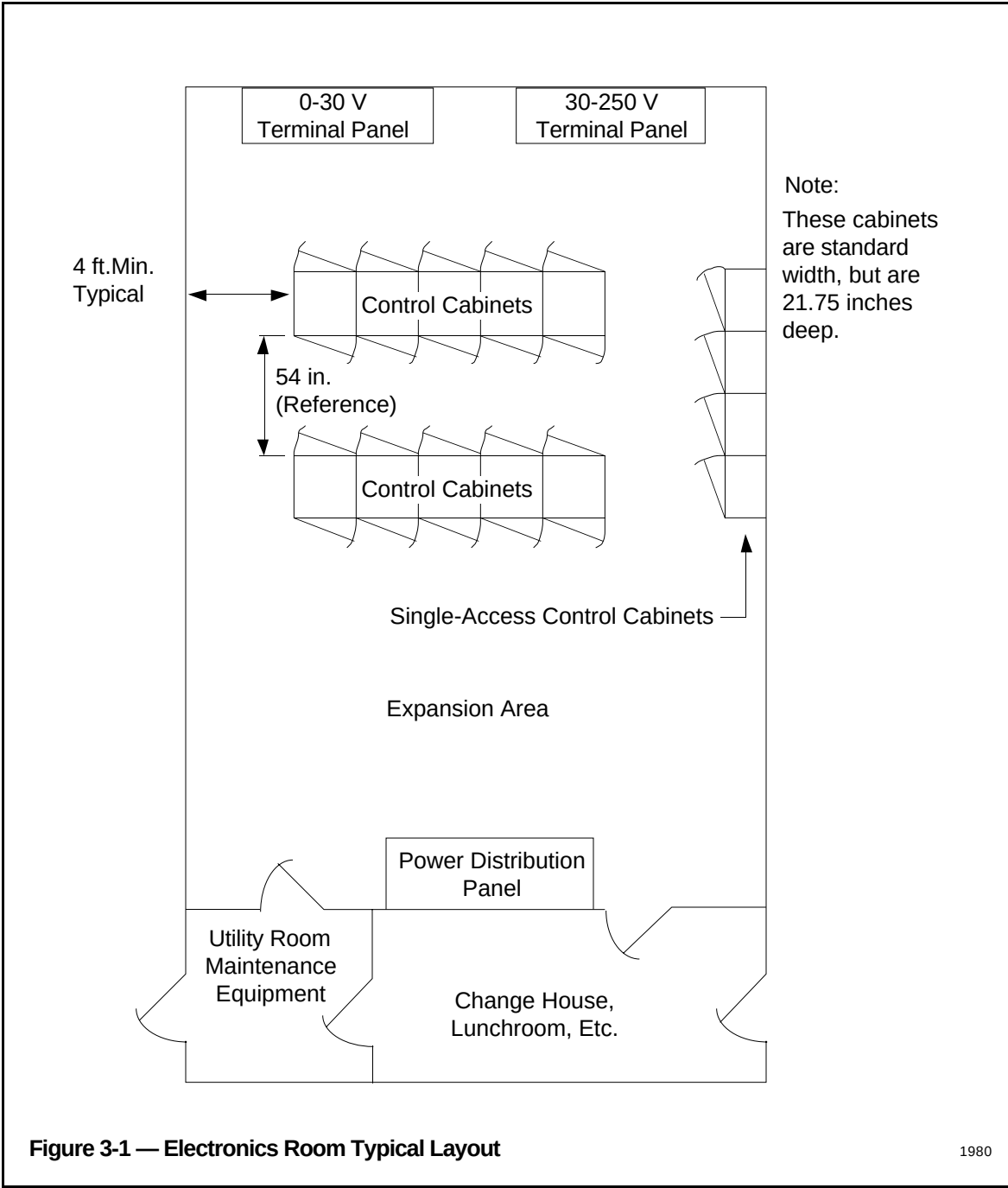
- Chart the equipment travel route
 - loading door—height and location
 - elevators—size, capacity, and availability
 - aisles—number, size, and layout
 - doorways—number and size
 - hallways—size, restrictions, and bends
 - stairs and ramps—may require some type of lifting
 - rigging equipment if needed
- Space—Is there enough space to hold the equipment?
- Floor Loading—Will the floor in all areas be strong enough?
- Power Distribution—Check for proper size transformers, distribution panels, and sufficient outlets for support, test, and maintenance equipment. Information can be found in the *Logic Manager Site Planning* manual in the *System Site Planning - 2* binder.
- Be sure that proper grounds are available. Information can be found in the *Logic Manager Site Planning* manual in the *System Site Planning - 2* binder.
- Conduits and Cable Trays—Make sure that conduits and cable trays are large enough and the proper type to accommodate all of the required cables.
- Environment—Lighting and other environmental requirements must be adequate. Information can be found in the *Logic Manager Site Planning* manual in the *System Site Planning - 2* binder.
- Security—Provide clearance for the installation crew if it is not composed of regular employees.
- Responsibilities—To eliminate work delays, make sure that all responsibilities are clearly defined and understood.

- Communications—At least three conveniently-located phones are needed for:
 - Honeywell Technical Assistance Center (TAC) data terminal
 - Honeywell TAC voice
 - Operator Emergency

3.2 PREPARATION

Prepare as follows to help ensure that the Logic Manager System will operate when power is first applied.

- Static Electricity
 - Strip the floors of any wax to prevent static charge buildup.
 - Use antistatic spray on floors, furniture, and fabric in the work area.
 - Place a sign on or near the equipment that states circuit boards are not to be handled without wearing a ground strap.
- Place tack rugs at all entrances to catch dust, grit, and abrasives from other areas.
- Consult with the local fire prevention authority to select fire extinguishers suitable for electrical fires.
- Study the room layout diagram for cabinet placement—see Figure 3-1. If this Logic Manager cabinet is to be bolted down, Figure 3-2 contains the outline dimensions for drilling holes in the floor.



3.3 CABINET ENTRY

The process signal cables can enter the Logic Manager cabinet through the top or bottom of the cabinet. The Logic Manager cabinet is special, because for bottom entry it has sliding floor plates that can be adjusted to provide entry slots. For top entry, use a hydraulic hole punch to minimize the generation of metal filings. For top or bottom entry, the cables should be clamped firmly to the cabinet. The clamping should be able to handle about a 45 kg (100 pound) pull. This prevents damage inside the cabinet when pulling on cables outside the cabinet.

The Logic Manager cabinet includes a cable clamp rail at the bottom. The rail can be remounted to the top if required.

UNPACKING AND PLACEMENT

Section 4

This section describes how the Logic Manager System (LM) is to be moved to its final location, positioned, and then bolted down.

4.1 UNPACKING

Unpack the equipment, check the shipment against the invoice, and notify Honeywell representative of any shortages immediately. When the Logic Manager System is comprised of individual card files, the component parts are packaged in labeled boxes. If a Product Registration Label containing the Model Number and Serial Number of the component is affixed to the shipping carton when received, remove and return the label to Honeywell at the noted address to ensure follow-up service and support.

4.2 MOVING

Equipment may be moved one of four ways:

- Fork Lift
- Mobile Lifter — See Figure 4-2
- Rollers — See Figure 4-3
- Crane — See Figure 4-4

Use lifters of 1100 kg (2500 pounds) or more capacity.

CAUTION

Tilting the cabinet more than 45° front or back may damage the doors; they are not recessed.

4.2.1 Fork Lift

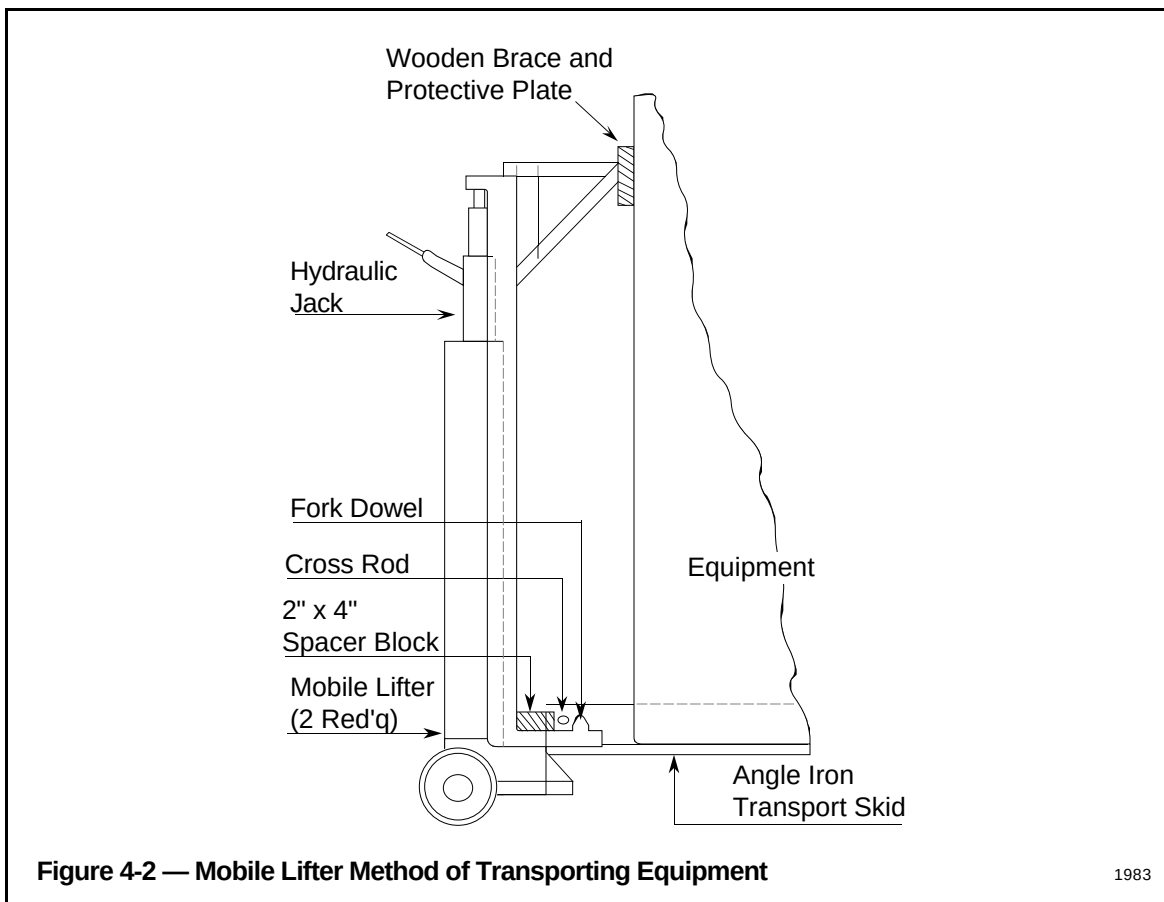
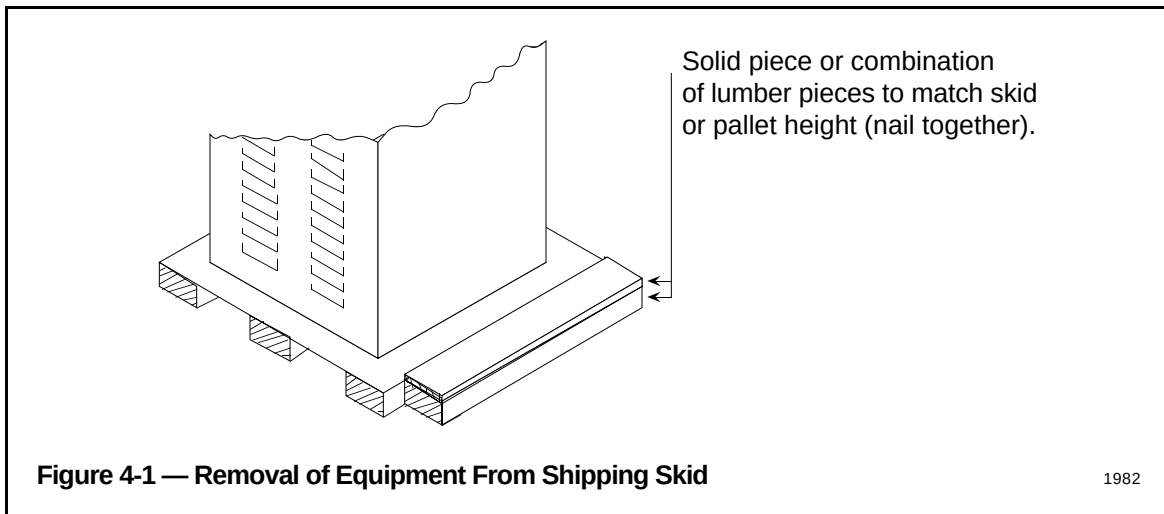
The equipment, on the original shipping skid, is moved by fork lift to the final location.

1. Move the equipment to the electronics room by fork lift; remove the fork lift.
2. Open the cabinet doors, locate the skid bolts, and remove them.
3. Stack lumber against the pallet to a matching height — see Figure 4-1.
4. Manually move the equipment onto the lumber by 10 cm (4 inches).
5. Tip the cabinet weight onto the lumber and pull out the skid.
6. Tip the cabinet off the lumber and onto the floor; remove the lumber.
7. Manually move the cabinet to the exact final position.

4.2.2 Mobile Lifter

The equipment is removed from the shipping skid and placed on a transport skid made from angle iron. It is then moved with a mobile lifter. See Figure 4-2.

1. Remove any steel bands, turnbuckles, and cables securing the equipment to the shipping skid.
2. Open the cabinet doors and locate the bolts holding the cabinets to the shipping skid; remove them and close the doors.
3. Tilt the equipment slightly and slide one of the angle irons used to make up a transport skid underneath the cabinet—see Figure 4-4.
4. Repeat the above step for the other angle.
5. Insert a strip of heavy cardboard between each angle iron and the equipment finish.
6. Tighten the angle irons against the cabinet by tapping slightly.
7. Insert and tighten two 12 mm (1/2-inch) minimum, threaded cross rods as in Figures 4-2 and 4-4.
8. Spread the lifter forks to 0.8 m (32-inch) centers and put them under each cross rod. Place the forks dowels (illustrated Figure 4-2) between the cross rods and the cabinet, just touching the cross rods.
9. Place wooden spacer blocks along the forks, long enough to press the cross rods against the fork dowels.
10. Position a protective plate using braces as in Figure 4-2. Be sure that the plate does not press against handles or other cabinet hardware.
11. Raise both lifter forks evenly until the equipment clears the shipping skid; pull it out of the way.
12. Wheel the equipment to the final position in the electronics room; lower it to the floor.
13. Pull out the mobile lifters. Dismantle and remove the angle iron transport skid.
14. Manually place the cabinet in the exact final position.



4.2.3 Roller Method

The cabinet is moved to its final destination, using pipe rollers under a transport skid made from angle iron.

1. Use one of the previous methods to move the cabinet to a location where rollers are to be used.
2. Install the angle iron transport skid as in the Mobile Lifter Method.
3. Place 12 mm (1 inch) pipes under the transport skid as in Figure 4-3. The load must be on the angle irons, not the cabinet.
4. Roll the cabinet to the final location. At least three pipes must be under the transport skid at all times.
5. When the cabinet is near the final location, tilt it and pull out the pipes, the middle one last.
6. Dismantle the transport skid and position the cabinet manually to the exact final location.

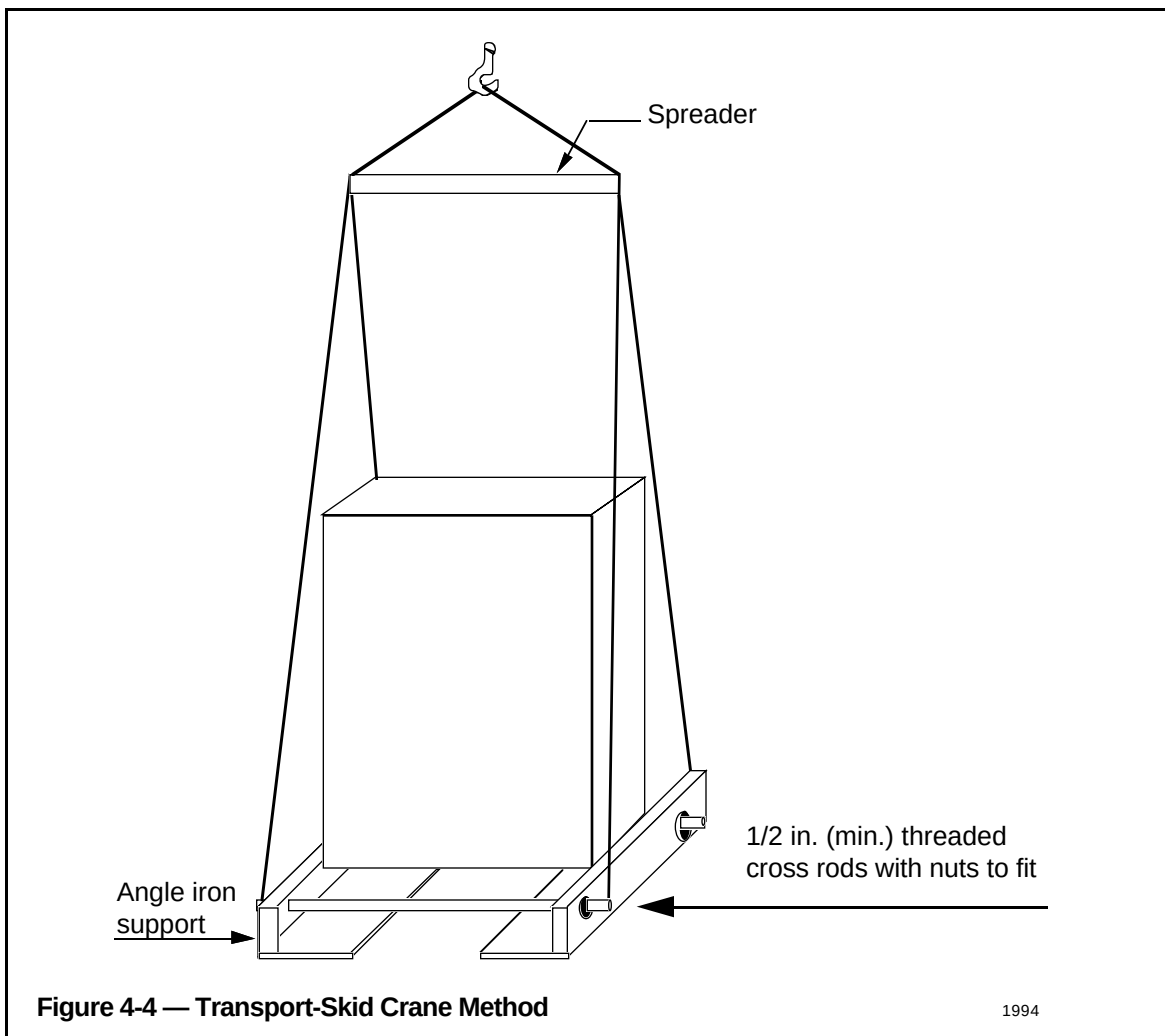
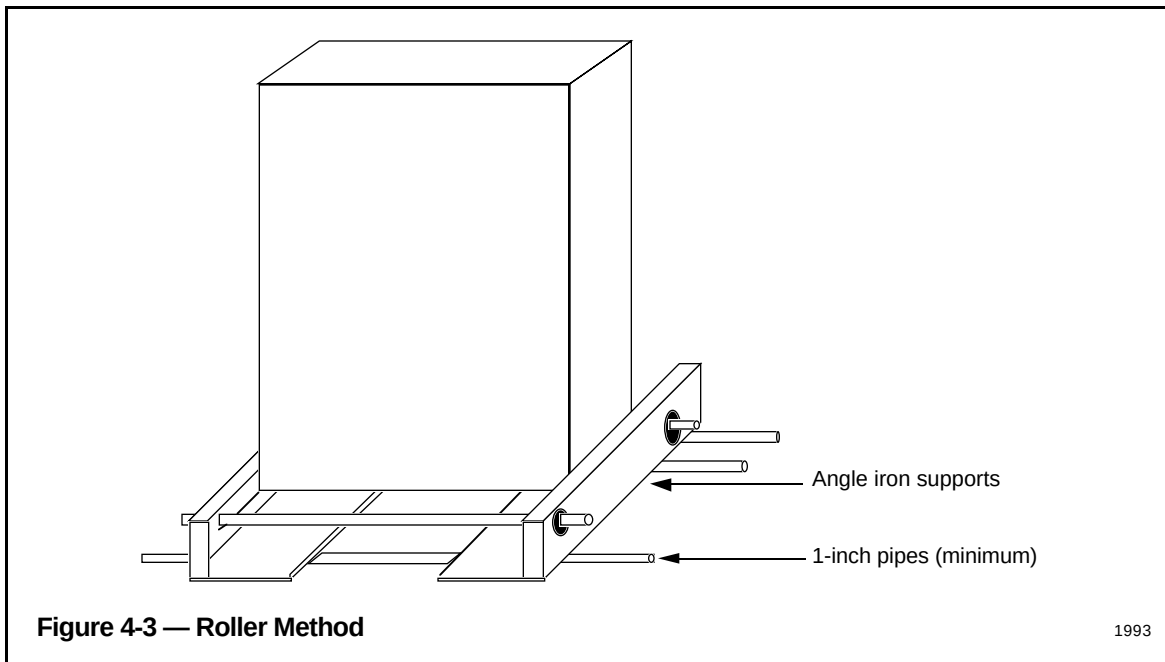
4.2.4 Crane Method

The cabinet is lifted into place with a crane. Use either an angle iron transport skid (Figure 4-4) or the lifting eyebolts installed on each cabinet (Figure 4-5).

4.3 LEVELING AND BOLT DOWN

LM cabinets do not include levelers; it may be necessary to hand-fit shims underneath the cabinet in order to make the cabinet sit solidly.

The LM cabinet has six holes in the bottom for bolting it to a floor. A partially filled cabinet should be bolted down. Cabinets that are bolted together are not easily moved and do not ordinarily require bolting down. See Figure 3-2 for bolt-down dimensions.



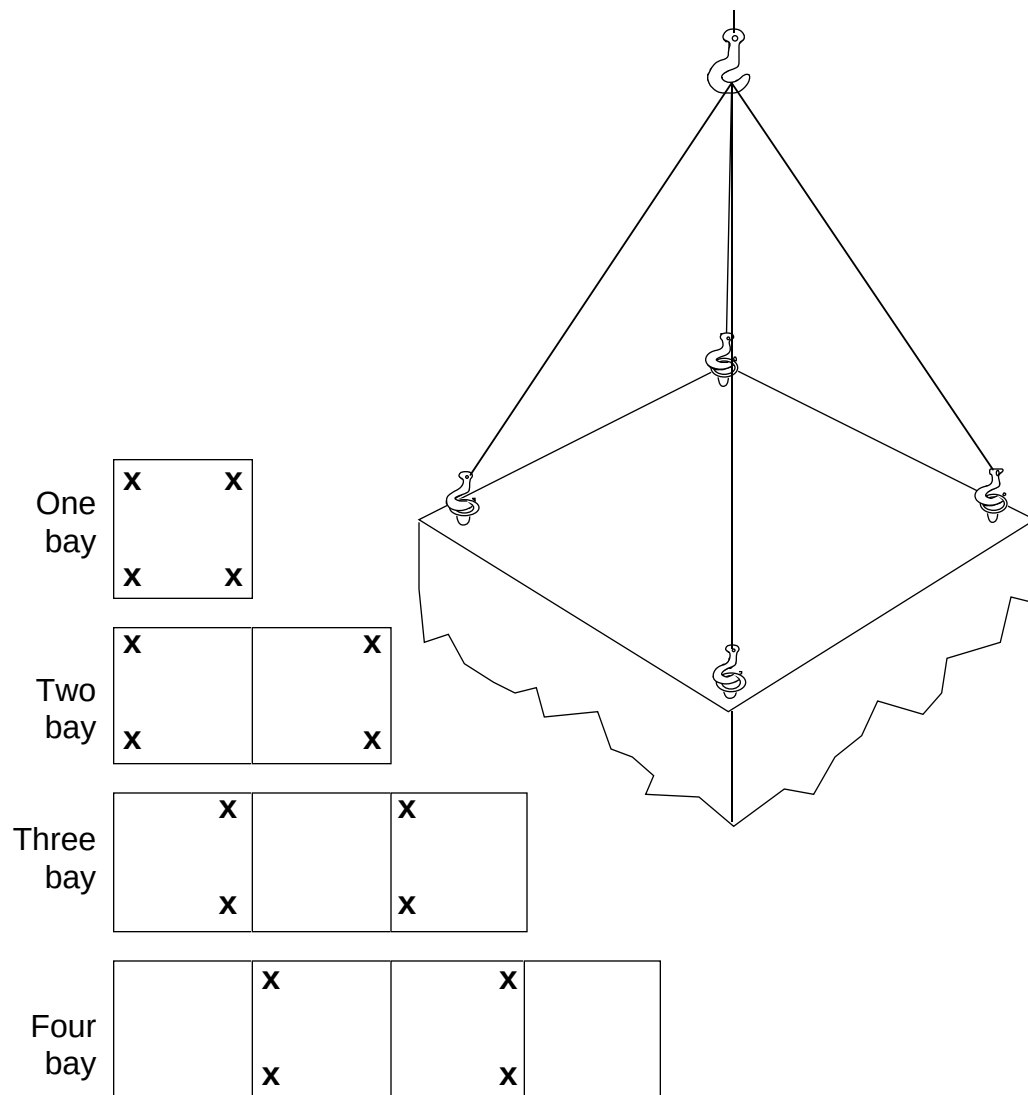


Figure 4-5 — Eyebolt Crane Method

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SYSTEM COMPONENT INSTALLATION

Section 5

This section describes the steps for safe and proper installation of the Logic Manager System.

5.1 INSTALLATION CONSIDERATIONS

The Logic Manager System may encompass a single cabinet, a multi-cabinet complex of up to four cabinets, individual card files and supporting power supply modules, or a combination of such hardware. If the system involves only a cabinet or cabinets, power, ground, and cabling is brought to the system and connected. The system is configured and installation is complete. However, should the system be composed of individual components, the components must be mounted to a structure and power and ground supplied by a choice of methods. In any case, consider the following installation checklist:

- Carefully review all documentation available for the equipment, paying close attention to cautionary notes.
- Install the Logic Manager System equipment in accordance with the current edition of the National Electric code ANSI/NFPA 70.
- Use technically qualified personnel to service the Logic Manager System components.

5.2 CABINET INSTALLATION

After placement as described in the *Logic Manager Site Planning* manual in the *System Site Planning* binder, the cabinet or cabinet complex requires the attachment of power and ground, cabling installation, and if necessary, configuration of the certain card file modules.

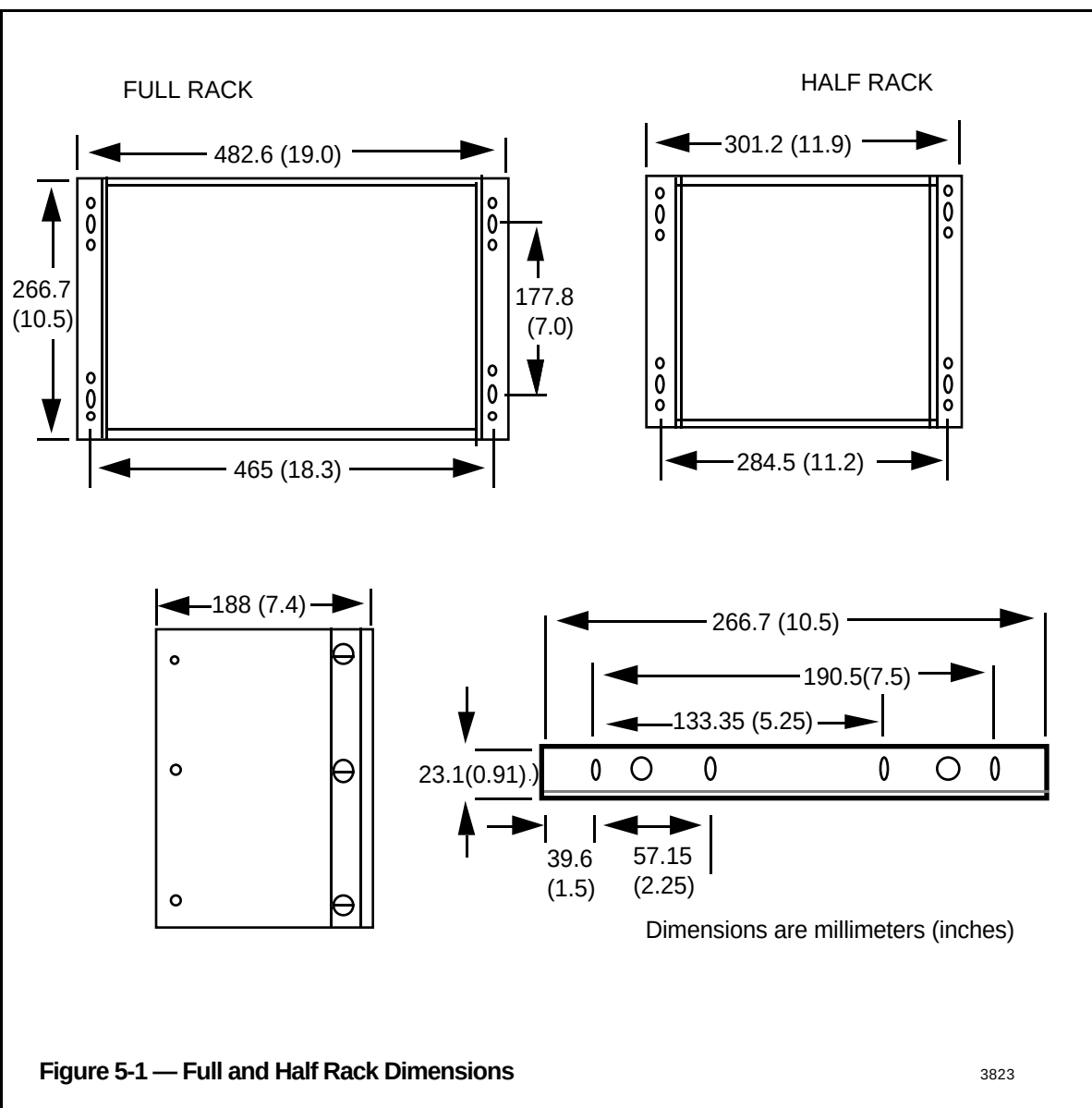
5.3 CARD FILE INSTALLATION

5.3.1 Rack Assembly Mounting

The external dimensions of all 19 inch full racks (Logic Manager Processor and I/O) are identical. All half racks (I/O) also have identical external mounting dimensions. See Figure 5-1 for full/half rack dimensions. A label located on the left inside wall of the rack assembly provides the rack model number, revision level, and serial number information.

Each Logic Manager rack fits into an 8 inch deep NEMA 12 enclosure or a 19 inch instrumentation rack. See Figure 5-2. The rack conforms to the European HE standard. Reversible mounting brackets allow the rack to be panel or rack mounted. When the brackets are attached to the rack front, the unit mounts in a standard 19 inch rack. When the brackets are rotated 180° and mounted to the rear, the rack can be panel mounted.

The rack assemblies are shipped with the mounting bracket installed for the panel mounting configuration. Each rack is marked to reference top and bottom.



5.3.1.1 Panel-Mounted Installation

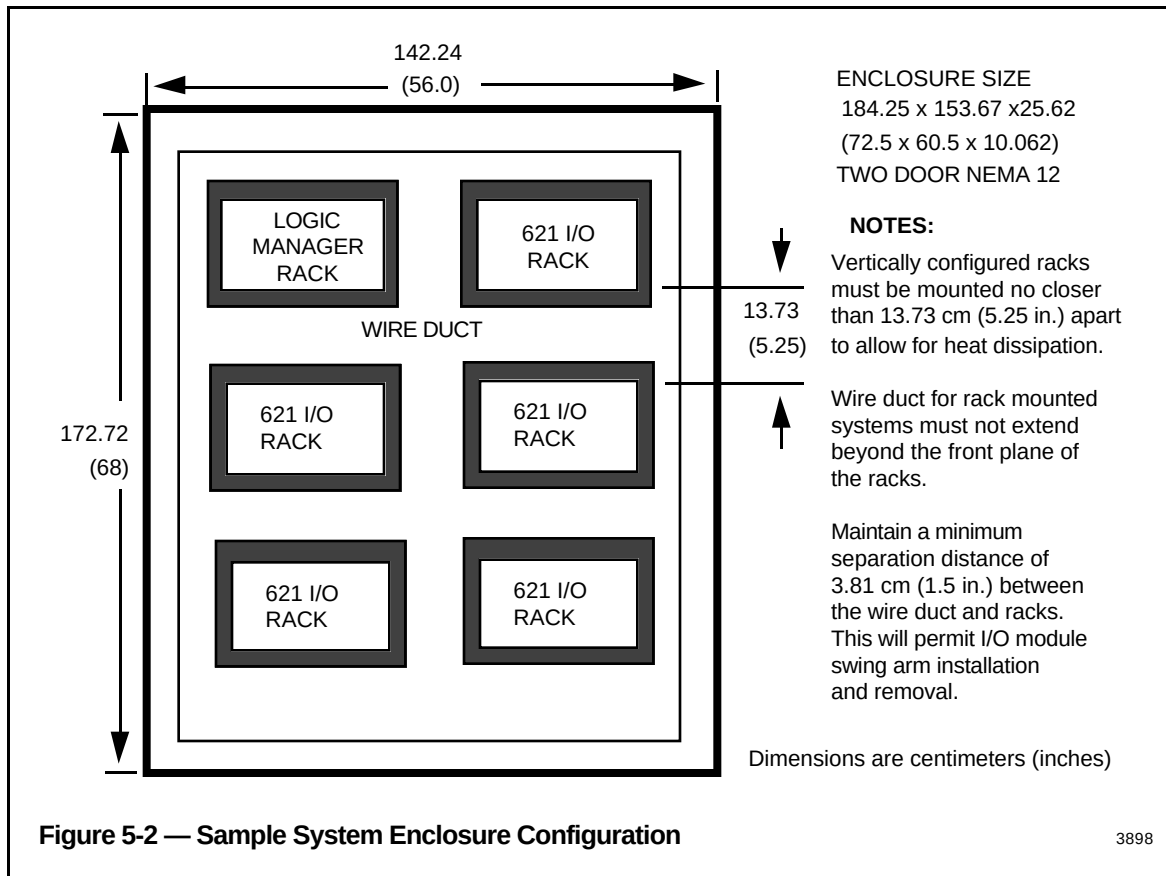
1. Leaving the mounting brackets attached to the rack chassis, place the chassis in the appropriate position in the rack structure.
2. If drilling is required, use the mounting bracket as a drilling template to mark the drilling locations. Drill and tap the required mounting holes.
3. Inspect the rack assembly and remove any paint particles or metal chips resulting from rack installation.
4. Install a 1/4 x 20 machine screw with external tooth star washer through the four holes in each mounting bracket.

WARNING

Mount vertically configured racks no closer than 5.25 inches apart to allow for heat dissipation.

5.3.1.2 Rack-Mounted Installation

1. Remove the mounting brackets from the sides of the chassis by removing the three mounting screws.
2. Align each bracket at the front of the chassis with the flange facing toward the front and install the three screws using external tooth star washers.
3. Position the chassis in the appropriate position in the panel.
4. If drilling is required, use the mounting bracket as a drilling template to mark the drilling locations. Drill the required mounting holes.
5. Inspect the rack assembly and remove any paint particles or metal chips resulting from rack installation.
6. Using external tooth star washers, install the four mounting screws through each bracket into the panel.



5.4 CABLE CONDUIT AND TRAYS

5.4.1 Conduit Shielding

Conduit or cable trays should be used to route the network trunkline within the plant for short distances or for several miles. The same conduit should not carry both data and power wiring because of problems created by electromagnetic interference. The choice of materials and configuration for the cable conduit can determine the degree of protection from this interference. Information can be found in the *Logic Manager Site Planning* manual in the *System Site Planning* binder.

Table 5-1, showing raceway shielding, is taken from IEEE Standard 518 for the installation of electrical equipment. It compares different types of conduits and raceways for magnetic field attenuation at 60 Hz and for electrostatic attenuation at 100 kHz.

Table 5-1 — Raceway Shielding

RACEWAY TYPE	THICKNESS mm (in.)	60 Hz Magnetic Field Attenuation		100 kHz Electric Field Attenuation	
		Ratio	db	Ratio	db
Free air		1:1	0	1:1	0
2 inch aluminum conduit	3.91 (0.154)	1.5:1	3.3	2,150:1	66.5
#16 gauge aluminum tray	1.52 (0.060)	1.6:1	4.1	15,500:1	83.9
#16 gauge aluminum tray	1.52 (0.060)	3:1	9.4	20,000:1	86.0
#16 gauge galvan. iron tray	1.52 (0.060)	3.2:1	10.0	22,000:1	86.8
2 inch IPS copper pipe	3.96 (0.156)	3.3:1	10.2	10,750:1	80.6
#16 gauge aluminum tray	1.52 (0.060)	4.2:1	11.5	29,000:1	89.3
#14 gauge aluminum tray	1.90 (0.075)	6:1	15.5	23,750:1	87.5
2 inch metallic tubing	1.65 (0.065)	6.7:1	16.5	3,350:1	70.5
2 inch rigid galvanized conduit	3.91 (0.154)	40:1	32.0	8,850:1	78.9

5.4.2 Tray Spacing

Tables 5-2, 5-3, and 5-4 list the recommended minimum distances between trays, trays and conduits, and conduits, respectively. The dimensions provided in the tables are between the top of one tray/conduit and the bottom of the tray/conduit above, or between sides of adjacent trays/conduits.

Trays containing DATA 1 or DATA 2 cables should have solid bottoms and covers to complete shielding. Ventilation slots or louvers may be used only in trays containing DATA 3 cables.

Table 5-2 — Tray Spacing (Inches)

CABLE CLASS	DATA 1	DATA 2	DATA 3	DATA 3A	DATA 4	DATA 4A
DATA 1		NOTE 2	6	6	26	26
DATA 2	NOTE 2		6	6	18	26
DATA 3	6	6		0	NOTE 1	12
DATA 3A	6	6	0		8	18
DATA 4	26	18	NOTE 1	8		0
DATA 4A	26	26	12	18	0	

NOTE 1: DATA 3 and DATA 4 cables may be run in a common tray but should be separated by a barrier. This barrier does not have to be grounded.

NOTE 2: DATA 1 and DATA 2 cables may be in a common tray, provided they are separated by a grounded steel barrier.

Table 5-3 — Tray/Conduit Spacing (Inches)

CABLE CLASS	DATA 1	DATA 2	DATA 3	DATA 3A	DATA 4	DATA 4A
DATA 1		1	4	4	18	18
DATA 2	1		4	4	12	18
DATA 3	4	4		0	0	8
DATA 3A	4	4	0		6	12
DATA 4	18	12	0	6		0
DATA 4A	18	18	8	12	0	

Table 5-4 — Conduit Spacing (Inches)

CABLE CLASS	DATA 1	DATA 2	DATA 3	DATA 3A	DATA 4	DATA 4A
DATA 1		1	3	3	12	12
DATA 2	1		3	3	9	12
DATA 3	3	3		0	0	6
DATA 3A	3	3	0		6	9
DATA 4	12	9	0	6		0
DATA 4A	12	12	6	9	0	

5.5 CABLE AND CONDUIT SELECTION

Select proper cables for each type of system connection. All programmable controller system wires and cables can be classified into two categories: high level conductors and low level conductors.

High level conductors carry high voltage and/or high current. Examples include ac I/O lines; high level dc I/O lines connected to such devices as switches, relays, or solenoids; and ac power lines. These are typically IEEE 518 DATA 2, DATA 3, and DATA 4 categories. See the Cable Spacing subsection, 5.5.2, for more information.

Low level conductors carry low voltage and/or low current. Typical low level conductors are serial communication cables (those that carry data to and from remote I/O chassis, intelligent I/O modules, programmable controllers, programming terminals, computers, or printers); dc I/O lines connected to solid state devices such as proximity switches or photoelectric sensors; TTL signal lines; analog signal lines; encoder signal lines; motion control signal lines; parallel I/O communication cables. These are typically IEEE 518 DATA 1 category. See the Cable Spacing subsection, 5.5.2, for more information.

5.5.1 Cabling Considerations

Observe the following guidelines when you route cables to the Logic Manager System and remote enclosures. Pay special attention to separating data cables from power cables to avoid electrical noise problems.

- Maintain a 60°C, or less, ambient air temperature at the bottom of each Logic Manager System card file.
- Do all wiring according to IEEE Standard 518-1977.
- When you use a wire duct less than 3.5 inches deep, allow at least 1.5 inches between it and the rack assembly. For a wire duct greater than 3.5 inches deep, allow clearance of 2 inches.
- Separate all 440 Vac (or higher) electrical supply lines as far from Logic Manager System components as possible. See Tables 5-2, 5-3, and 5-4 for the recommended minimum distances.
- Run all 440 Vac lines in separate conduits from I/O wiring ducts.
- Separate all ac and low level dc wiring. See Tables 5-2, 5-3, and 5-4 for the recommended minimum distances.
- When 8 inch deep enclosures are used, door stiffeners are not permitted in front of the Logic Manager System components.
- Shields of data communications cables must be terminated at both ends. Shields of analog signal cables must be terminated to ground only at the signal source. If a termination is not made, or is intermittent, noise immunity is lost.
- Module thumbscrews must be tightened to ensure proper grounding for the signal shield connections.

- Data and power cables should cross at right angles and not run in parallel for any distance.
- Ensure that the entire communications network system is grounded to earth ground.
- Maintain shield continuity. Ensure that shield leads are not broken.
- Route cables around rather than through high noise areas.

5.5.2 Cable Spacing

It is recommended that noise creating sources (those with time varying voltage or time varying current) be separated from data signal cable. Cables with similar levels of noise susceptibility should be grouped together, and those with similar levels of noise generation should be separately grouped in trays and conduits.

IEEE 518 defines four classes of wiring that differ in signal level and noise susceptibility:

DATA 1: High noise susceptibility. Analog signals of less than 50 V and digital signals of less than 15 V.

Examples: 628-2000 series parallel cable
Belden 9182 and 9729 serial I/O cable

DATA 2: Medium susceptibility. Analog signals greater than 50 V, regulating signals of 50 V with currents less than 20 A; ac feeders less than 20 A.

Example: Power cable to a card file

DATA 3: Low susceptibility. Switching signals greater than 50 V, analog signals of 50 V with currents less than 20 A; ac feeders less than 20 A.

DATA 4: Power ac and dc buses of 0-1000 V with currents of 20-800 A.

DATA 3 and DATA 4 classes have one subclass each for applications that require special cable and spacing. In the tables 5-2, 5-3, and 5-4, the subclass for DATA 3 is listed as DATA 3A; the subclass for DATA 4 is listed as DATA 4A. See IEEE 518 for complete information on wiring classes and subclasses. Examples of DATA 3A and DATA 4A are as follows:

- Signals from communicating fields and line resistors.
- Signals from shunts to regulators.
- Power greater than 1000 volts or greater than 800 amperes or both.

5.6 WIRING

Wiring connections to the individual modules is the last step in the Logic Manager System installation process. Depending on the system configuration, parallel I/O cables may have to be wired to the appropriate PIOMs, or serial I/O cables may have to be wired to the appropriate SIOMs. I/O terminal blocks must be installed and wired. Power cables must be connected to the power supply modules. See this section, the Power and Ground Installation section, and the Process I/O Wiring Installation section for wiring information.

5.6.1 Parallel I/O System Cabling

The parallel I/O system is designed for applications where long distances between I/O card files and the processor are not required. Parallel data cable is in the DATA 1 category. The Logic Manager System permits a total maximum data cable length of 30 meters (98 feet) for parallel I/O. Refer to Table 5-5 for parallel I/O cable lengths available. Figure 5-3 illustrates the parallel I/O system wiring from the Parallel Link Driver Module (PLDM) to the first Parallel I/O Module (PIOM). Standard D type connectors are supplied on the modules.

Install the parallel cables as follows:

1. Route the appropriate cable from the PLDM in the Logic Manager card file (slot N) to the PIOM in the first parallel I/O card file in the system.
2. Route the appropriate cable from the PIOM of the first parallel I/O card file to the PIOM of the second parallel I/O card file.
3. Repeat this process, connecting all parallel I/O card files.
4. Do not exceed a total cable length of 30 meters (98 feet).

NOTE

The slide latch devices on the D type connectors must be latched down to secure the connection.

Inside a cabinet or cabinet complex, parallel I/O cable ML-KLIP00 is used and is tailored to the required length.

Between noncomplexed cabinets, parallel I/O connections must be made with the standard 628-2000 series cables supplied by Honeywell. Cables range in length from 20 centimeters (9 inches), model number 628-2000, to 30 meters (98 feet), model number 628-2026. See Table 5-5 for a listing of the compatible cables.

Parallel I/O cables do not require any terminating resistors.

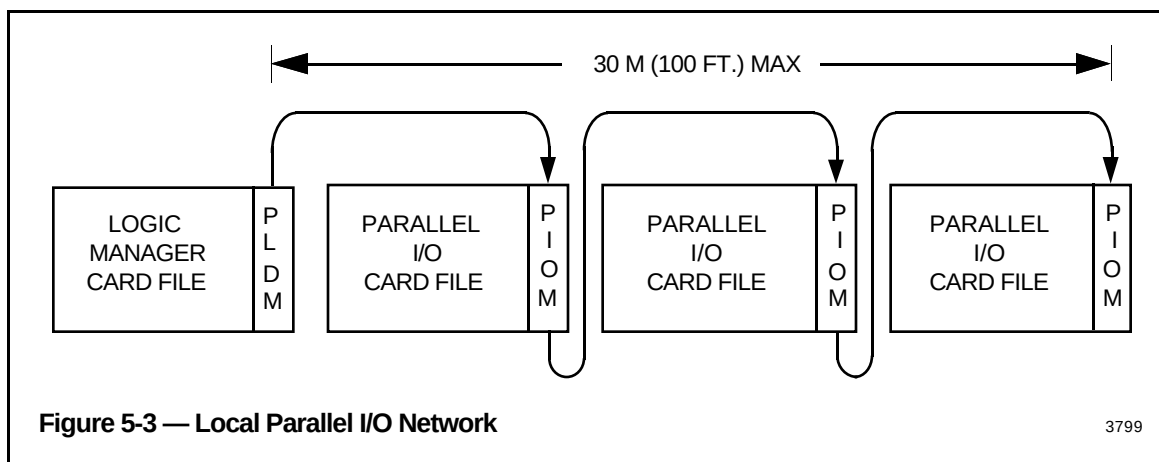


Table 5-5 — Parallel I/O Cable Length

PARALLEL CABLE MODEL NUMBER	LENGTH	PARALLEL CABLE MODEL NUMBER	LENGTH
628-2000	20 cm (9 in.)	628-2017	12 m (39 ft.)
628-2001	35 cm (14 in.)	628-2018	14 m (46 ft.)
628-2002	120 cm (4 ft.)	628-2019	16 m (52 ft.)
628-2003	180 cm (6 ft.)	628-2021	20 m (65 ft.)
628-2004	3 m (10 ft.)	628-2022	22 m (72 ft.)
628-2010	4 m (13 ft.)	628-2023	24 m (78 ft.)
628-2011	5 m (16 ft.)	628-2024	26 m (85 ft.)
628-2012	4 m (13 ft.)	628-2026	30 m (100 ft.)
628-2016	10 m (32 ft.)		

5.6.2 Serial I/O System Cabling

Remote serial I/O configurations are available with the Logic Manager. A serial I/O configuration has the advantage of permitting the I/O card files to be located near the process being controlled, reducing the length of costly control circuit wiring. Serial I/O is accomplished with one or two SLMs (Serial Link Module) located in the Logic Manager card file and an SIOM (Serial I/O Module) in each of the remote serial I/O card files. The system is installed in a multidrop configuration (see Figures 5-4 and 5-5 for examples). Terminating resistors are required at the ends of the multidrop communications lines. The rules for installing terminating resistors are as follows:

- If an SIOM is at the end (either end) of the communications line, install two terminating resistors in that SIOM. Do not install resistors in SIOMs which are at intermediate locations on the communications circuit. (See Figures 5-4 and 5-5 for examples). Refer to Figure 7-8 for information on how to install the resistors in an SIOM board.
- If an SLM channel is at the end of a communications circuit, install one terminating resistor for that channel. There are two channels available in an SLM, and each channel has a place for an optional terminating resistor. (See Figure 5-4 for an example of an SLM channel requiring a terminating resistor). Do not install a terminating resistor in an SLM channel if that SLM channel connects to the communications cable at an intermediate location (see Figure 5-5 for an example). Refer to Figure 7-4 for information on how to install the resistor(s) in an SLM board.

A Logic Manager card file can have two SLMs, and each SLM can support two communications circuits. Each circuit can have up to 16 remote I/O card files.

A single cable, consisting of two individually shielded twisted pairs, may be used in configurations that require up to 4000 feet of cable. We recommend Belden Datalene 9729. Two twin-axial (single twisted pair) cables are used in applications requiring up to 10,000 feet (max.); Belden Datalene 9182 is recommended. See Figure 5-6. Use 200 ohm terminating resistors when using the 9729 cable, and use 300 ohm terminating resistors when using the 9182 cable. Serial data cable falls under the "DATA 1" category.

NOTE: Any cable other than Belden 9729 or Belden 9182 must be approved by Honeywell.

The cable is connected from the SLM, housed in the processor card file, to the SIOM of the first serial I/O card file in the system. The first card file SIOM is then connected to the SIOM of the second card file. Up to 16 I/O card files may be used in each serial link, providing the processor addressing capacity is not exceeded.

The front of the 621-9939 SLM contains a 19-pole terminal block for cable connection. The 621-9940 SIOM contains a 12-pole terminal block on the upper portion of the module front. The terminal assignments and wiring connections are shown in Figure 5-7.

Install the 621-9939 SLM wiring as follows:

1. Route the cable from the SLM to the SIOM of the first remote serial I/O card file in the system.
2. Strip approximately 1 inch of the outer jacket of the cable exposing the wires.
3. Connect the SLM terminals as shown in Figure 5-7.

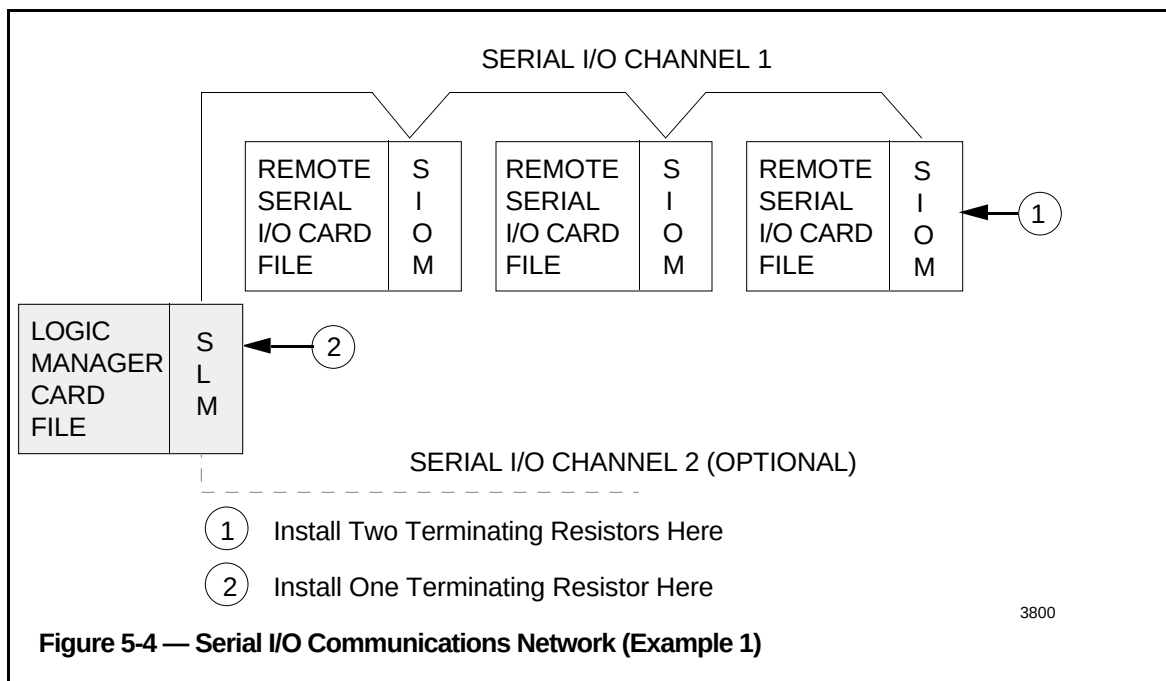
4. Strip the cable and make the SIOM connections.
5. Connect the SIOM of the first serial I/O card file to the SIOM of the second serial I/O card file.
6. Continue to connect SIOM to SIOM until all the remote serial card files in the channel have been connected.
7. Repeat the process for each additional channel.

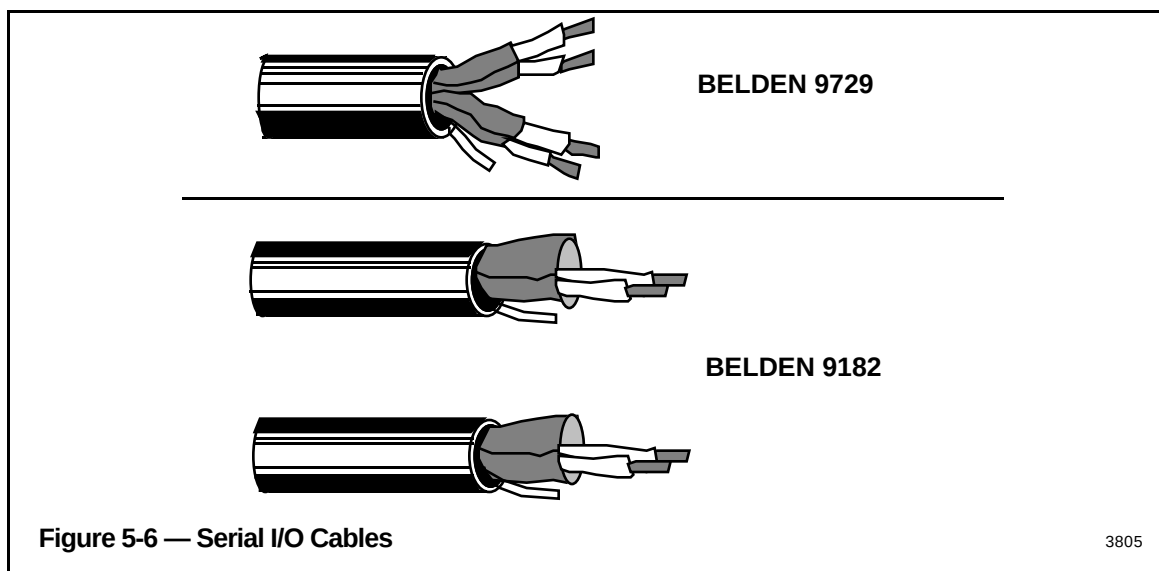
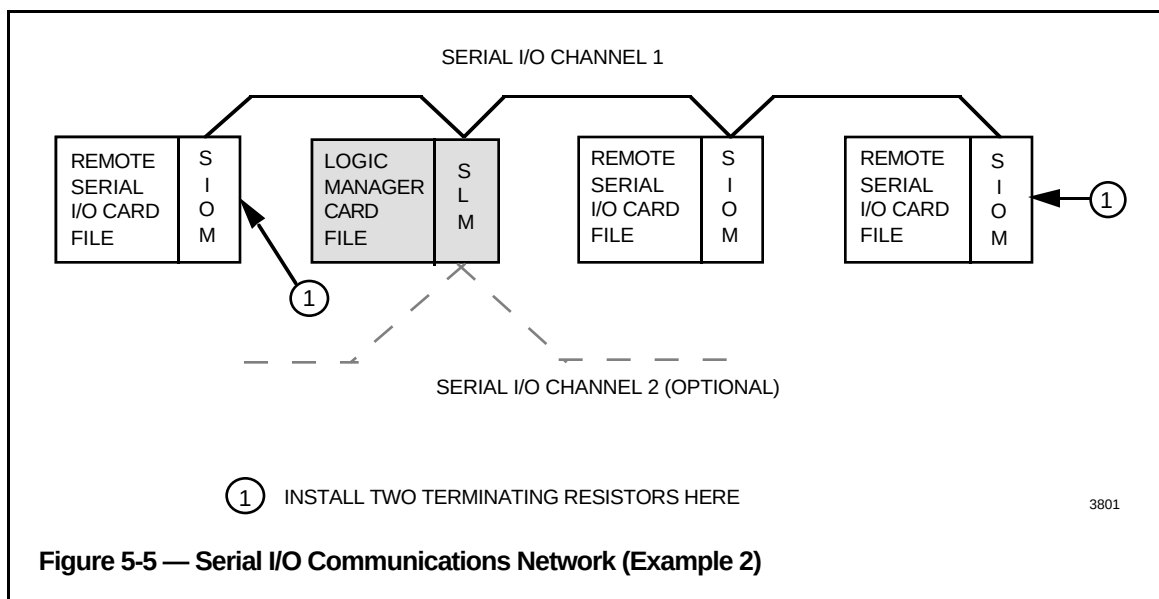
For convenience, terminal blocks may be used near the remote SIOMs (within 10 feet) to act as cable junction points for the inbound cable, the outbound cable, and a short cable to the SIOM. This is shown in the lower portion of Figure 5-4.

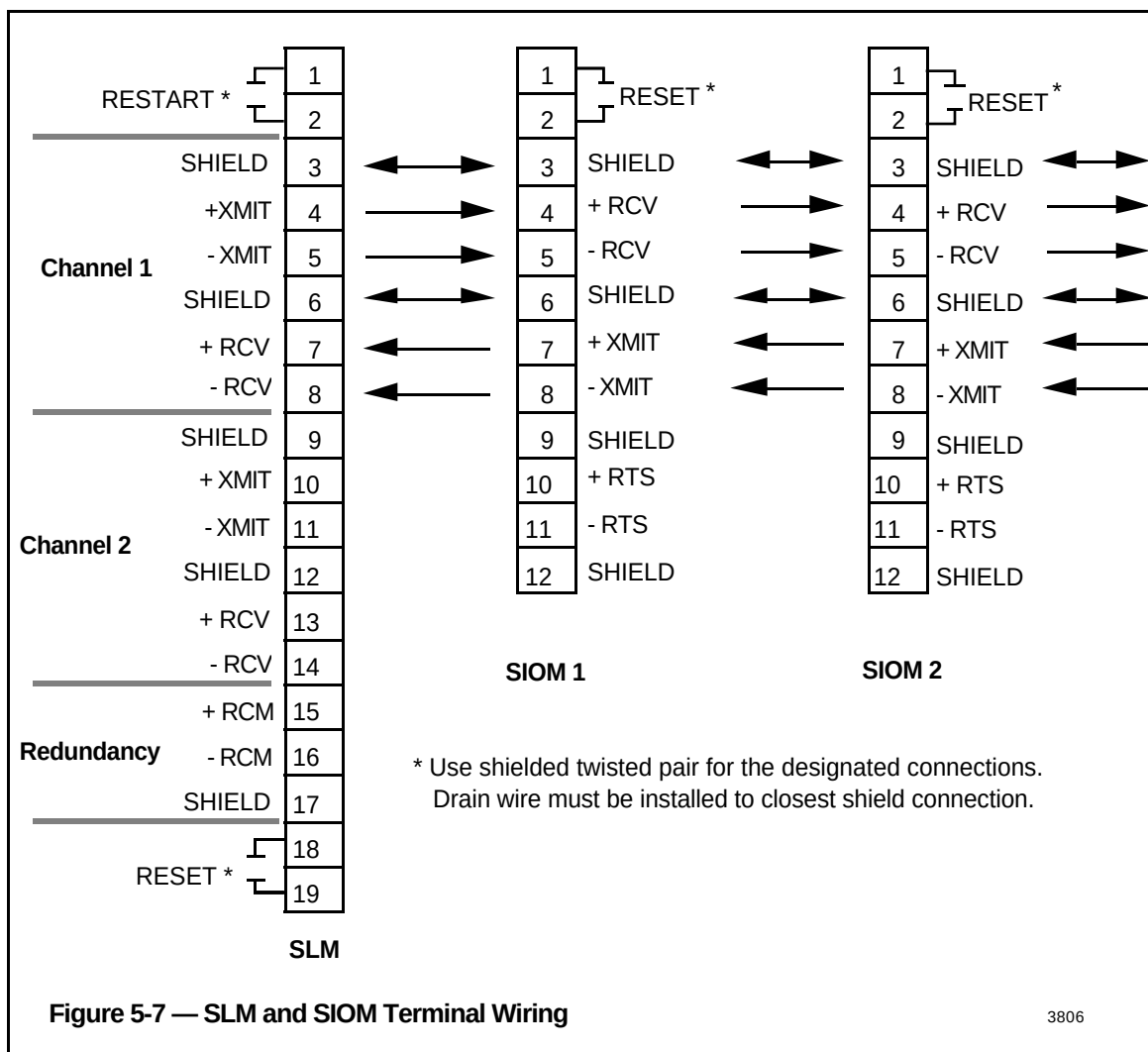
Table 5-6 — Serial I/O Cable Specifications

	BELDEN DATALENE 9182	BELDEN DATALENE 9729
STANDARD LENGTH	500 and 1000 feet	100, 500, and 1000 feet
WIRE DIAMETER	22 AWG (0.79 mm) tinned copper drain wire	24 AWG (0.61 mm) tinned copper drain wire
IMPEDANCE	150 ohm nominal	100 ohm nominal
PROPAGATION	78% speed of light 4.27 nsec/m or 1.3 nsec/ft	78% speed of light 4.27 nsec/m or 1.3 nsec/ft
CAPACITANCE	28.9 pf/m (8.8 pf/ft)	41.0 pf/m (12.5 pf/ft)
TYPE	One pair with shield and tinned copper drain wire	Two shielded pairs with individual tinned copper drain wire

NOTE: Any cable other than the Belden cables noted above must be approved by Honeywell.







5.6.3 Redundant Control System Wiring

The Logic Manager Redundant Control System provides a means of incorporating a backup processor in the Logic Manager system. The redundancy system consists of two identically configured Logic Manager processors, each containing a model 620-0059 Redundancy Control Module (RCM). The RCMs exchange data through a model 628-5000 RCM Data Cable that is supplied with the RCMs. The cable is 3 meters (10 feet) long and has 25-pin D-type connectors on each end. The RCMs initiate an orderly switchover of system I/O from the lead (primary) to the backup (secondary) processor if a critical processor or serial link fault occurs.

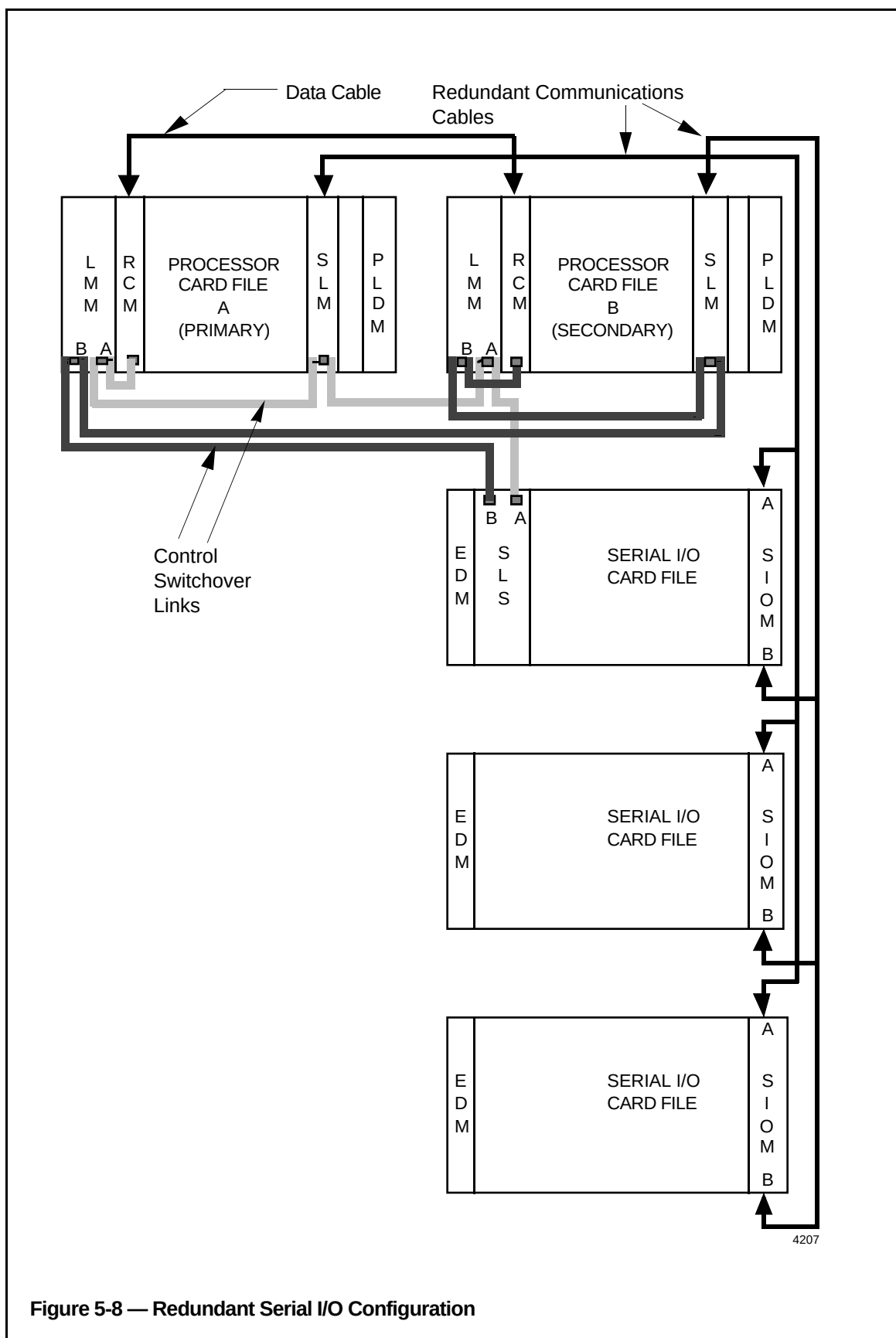
One processor must be designated and set up by the user as the primary or lead processor, and the other processor as the secondary or backup processor. This configuration is set up by position 3 of switch SW1 on each RCM. On the primary processor (Processor A in Figure 5-8), SW1-3 of the RCM should be in the open (off) position, and on the secondary or backup processor (Processor B in Figure 5-8), SW1-3 on the RCM should be in the closed (on) position.

Both processors are connected to a common I/O system that may be parallel I/O, serial I/O, or a combination of the two. Serial I/O systems only are recommended. Figure 5-8 shows the configuration and wiring for a redundant system using serial I/O.

Serial I/O redundancy requires the model 621-9938R redundant Serial I/O Module (SIOM) in each serial I/O card file. This version of the SIOM has dual communication ports so that it can attach to the redundant serial I/O lines as shown in Figure 5-8. The optional model 621-9928R Serial Link Selector (SLS) is available to switch a communication link so that it is always connected to the active processor. This serial link would normally be used for the Loader/Terminal.

Each RCM originates a control signal, called a control switchover link, which is wired to corresponding switch control inputs on the LMMs, SLMs, and the SLS if used. These links are wired using shielded twisted pair with characteristic impedance greater than 60 ohms, such as Belden 8761 cable. The cable is wired beginning at the RCM, following the pattern shown in Figure 5-8. At the RCM, LMM, and optional SLS, connections are made at 3-position terminal blocks. The three positions are labeled “+”, “-”, and “S.” The “+” and “-” terminals are for the twisted pair connections, and the “S” terminal is for the shield drain wire connections. The processor that is configured as the lead or primary processor must be connected to the terminal blocks labeled “A”. At the SLM, connection is made to positions 15 (“+”), 16 (“-”), and 17 (“S”) of the 19-position terminal block as shown in Figure 5-7. When the wiring is completed, all of the “+” terminals should be connected together, all of the “-” terminals should be connected together, and the shield drain wires should be connected to the “S” terminals at each end of each section.

The last connection point in each link requires a 100 ohm 1/2 watt terminating resistor. In Figure 5-8, this would be at the SLS. On the SLS, terminating resistors can be switched onto the line with switch SW1. To add a resistor to link A, turn on SW1-1. To add a resistor to link B, turn on SW1-2. If the SLS is not used, the SLM would be at the end of the control switchover link. The 100 ohm resistor supplied in the terminating resistor kit may be installed in the SLM (location R35) for this purpose. The maximum length of each control switchover link is 6 meters (20 feet).



5.7 I/O TERMINAL BLOCK INSTALLATION

5.7.1 8- and 16-Point Terminal Block Installation

The I/O module terminal block set (8 or 16 point) consists of a top and bottom half. Each is labeled TOP or BOTTOM to prevent improper installation. The TOP terminals are always L1 or +Vdc, and the BOTTOM terminals are always L2 or -Vdc. The 8-point terminal block models are factory jumpered at T1 and T2 and B1 and B2, providing a 12 A capacity. The 16-point terminal block models are factory jumpered at T1, T2, and T3, and B1, B2, and B3, providing an 18 A capacity. The terminals are screw terminals with pressure pads.

CAUTION

Do NOT remove terminal block jumpers unless specified by the module wiring instructions in Section 8 of this manual.

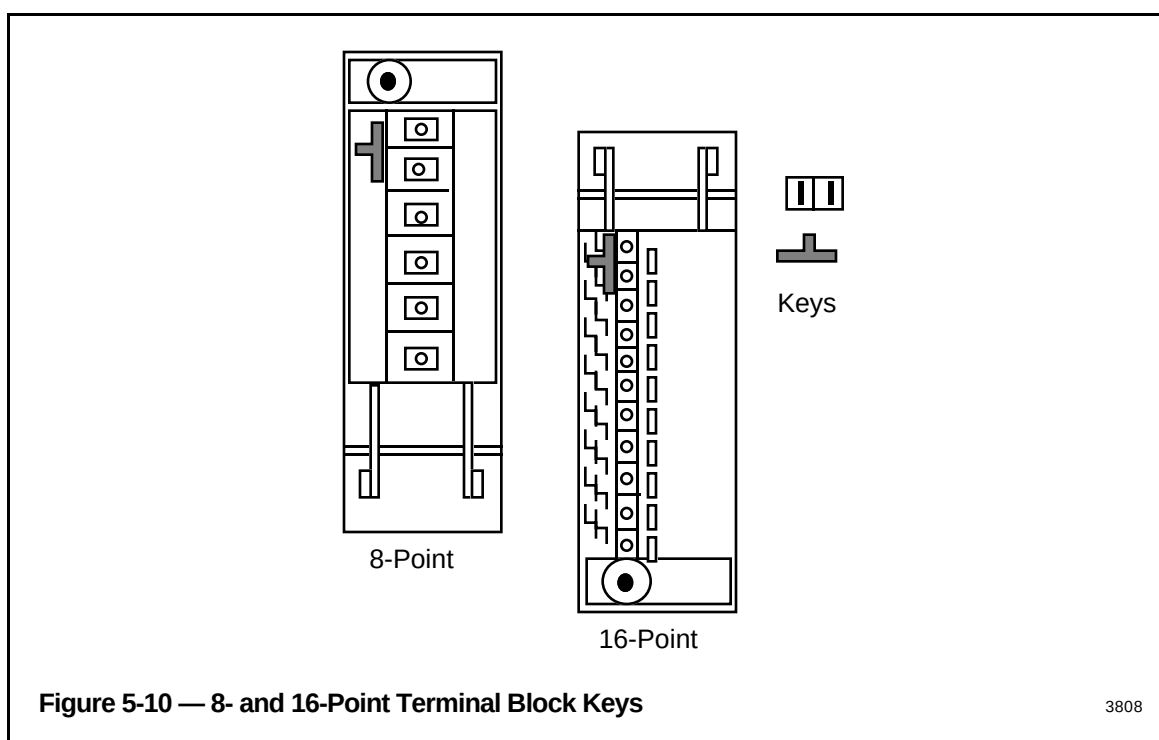
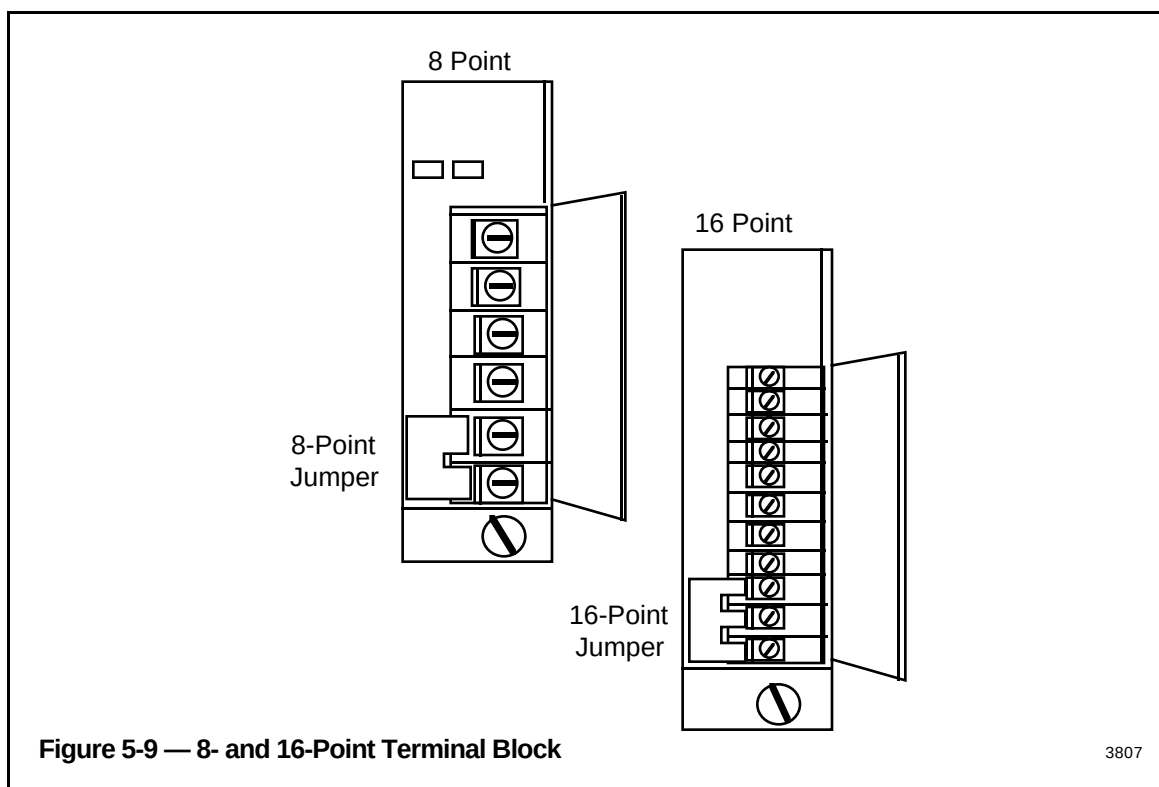
Each terminal block set includes a hinged terminal cover plate to protect the terminals and provide for placement of I/O address labels. The terminal block is inserted into the T-shaped slots on the top and bottom of the rack chassis. A captive panel screw and two locating pins on each terminal block enable it to be secured to the I/O module and the rack chassis. See Figure 5-9.

The terminal blocks can be separated from the I/O module and locked in an open position without disconnecting the I/O module wiring. This enables I/O module removal and replacement with a minimum of maintenance effort.

The I/O module circuit boards contain slots positioned to differentiate between the various module voltage levels available in the Logic Manager I/O System. The user should install terminal block keys that match the module circuit board slots to ensure that modules are replaced in the appropriate rack slots, preventing module damage. Four keys are supplied with each terminal block set. See Figure 5-10.

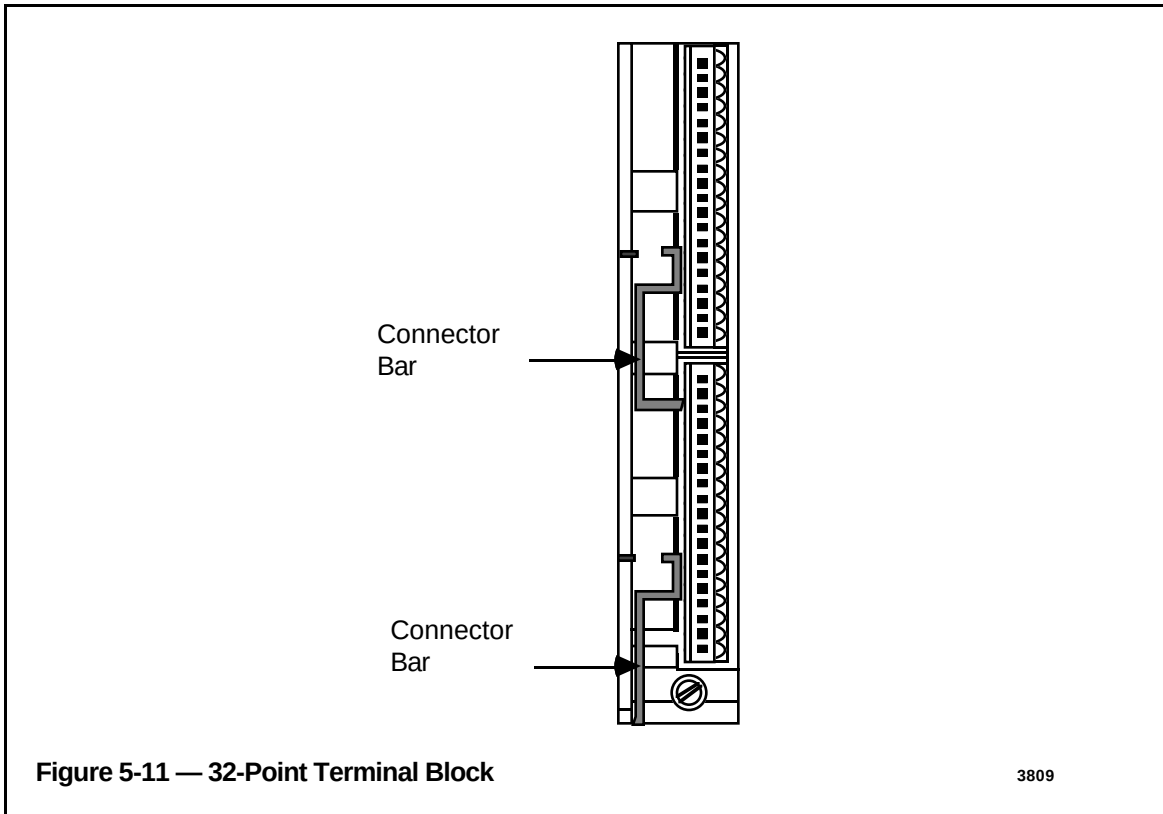
Install the terminal blocks as follows:

1. Insert the keys in the terminal blocks to match I/O module circuit board slots.
2. For 8-point modules, insert the tie wrap down through the left hole in the terminal.
3. Insert terminal blocks into the T-shaped slots in the top and bottom of the card file chassis.
4. Slide the blocks forward and pull the BOTTOM block down and the TOP block up over the I/O module circuit board edge.
5. Align the locating pins and captive panel screw with their mating holes and tighten the panel screw to secure the terminal block to the I/O module and card file chassis.



5.7.2 32-Point Terminal Block Installation

The 32-point I/O modules use two 19-position terminal blocks that attach to the front of the I/O module. The field wiring can be installed in the front of the module by using set screws that can be accessed without removing the connector from the module. If a connector must be removed, metal connector bars eject the connectors from the module. See Figure 5-11.



5.7.3 I/O Wiring

Refer to the I/O module wiring diagrams in Section 8 of this manual and use the following procedure:

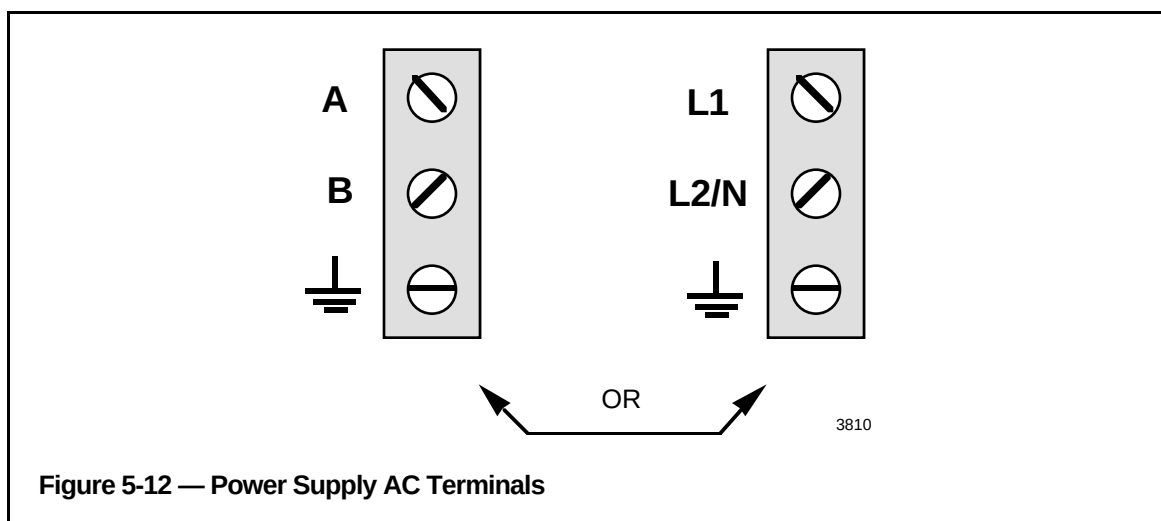
1. Strip approximately 6 mm (1/4 inch) of wire.
2. Open the terminal block cover by placing a screwdriver between the notched terminal partitions and the left edge of the cover, popping the cover open.
3. Wire each terminal.
4. For 8-point modules, fasten the wire wrap around the wires and cut the wrap.
5. Close the terminal block cover and place the I/O address labels on the terminal cover.

NOTE

AC output modules may be wired without regard to phase relationships. AC input modules should have L1 connected to terminals T1 and T2 when used in ungrounded control circuits.

5.8 POWER SUPPLY AC WIRING

If installing the LM without a cabinet you will have to wire the power wiring as shown below. The terminal block on the front of each power supply module is labeled, from top to bottom:



Connect the wiring as follows:

1. Strip approximately 1/4 inch (6 mm) from the end of each wire.
2. Connect the hot lead to the terminal marked “A” or “L1”.
3. If 115 Vac, connect the neutral lead to the terminal marked “B” or “L2/N.”
If 230 Vac, connect the other hot lead to the terminal marked “B” or “L2/N.”
4. Connect the safety ground lead to the terminal marked with the ground symbol.
5. Snap the yellow safety cover over the terminal block.

CAUTION

Voltage selection jumper and fuse must agree with the line power being used. Refer to the Logic Manager Service.

POWER AND GROUND INSTALLATION

Section 6

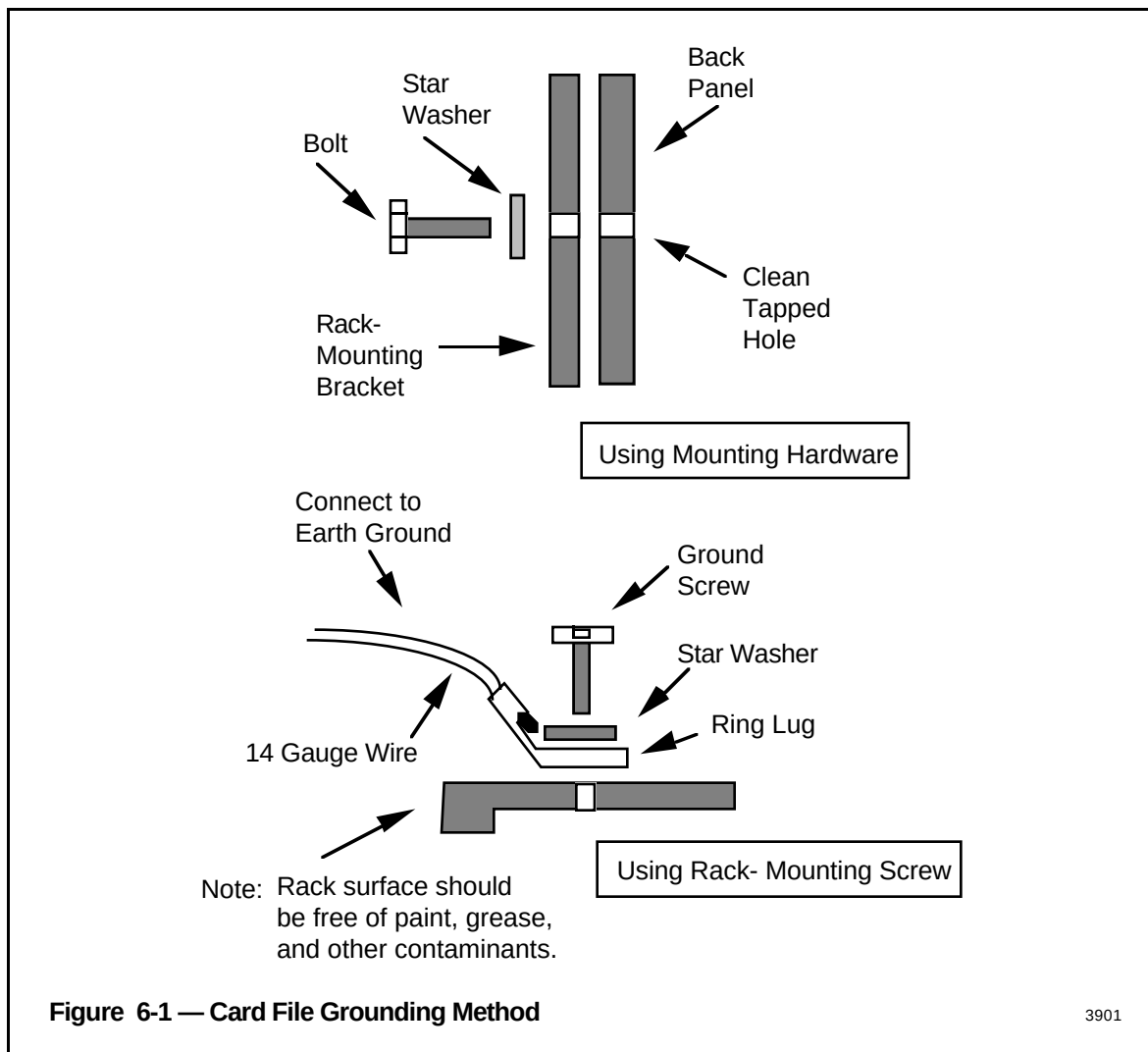
This section details the installation of the cabinet and component connections to ac wiring and ground.

6.1 GROUND CONNECTION

6.1.1 System Ground Considerations

Adequate grounding is important for safety considerations and for reducing electromagnetic noise interference. All earth ground connections must be permanent and provide a continuous low impedance path to earth ground for induced noise currents and fault currents. Follow these suggested steps:

- Use the system grounding setup as shown in Figures 6-1 and 6-2. System ground is established in one of two ways with the card file mounting hardware.
- Establish system ground by mounting all Logic Manager Processor and I/O card files on a metal back panel and connecting the back panel to earth ground, or by using the special rack mounting screws and star washers provided with each system card file, connect a screw to the earth ground connection point using 14 gauge copper wire. See Figure 6-1.
- Reference to earth ground all I/O card files that are installed in separate enclosures remote from the Logic Manager card file.
- Ensure that all Logic Manager Processor card file power supply ground terminals are connected to earth ground as shown in Figure 6-2.
- Securely fasten all cover plates in order to maintain system integrity and to ensure system grounding.
- Select the proper voltage (115 Vac or 230 Vac) before supplying power to the system. Some Logic Manager System components allow voltage selection.
- Install and remove modules only when power is removed from the system. Do not attempt to install or remove modules from the rack while the card file is powered; doing so may cause damage to the module, may possibly affect the backpanel communications and may turn on outputs unintentionally for one program scan.
- Ensure that cover plates are installed over all blank slots. (The cover plates supplied with Logic Manager System card files and modules shield the electronics from electromagnetic noise.)
- Firmly tighten all thumbscrews on module front plates and cover plates.
- Consult IEEE Standard 518-1977, the National Electric Code, and the Cabling Considerations, subsection 5.5.1 of this manual, before wiring the system components.



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6.1.2 Cabinet Ground

Ground wires from each card file in the cabinet are collected at a bus bar located at the bottom of the cabinet. This bus bar must be connected to building safety ground with a 1.67 mm (14 AWG) wire in conduit or a 5.3 mm (4 AWG) wire without conduit. **Each cabinet must be connected individually to building safety ground and not daisy-chained with other cabinets or equipment.** See Figures 6-1 and 6-2.

6.1.3 Equipment Card File Ground

In situations where there is no cabinet and individual equipment racks are to be panel mounted, grounding must be provided between the rack and the building safety ground. The same method applies. Use 1.67 mm (14 AWG) wire in conduit or 5.3 mm (4 AWG) wire without conduit. Each equipment rack must be connected individually to building ground and not daisy-chained with other equipment. See Figures 6-1 and 6-2.

WARNING

It is imperative that the entire **TotalPlant** Solution (TPS) ground system be in compliance with wiring instructions in this section of the manual. Failure to do so can result in injury to personnel, damage to equipment, or ground loops that may cause errors in the operation of the equipment.

6.2 AC POWER CONNECTION

6.2.1 Cabinet Power

Conductors and breakers are sized using rules in the *Logic Manager Site Planning* manual in the *System Site Planning* binder. Install power as specified by your site planning drawings; also see Figure 6-2.

Do not use circuit breakers larger than 15 amperes; the power supply subassembly is not rated for larger circuit breakers.

A cabinet will have a circuit breaker box with either two or six dual 15 ampere circuit breakers boxes at the bottom of the cabinet. Each circuit breaker services a single card file.

6.2.2 Equipment Card File Power

When no cabinet exists to house the equipment card files, and they are to be panel mounted, individual circuit breakers rated no greater than 15 amperes must be provided for each equipment rack. These circuit breakers should be located near the equipment to conveniently control power to the equipment.

You can find the recommended size of conductors and circuit breakers in the *Logic Manager Site Planning* manual in the *System Site Planning* binder. Install power as specified by your site planning drawings; also see Figure 6-2.

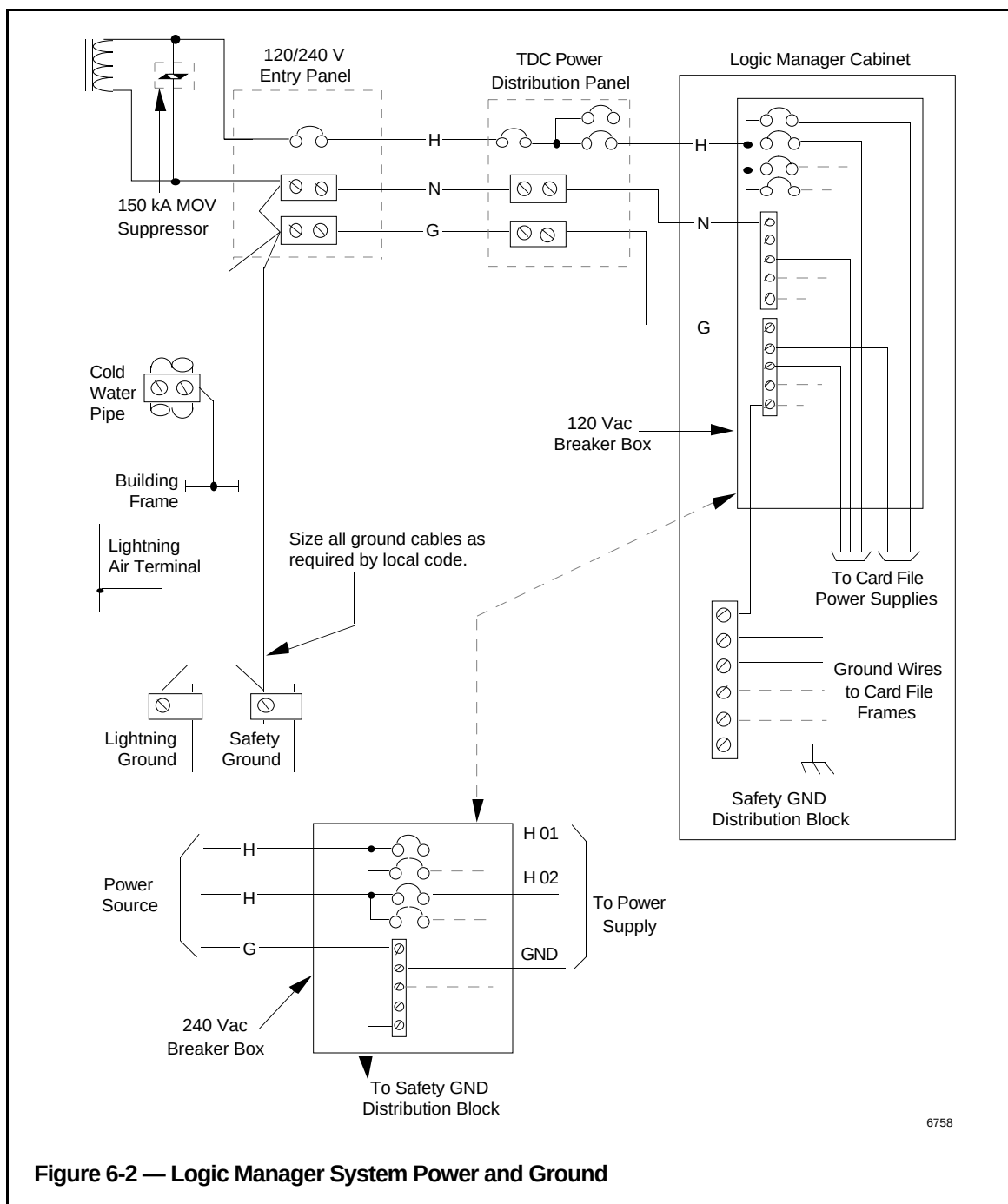


Figure 6-2 — Logic Manager System Power and Ground

SYSTEM CONFIGURATION

Section 7

This section details the acceptable configuration of the Logic Manager System and the module switch settings that need to be confirmed as a part of the installation process.

7.1 SYSTEM CONFIGURATION OVERVIEW

The Logic Manager System (LM) consists of a single or redundant Logic Manager Processor card file(s) and either, but not both, serial or parallel I/O card file(s). That is, serial and parallel I/O will not be mixed in the same “control group” of any Logic Manager System. However, although this serial/parallel mix is not permitted by the automatic configurator, it is technically permissible and can be accomplished as a “Special System.”

Due the inherent nature of the risks associated with parallel I/O, these configuration rules do not support a redundant LM used with parallel I/O. Only serial I/O is supported with a redundant LM. A redundant LM with parallel I/O is a “Special System.”

Redundancy in the Logic Manager System involves a single I/O system controlled by dual (redundant) processors and is compatible only with the Logic Manager Processors. Redundancy is managed by two Redundant Control Modules (RCM), one located in each Logic Manager card file. Any optional module (including the LMM) which resides in the Logic Manager card file, must also be redundant.

A Logic Manager System is not allowed to contain a Hiway Interface Module (HIM).

7.2 LOGIC MANAGER SYSTEM CARD FILES

7.2.1 Logic Manager Nonredundant Processor Card File

The basic nonredundant Logic Manager Processor card file contains the standard card modules listed below and is designated model ML-LMSA35 if it has an ac power supply in slots J/K. If it has a dc power supply in slots J/K, it is designated model ML-LMSD35.

Option modules listed in subsection 7.2.3 can be added to the card file as required.

Table 7-1 — Nonredundant Logic Manager Processor Card File (Front View)

SLOT													
A	B	C	D	E	F	G	H	I	J	K	L	M	N

SLOT	MODEL NUMBER	MODULE DESCRIPTION	NOTES
A & B	620-0071	Logic Manager Module	6,8
C	629-4004	Blank Metal Coverplate	1,2,4
D	629-4004	Blank Metal Coverplate	1,2,4
E	620-0024	Memory Module (24 K words)	
F	620-0056	Register Module (4 K by 4 K)	
G	620-0054	System Control Module	
H	620-0080	Processor Module	
I	620-0085	I/O control Module (2048 I/O)	
J&K	620-0036	Power Supply Module (115/230 Vac)	6
J&K	620-0047	Power Supply Module (24 Vdc)	6
L	621-9939	Serial Link Module	
M	621-9900	Blank Plastic Coverplate	2,3,5
N	620-0088	Parallel Link Driver Module	7

- Notes:
1. This slot is for option modules.
 2. Check DIP switch settings for option modules.
 3. This slot is for the optional second Serial Link Module, model 621-9939.
 4. Install the blank metal coverplate 629-4004 if this slot is empty.
 5. Install the blank plastic coverplate 621-9900 if this slot is empty.
 6. The module is double width.
 7. The Parallel Link Driver Module is required even when serial I/O is used exclusively, because the PLDM performs some system control functions in addition to supporting parallel I/O.
 8. The Logic Manager Module has UCN drop cable adaptors mounted on the front panel.

Some of the modules shown above in the basic card file configuration have DIP switch settings that are preset at the factory. Refer to individual module descriptions in the Logic Manager Card Modules section for information.

DIP switch settings on option modules added to the card file must be set at the time of installation. As two option slots are available, the option modules must be set to backpanel address 2 or address 3. If two option modules are implemented, one must be set to address 2 and the other set to address 3.

7.2.2 Logic Manager Redundant Processor Card File

A basic redundant Logic Manager System contains two processor card files with standard card modules installed, as shown below, and is designated a model ML-LMRA35 if it has an ac Power Supply Module in slots J/K. If a dc Power Supply Module is installed in slots J/K, the designation is model ML-LMRD35. A three meters (10 feet) long model 628-5000 Redundancy Control Data Cable is also included to connect the Redundancy Control Modules in each card file.

Option modules listed in subsection 7.2.3 can be added to the card file as required.

Table 7-2 — Redundant Logic Manager Processor Card File (Front View)

SLOT													
A	B	C	D	E	F	G	H	I	J	K	L	M	N

SLOT	MODEL NUMBER	MODULE DESCRIPTION	NOTES
A & B	620-0071	Logic Manager Module	6,8
C	620-0059	Redundancy Control Module	
D	629-4004	Blank Metal Coverplate	1,2,4
E	620-0024	Memory Module (24 K words)	
F	620-0056	Register Module (4 K by 4 K)	
G	620-0054	System Control Module	
H	620-0080	Processor Module	
I	620-0085	I/O control Module (2048 I/O)	
J&K	620-0036	Power Supply Module (115/230 Vac)	6
J&K	620-0047	Power Supply Module (24 Vdc)	6
L	621-9939	Serial Link Module	
M	621-9900	Blank Plastic Coverplate	2,3,5
N	620-0088	Parallel Link Driver Module	7

- Notes:
1. This slot is for option modules.
 2. Check DIP switch settings for option modules.
 3. This slot is for the optional second Serial Link Module, model 621-9939.
 4. Install the blank metal coverplate 629-4004 if this slot is empty.
 5. Install the blank plastic coverplate 621-9900 if this slot is empty.
 6. The module is double width.
 7. The Parallel Link Driver Module is required even when serial I/O is used exclusively, because the PLDM performs some system control functions in addition to supporting parallel I/O.
 8. The Logic Manager Module has UCN drop cable adaptors mounted on the front panel.

Some of the modules shown above in the basic card file configuration have DIP switch settings that are preset at the factory. Refer to individual module descriptions in the Logic Manager Card Modules section for information.

DIP switch settings on option modules added to the card file must be set at the time of installation. As only one option slot is available, the option module must be set to backpanel address 2 or address 3.

7.2.3 Logic Manager Processor Card File Option Modules

The following modules may be used in the option slots C and D of the nonredundant processor card file, or slot D of the redundant processor card file:

620-0027	Memory (8 K) Module
620-0038	Control Network Module (CNM)
620-0043	Mod/bus Computer Interface Module (CIM)
620-0048	ABC Data Collection Module (DCM)
620-0052	Dual Port ABC Data Collection Module (DCM)

If an optional communications module is used in a redundant serial I/O system, each of the redundant processor card files must contain an identical option module. Each option module must be linked with an additional model 621-9928R Serial Link Selector Module (SLS).

To utilize a model 620-0038 CNM or model 620-0043 CIM in a redundant serial I/O Logic Manager System, a DCM or CIM must be installed in both processor card files and connected to an additional SLS in an I/O card file.

To utilize a dual port model 620-0052 DCM, an identical DCM must be installed in both processor card files and connected to two additional SLS modules in an I/O card file.

7.2.4 Logic Manager Serial I/O Card File

The basic Serial I/O card file contains the standard card modules shown below and is designated ML-LIOS01.

Table 7-3 — Serial I/O Card File (Front View)

SLOT													
A	B	C	D	E	F	G	H	I	J	K	L	M	N

SLOT	MODEL NUMBER	MODULE DESCRIPTION	NOTES
A	621-0021	Enhanced Diagnostic Module	1,3
*B&C	621-9928R	Serial Link Selector Module (1st card file only)	3,4
D		Any valid 621-xxxx I/O Module	2,3,5
E		Any valid 621-xxxx I/O Module	2,3,5
F		Any valid 621-xxxx I/O Module	2,3,5
G		Any valid 621-xxxx I/O Module	2,3,5
H		Any valid 621-xxxx I/O Module	2,3,5
I		Any valid 621-xxxx I/O Module	2,3,5
J		Any valid 621-xxxx I/O Module	2,3,5
K		Any valid 621-xxxx I/O Module	2,3,5
L		Any valid 621-xxxx I/O Module	2,3,5,7
L&M	621-9933	Power Supply Module (115/230 Vac 15 A)	2,6
M	621-9934	Power Supply Module (115/230 Vac 8 A)	
M	620-9932	Power Supply Module (24 Vdc 8 A)	2
N	621-9940	Serial I/O Module	3
N	621-9938	Serial I/O Module (Dual Port)	2,3,8

- Notes:
1. This is a standard card module.
 2. This is an option card module slot.
 3. Check DIP switch settings.
 - *4. One SLS per redundant system. Other I/O card files cannot contain an SLS but may contain other valid 621-xxxx I/O module in slots B and C.
 5. Install the blank plastic coverplate 621-9900 if this slot is empty.
 6. The module is double width.
 7. Not available for I/O module if occupied by double width 621-9933 power supply.
 8. Model 621-9938 has dual ports for communication with redundant processors.

All low level signal I/O modules (analog, TTL, PIM, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal I/O modules must be grouped together, starting at the rightmost slot and proceeding left.

All I/O slots must be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the slot and set via the appropriate SIOM. Each empty slot must be assigned a quantity of 16 addresses to accommodate spares and allow future expansion. The only slots to be assigned an address of zero are those slots occupied by the left half of a double width module that requires a quantity of eight addresses for its right half portion.

- * The purpose of SLS is to switch a serial device such as the MS-DOS Loader between redundant LMs. The SLS is not needed to make the redundant LMs work.

I/O addresses 2040 through 2047 must not be assigned to any parallel or serial slot of a system containing serial I/O.

The power supply module selected for the card file must be compatible with the power input (115/230 Vac or 24 Vdc) and must supply enough current to the card file to satisfy the total load requirement. On the 115/230 Vac power supplies, change the voltage selection jumper if 230 Vac is used. The voltage jumper must be correct before power is applied.

All systems configured with a MiniCOP (double width 627-1002R or triple width 627-1003R) must contain a 620-0048/0052 DCM installed in the processor card file option slots. The only exception to this rule is when the customer is fully knowledgeable of the attendant required complexity and the amount of relay ladder logic required to access information from the MiniCOP without a DCM.

7.2.5 Logic Manager Parallel I/O Card File

The basic Parallel I/O card file contains the standard card modules shown below and is designated ML-LIOP01.

Table 7-4 — Parallel I/O Card File (Front View)

SLOT													
A	B	C	D	E	F	G	H	I	J	K	L	M	N

SLOT	MODEL NUMBER	MODULE DESCRIPTION	NOTES
A	621-0021	Enhanced Diagnostic Module	1,3
B		Any valid 621-xxxx I/O Module	2,3,4
C		Any valid 621-xxxx I/O Module	2,3,4
D		Any valid 621-xxxx I/O Module	2,3,4
E		Any valid 621-xxxx I/O Module	2,3,4
F		Any valid 621-xxxx I/O Module	2,3,4
G		Any valid 621-xxxx I/O Module	2,3,4
H		Any valid 621-xxxx I/O Module	2,3,4
I		Any valid 621-xxxx I/O Module	2,3,4
J		Any valid 621-xxxx I/O Module	2,3,4
K		Any valid 621-xxxx I/O Module	2,3,4
L		Any valid 621-xxxx I/O Module	2,3,4,6
L&M	621-9933	Power Supply Module (115/230 Vac 15 A)	2,5
M	621-9934	Power Supply Module (115/230 Vac 8 A)	2
M	620-9932	Power Supply Module (24 Vdc 8 A)	2
N	621-9937	Parallel I/O Module	1,3

Notes: 1. This is a standard card module.
 2. This is an option card module slot.
 3. Check DIP switch settings.
 4. Install the blank plastic coverplate 621-9900 if this slot is empty.
 5. The module is double width.
 6. Not available for I/O module if occupied by double width 621-9933 power supply.

All low level signal I/O modules (analog, TTL, PIM, etc.) must be grouped together, starting at the leftmost slot and proceeding to the right. All high level signal I/O modules must be grouped together, starting at the rightmost slot and proceeding left.

All I/O slots must be assigned a quantity of 0, 8, 16, or 32 addresses, depending upon the module installed in the slot and set via the appropriate SIOM. Each empty slot must be assigned a quantity of 16 addresses to accommodate spares and allow future expansion. The only slots to be assigned an address of zero are those slots occupied by the left half of a double width module that requires a quantity of eight addresses for its right half portion, or both slots occupied by a double width module that requires no addresses, such as the Parallel Bus Selector Module. I/O addresses 2040 through 2047 must not be assigned to any parallel or serial slot of any system.

All empty I/O slots must be covered with a model 621-9900 plastic coverplate.

The power supply module selected for the card file must be compatible with the power input (115/230 Vac or 24 Vdc) and must supply enough current to the card file to satisfy the total load requirement. On the 115/230 Vac power supplies, change the voltage selection jumper if 230 Vac is used. The voltage jumper must be correct before power is applied.

All systems configured with a MiniCOP (double width 627-1002R or triple width 627-1003R) must contain a 620-0048/0052 DCM installed in the processor card file option slots. The only exception to this rule is when the customer is fully knowledgeable of the attendant required complexity and the amount of relay ladder logic required to access information from the MiniCOP without a DCM.

Although not permitted by the automated configurator, a “special” redundant Logic Manager System with parallel I/O would use a model 621-9928R Serial Link Selector module in slots B and C of Parallel I/O card file (one per redundant system, located in one card file) for the purpose of loader/terminal switch over. Also, there would be a model 621-9929 Parallel Bus Selector Module in slots D and E (one per redundant system, located in one card file). Additional SLSs may be used as option modules.

7.3 LOGIC MANAGER CARD MODULES

7.3.1 Logic Manager Module

The Logic Manager Module (LMM), model 620-0071, is a two card assembly and is installed in slots A and B on the left-front side of the Logic Manager Processor card file.

The LMM has a battery for backup power to protect the contents of the module's RAM in the event of a power failure. The module is shipped with an insulating wafer installed to isolate the battery in the battery holder. To activate the battery, remove the battery coverplate from the front of the module. Remove the battery and insulating wafer. Install the battery in the holder again and replace the coverplate on the front of the module. See Figure 7-2.

Four switch-selectable functions must be configured on the LMM Daughter Board. All are found on the eight position DIP switch module, SW2. See Table 7-5. The first position selects the LMM as the primary or secondary module. Positions 2 through 6 determine the UCN node address of the LMM. Refer to Table 7-3 for a detailed table of the UCN node address settings. Position 7 enables or disables the Network Test function. Position 8 is Factory Test Enable and should always be disabled.

Table 7-5 — LMM Daughter Board SW2 Settings

SW2 POSITION	STATE	FUNCTION
1	Open/Off/Up	This LMM is the primary LMM
	Closed/On/Down	This LMM is the secondary LMM
2	Open/Off/Up	Positions 2 through 6 are used to determine the LMM's node address on the UCN. See Table 7-7.
	Closed/On/Down	
3	Open/Off/Up	
	Closed/On/Down	
4	Open/Off/Up	
	Closed/On/Down	
5	Open/Off/Up	
	Closed/On/Down	
6	Open/Off/Up	
	Closed/On/Down	
7	Open/Off/Up	Network Test Enable Off = Disable (normal setting)
	Closed/On/Down	Network Test Enable On = Enable
8	Open/Off/Up	Factory Test Enable Off = Disable (normal setting)
	Closed/On/Down	Factory Test Enable On = Enable

All switches are preset to the Open/Off/Up position by the factory.

There is a 4-position DIP switch, SW1, located on the LMM Mother Board, that must be configured to identify the address of the LMM. Only the first two positions are used. Switch positions 3 and 4 are used for factory test and must always be in the off position for proper LMM operation. Table 7-6 indicates the four possible settings of switches 1 and 2. **The LMM should always be configured for an address of 1.**

In a redundant Logic Manager System, the LMM Primary/Secondary DIP switch setting must agree with the Primary/Secondary DIP switch setting of the RCM in the same card file. In a nonredundant Logic Manager System, the Primary/Secondary DIP switch setting must be set to primary.

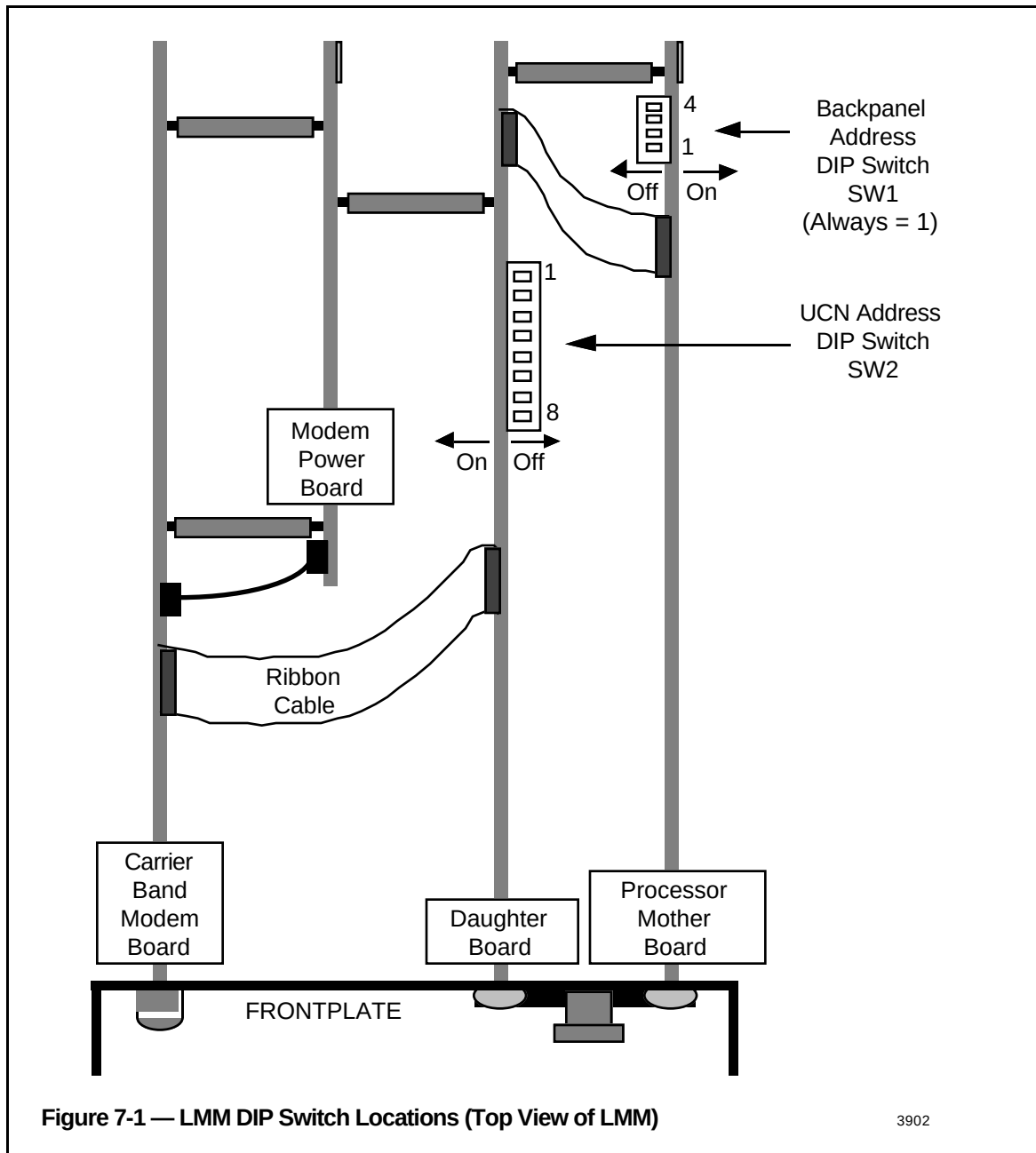


Table 7-6 — LMM Mother Board SW1 Settings

SW1 POSITION/STATE		BACKPANEL ADDRESS
1	2	
Closed/On/Down	Closed/On/Down	0
Open/Off/Up	Closed/On/Down	1 *
Closed/On/Down	Open/Off/Up	2
Open/Off/Up	Open/Off/Up	3

* Factory setting

Table 7-7 — LMM SW2 DIP Switch Settings For UCN Addresses

ADDRESS (Redundant Pairs)	SW2 POSITION				
	6	5	4	3	2
1,2	Open/Off *	Open/Off *	Open/Off *	Open/Off *	Open/Off *
3,4	Open/Off	Open/Off	Open/Off	Open/Off	Closed/On
5,6	Open/Off	Open/Off	Open/Off	Closed/On	Open/Off
7,8	Open/Off	Open/Off	Open/Off	Closed/On	Closed/On
9,10	Open/Off	Open/Off	Closed/On	Open/Off	Open/Off
11,12	Open/Off	Open/Off	Closed/On	Open/Off	Closed/On
13,14	Open/Off	Open/Off	Closed/On	Closed/On	Open/Off
15,16	Open/Off	Open/Off	Closed/On	Closed/On	Closed/On
17,18	Open/Off	Closed/On	Open/Off	Open/Off	Open/Off
19,20	Open/Off	Closed/On	Open/Off	Open/Off	Closed/On
21,22	Open/Off	Closed/On	Open/Off	Closed/On	Open/Off
23,24	Open/Off	Closed/On	Open/Off	Closed/On	Closed/On
25,26	Open/Off	Closed/On	Closed/On	Open/Off	Open/Off
27,28	Open/Off	Closed/On	Closed/On	Open/Off	Closed/On
29,30	Open/Off	Closed/On	Closed/On	Closed/On	Open/Off
31,32	Open/Off	Closed/On	Closed/On	Closed/On	Closed/On
33,34	Closed/On	Open/Off	Open/Off	Open/Off	Open/Off
35,36	Closed/On	Open/Off	Open/Off	Open/Off	Closed/On
37,38	Closed/On	Open/Off	Open/Off	Closed/On	Open/Off
39,40	Closed/On	Open/Off	Open/Off	Closed/On	Closed/On

(Continued)

* Factory setting; Open/Off = Up; Closed/On = Down

Table 7-7 — LMM SW2 DIP Switch Settings For UCN Addresses (Continued)

ADDRESS (Redundant Pairs)	SW2 POSITION				
	6	5	4	3	2
41,42	Closed/On	Open/Off	Closed/On	Open/Off	Open/Off
43,44	Closed/On	Open/Off	Closed/On	Open/Off	Closed/On
45,46	Closed/On	Open/Off	Closed/On	Closed/On	Open/Off
47,48	Closed/On	Open/Off	Closed/On	Closed/On	Closed/On
49,50	Closed/On	Closed/On	Open/Off	Open/Off	Open/Off
51,52	Closed/On	Closed/On	Open/Off	Open/Off	Closed/On
53,54	Closed/On	Closed/On	Open/Off	Closed/On	Open/Off
55,56	Closed/On	Closed/On	Open/Off	Closed/On	Closed/On
57,58	Closed/On	Closed/On	Closed/On	Open/Off	Open/Off
59,60	Closed/On	Closed/On	Closed/On	Open/Off	Closed/On
61,62	Closed/On	Closed/On	Closed/On	Closed/On	Open/Off
63,64	Closed/On	Closed/On	Closed/On	Closed/On	Closed/On

Open/Off = Up; Closed/On = Down

Connection to the Universal Control Network (UCN) is made at the front of the Logic Manager Module. The connections are labeled A and B for the redundant cable system. See Figure 7-2.

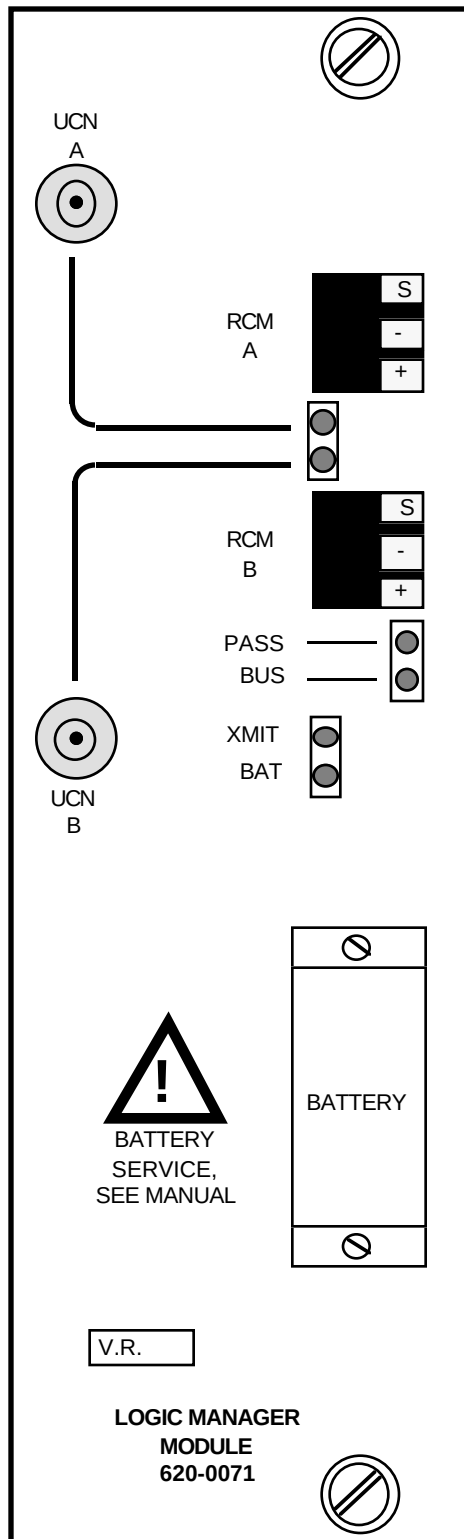


Figure 7-2 — Logic Manager Module Front Panel

3903

7.3.2 Redundancy Control Module

The Redundancy Control Module (RCM), model 620-0059, is the main component in a Logic Manager redundant system and is responsible for transferring LCS data from the primary to the secondary RCM. The RCM also monitors the status of both Logic Manager processors (primary and secondary). If this option is implemented, the module is installed in slot C of the Logic Manager Processor card file.

The RCM has two connectors that are mounted on its front panel that are labeled PORT1 and PORT2. PORT1 is a 25 pin, D-type connector which interfaces with the corresponding port on the RCM in the redundant Logic Manager Processor card file. The I/O data exchange occurs through this port. PORT2 is a 3 position, detachable screw terminal, which connects to the Parallel Bus Selector module (PBS), Serial Link Selector Module (SLS), or the Serial Link Module (SLM), and provides a switch-over control output to these modules.

The RCM has a five-position DIP switch module, SW1, that determines the backpanel address, primary or secondary selection, primary request enable/disable, and program down loading enable/disable. See Table 7-8.

Table 7-8 — RCM SW1 DIP Switch Settings

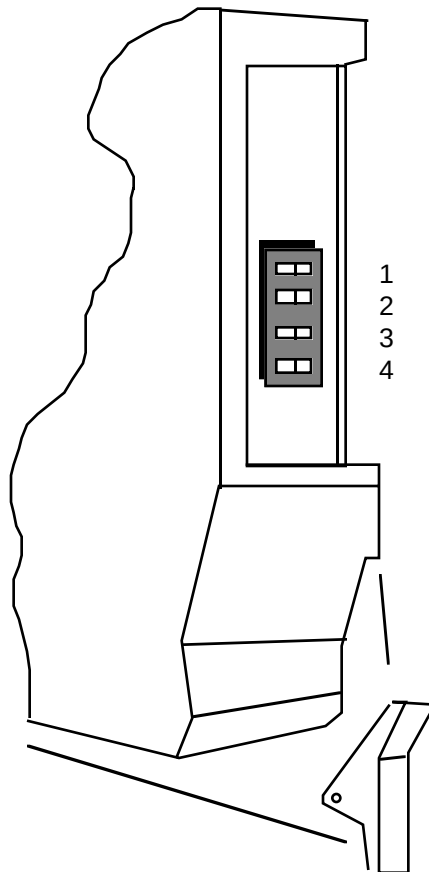
SW1 POSITION	STATE	FUNCTION
1	Closed/On	Address = 0. This is the factory setting. Only one RCM is permitted per rack. The RCM is always set to address zero (0).
2	Closed/On	
3	Closed/On*	RCM in the secondary position
	Open/Off	RCM in the primary position
4	Closed/On*	Prohibits program down load function into this processor
	Open/Off	Permits program down load function into this processor
5	Closed/On*	Enables primary request switch
	Open/Off	Disables primary request switch

* Factory Setting - Your RCM maybe set differently if it was configured by Honeywell as part of a **TotalPlant** Solution (TPS) system.

7.3.3 24 K Memory Module

The 24 K Memory Module, model 620-0024, is installed in slot E of the Logic Manager Processor card file. It contains a total of 24 K words of memory and requires a mandatory starting memory address of zero (0).

A four-position DIP switch is located on the front of the module and is used to set the module's starting memory address. All four positions on the switch are set to OFF for a starting memory address of zero (0). See Figure 7-3.



SWITCH POSITION	SWITCH VALUE	
	OFF	ON
1	0	2 K
2	0	4 K
3	0	8 K
4	0	16 K

Figure 7-3 — Memory Module Start Address DIP Switch Settings

3904

7.3.4 8 K Memory Module

The 8 K Memory Module, model 620-0027, is an optional memory addition to the 24 K Memory Module and is installed in slot D of the Logic Manager Processor card file. It contains 8 K-words of memory and requires a mandatory starting memory address of 24 K. The total memory available to the Logic Manager processor becomes 32 K, the maximum allowed.

A four-position DIP switch is located on the front of the module and is used to set the module's starting memory address. The DIP switch selections are additive. Set DIP switch positions 1 and 2 to Off and DIP switch positions 3 (8 K) and 4 (16 K) to On, resulting in a starting address of 24 K. See Figure 7-3.

7.3.5 Serial Link Module

The Serial Link Module (SLM), model 621-9939, controls serial I/O data transfer to and from remote I/O. This module has a separate processor with its own data table and operates asynchronously to the processor module. One or two SLMs can be installed in slots L and/or M.

Each SLM controls two serial I/O channels. Each operating channel must be terminated at the extreme ends using the terminating resistor provided. Three terminating resistor kits are supplied with each SLM: eight 200 ohm 1/2 W resistors, eight 300 ohm 1/2 W resistors, and a 100 ohm 1/2 W resistor for use with the Redundancy Control Module (RCM), model 620-0059, for redundant serial systems.

Each channel of the SLM has provision for an optional terminating resistor. For each channel, the resistor must be installed if the channel is at the extreme end of the communications link. It is not installed if the channel connects to the communications line at an intermediate point rather than at one end. Refer to subsection 5.6.2 for additional information, and refer to Figures 5-4 and 5-5 for examples. Use 200 ohm resistors if the communications cable is Belden 9729, and use 300 ohm resistors if the cable is Belden 9182.

1. Remove the two screws that secure the module side cover and remove the cover.
2. Locate the R31 and R32 lead receptacles on the SLM circuit board per Figure 7-4.
3. Terminate channel 1 by inserting the terminating resistor into the R31 lead receptacle if channel 1 is at one end of the communications link.
4. Terminate channel 2 by inserting the terminating resistor into the R32 lead receptacle if channel 2 is at one end of the communications link..
5. Replace the module side cover.

An eight-position DIP switch module, SW1, located near the top edge of the SLM circuit board, is used to configure the SLM. See Figure 7-4.

Positions 1 through 4 determine the SLM's shutdown response (positions 1 and 2 for channel 1, and positions 3 and 4 for channel 2).

Position 5 selects the number of PUSH/PULL operations that are updated per I/O scan. Position 6 determines if one or two SLMs will be configured in the Logic Manager processor card file. Position 7 selects a redundant or nonredundant serial I/O configuration. Position 8 provides an option for normal operation or auto-test of the serial I/O system. Auto-test is a diagnostic tool provided for bench test procedures. It is not applicable to routine system operation. When auto-test selection is made, the operation of switch positions 1 through 7 is redefined.

See Table 7-9 for the SLM SW1 DIP switch module functions.

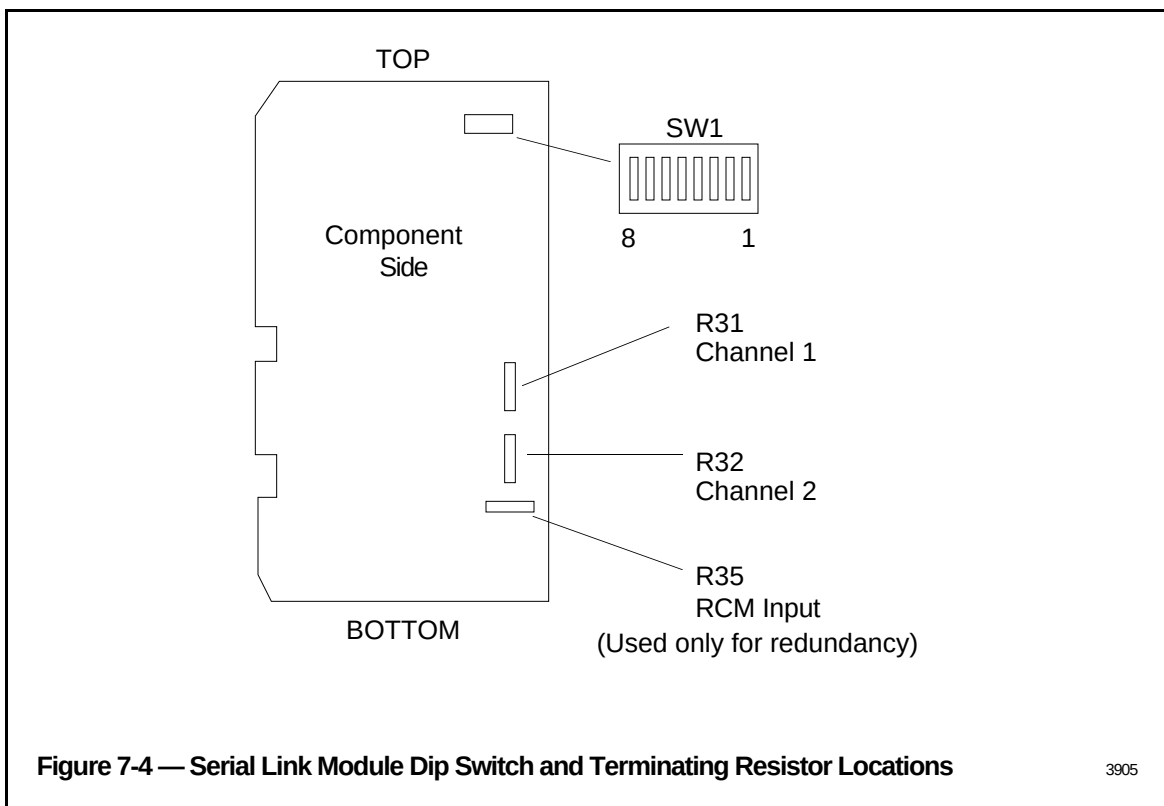


Table 7-9 — Serial Link Module DIP Switch Settings

SW1 POSITION/STATE		FUNCTION (SHUTDOWN RESPONSE)
1 and 3 *	2 and 4 *	
Open/Off **	Open/Off **	The channel continues to operate if a channel fault occurs. ***
Closed/On	Closed/On	The channel ceases operation if a channel fault occurs. ***
Open/Off	Closed/On	Both channels on SLM cease operation if a channel fault occurs. ***
Closed/On	Open/Off	Both SLM channels shutdown, plus the CLROL line on the I/O bus is set—resetting the parallel I/O and causing a second SLM, if present, to have its corresponding SIOMs clear or freeze data, depending on the SIOM DIP switch settings. ***

SW1 POSITION	STATE	FUNCTION
5	Closed/On	Service one PUSH/PULL card per I/O rack each scan.
	Open/Off**	Service all PUSH/PULL cards per I/O rack each scan.
6	Closed/On	Identifies the SLM as the second SLM in the card file.
	Open/Off**	The first or only SLM in the card file requires this setting.
7	Closed/On	Redundant serial I/O system
	Open/Off **	Nonredundant Serial I/O system
8	Closed/On	Factory Test enabled
	Open/Off **	Factory Test disabled (normal operation)

* Switches 1 and 2 apply to channel 1; switches 3 and 4 apply to channel 2.

The switches are set in pairs as shown in the table.

** Factory setting

*** The following events cause a channel fault:

- 1) Power failure
- 2) Serial Link timeout
- 3) Two consecutive serial data errors received

7.3.6 Parallel Link Driver Module

The Parallel Link Driver Module (PLDM), model 620-0088, installed in slot N of the Logic Manager Processor card file, controls I/O communications. It provides a D-type connector on the front coverplate of the module to enable I/O card files to be daisy-chained from the PLDM. The processor keyswitch is also located on the PLDM coverplate.

The PLDM contains two DIP switch modules that are set to perform the following functions:

- Enable or disable Forcing, Data Change, On-line Programming and Scan Loss functions.
- Clear or freeze outputs in the Program mode and during Scan Loss.
- Set Scan Loss time-out.

Two DIP switches, SW1 and SW2, are located on the top front edge of the PLDM printed circuit board under the module side cover. They can be set without removing the side cover. The Closed/On setting is achieved by pressing the switch toward the circuit board.

See Figure 7-5 and Table 7-10 for the location and function of each PLDM DIP switch module.

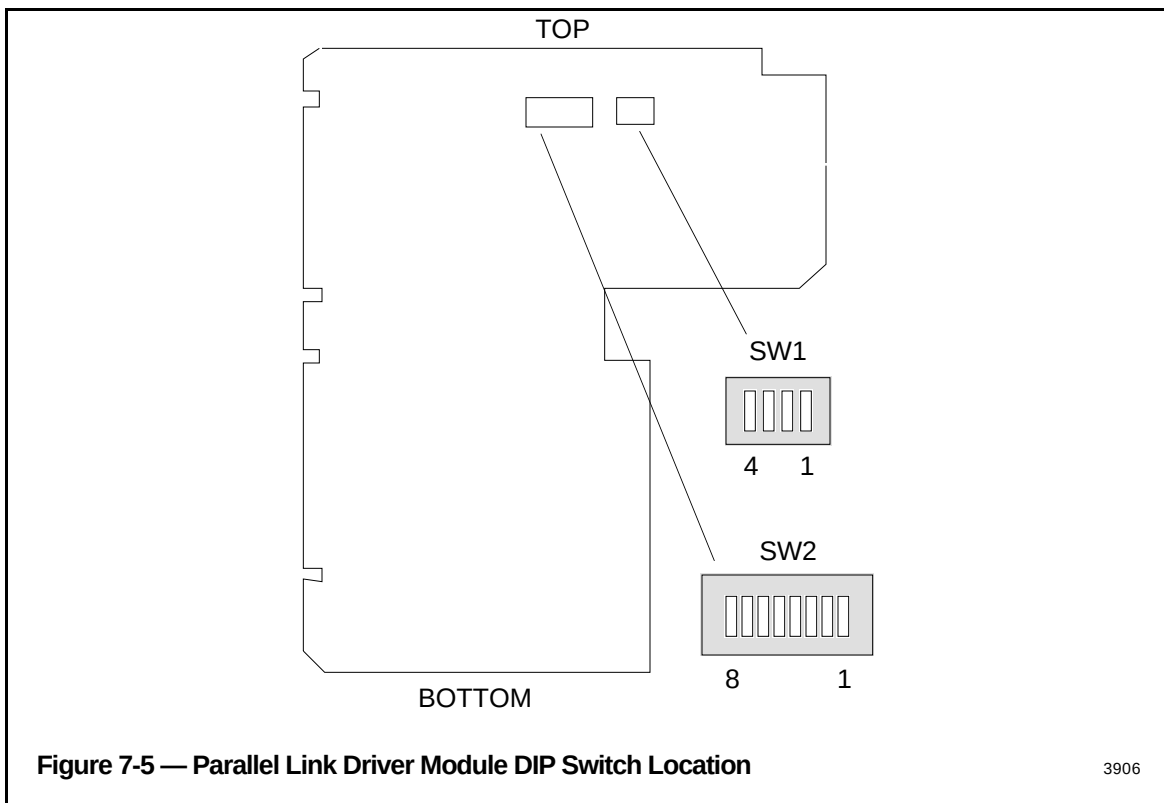


Table 7-10 — Parallel Link Drive Module DIP Switch Settings

SW1 POSITION	STATE	FUNCTION	
1	Closed/On	Disables Processor Scan function	
	Open/Off *	Enables Processor Scan function	
2	Closed/On *	Adds 20 ms	The user should set to <u>60 ms</u> in excess of processor scan time. The module is factory set to the maximum value.
	Open/Off	Adds 0 ms	
3	Closed/On *	Adds 40 ms	
	Open/Off	Adds 0 ms	
4	Closed/On *	Adds 80 ms	
	Open/Off	Adds 0 ms	

SW2 POSITION	STATE	FUNCTION	
1	Closed/On *	Must be set to Closed/On.	
2	Closed/On *	Must be set to Closed/On.	
3 **	Closed/On	I/O racks clear or freeze outputs according to PIOM or SIOM freeze/clear setting when the Software Program mode is used.	
	Open/Off	When in the Software Program mode, all outputs remain in their last state regardless of the PIOM or SIOM freeze/clear setting. See Note.	
4	Closed/On *	Enables Force function.	
	Open/Off	Disables Force function.	
5	Closed/On *	Enables the following data changes in Run mode: RM Table values and Sequencer Register constant values.	
	Open/Off	Disables the above data changes in the Run mode	
6	Closed/On *	Enables on-line programming in the Run/Program mode.	
	Open/Off	Disables on-line programming in the Run/Program mode.	
7		Not used.	
8		Not used.	

* Denotes factory setting

** NOTE: Ordinary coils (not retentive or latched) are turned Off at the beginning of the program scan when the processor is switched to Run mode.

7.3.7 Parallel I/O Module

The Parallel I/O Module (PIOM), model 621-9937, is the interface to the other I/O card files or the Parallel Link Driver Module. The PIOM is installed in slot N of the I/O card file.

The PIOM allows any combination of 8, 16, or 32 point I/O modules in the I/O card file, providing the number of I/O points stay within the boundaries set for each Logic Manager System.

The Parallel I/O Module has five DIP switch modules.

DIP switch modules SW1 and SW2 should be set before installing the PIOM. SW1 determines the starting address of the I/O card file and SW2, positions 1 and 2, determine card file output states if an output fault occurs.

DIP switch modules SW3, SW4, and SW5 are used to set the number of I/O addressed in slots A through L of the I/O card file. A slot also may be configured for 0 if it is occupied by a multi-slot special function module.

See Figures 7-6, 7-7, and Table 7-11 for the locations and a further explanation of the PIOM DIP switch selections.

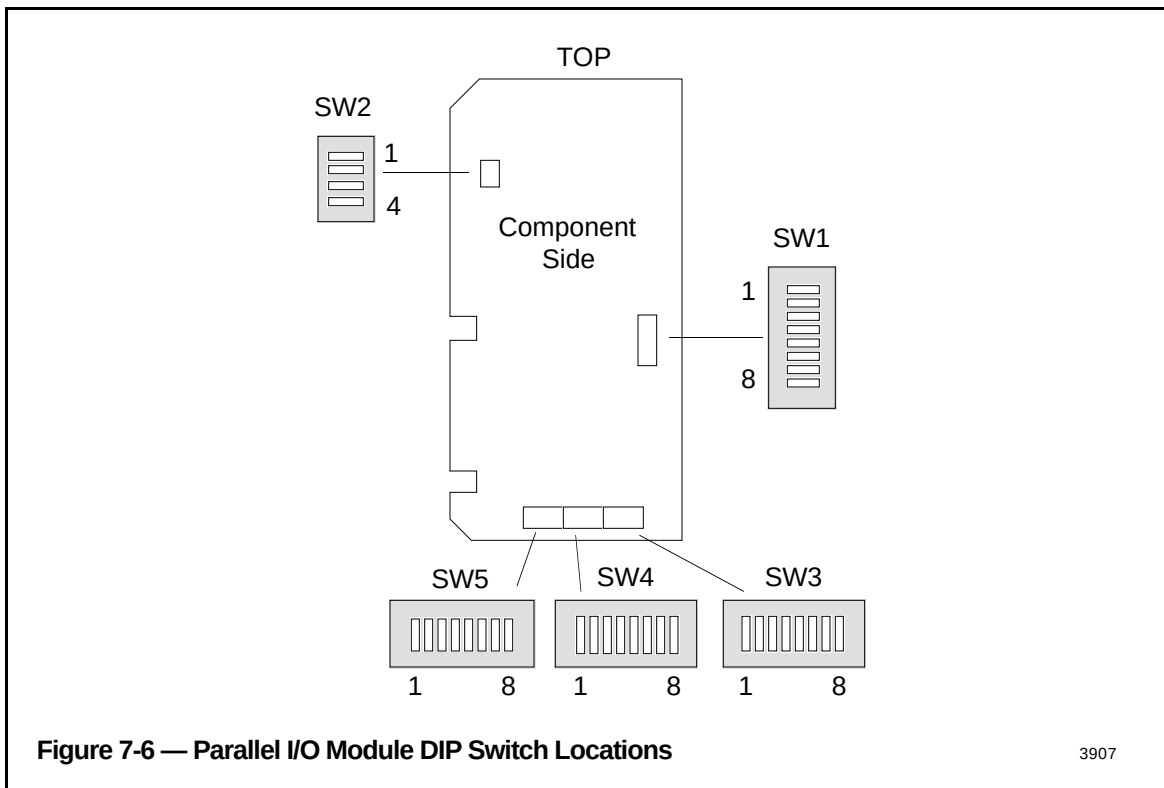
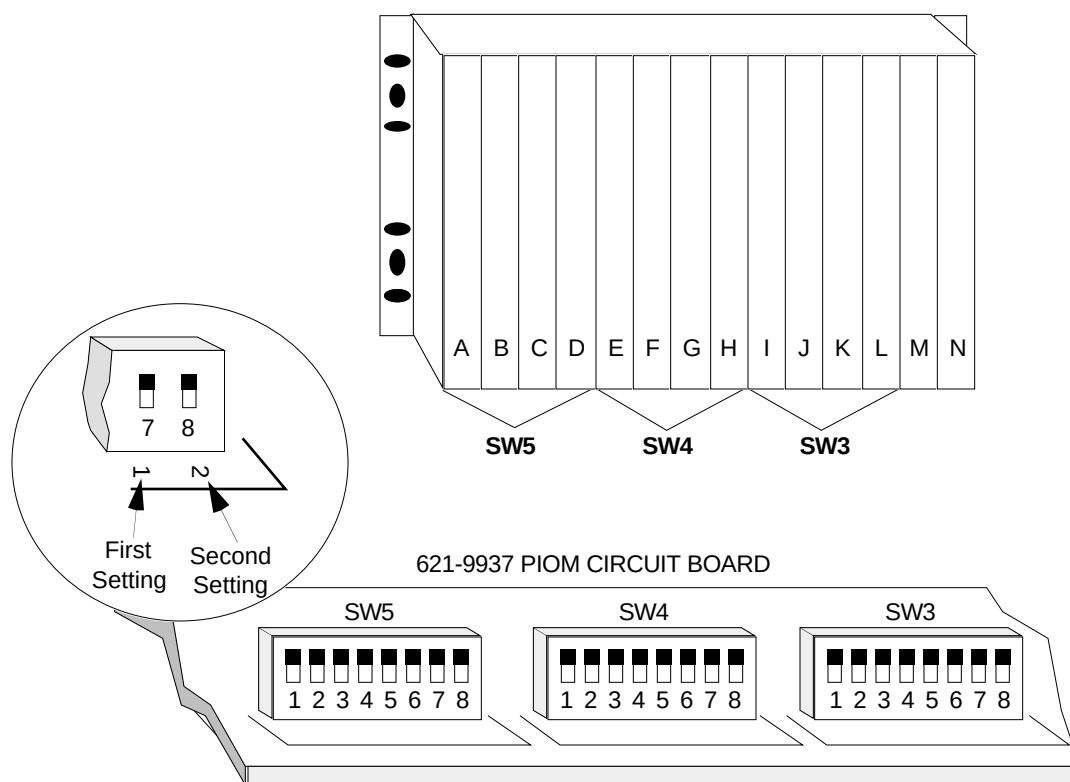


Table 7-11 — Parallel I/O Module Switch Settings

SWITCH	POSITION	STATE	SWITCH VALUE	
SW1	1	Closed/On	8	Determines the starting address for the parallel I/O card file containing this module. For example, closing positions 2, 4, and 6 yields a starting address of: $16 + 64 + 256 = 336$ NOTE: No two card files are allowed to have overlapping I/O addresses.
		Open/Off *	0	
	2	Closed/On	16	
		Open/Off *	0	
	3	Closed/On	32	
		Open/Off *	0	
	4	Closed/On	64	
		Open/Off *	0	
	5	Closed/On	128	
		Open/Off *	0	
	6	Closed/On	256	
		Open/Off *	0	
	7	Closed/On	512	
		Open/Off *	0	
	8	Closed/On	1024	
		Open/Off *	0	

SW2 POSITION	STATE	FUNCTION
1	Closed/On	Clears the outputs when: 1) SW2 Position 2 is closed and an I/O fault occurs, 2) the external cable is disconnected, 3) an external power failure, 4) the processor is in Program or Disable modes.
	Open/Off *	Outputs remain the same with the above conditions.
2	Closed/On	Recognize an output module fault.
	Open/Off *	Ignore an output module fault.
3		Not used.
4		Not used.

* Factory setting



SLOT CONFIGURATION SELECTION

MODULE TYPE	SETTING	
	FIRST	SECOND
0-Point	Closed/ON	Closed/ON
8-Point	Open/OFF	Closed/ON
16-point*	Closed/ON	Open/OFF
32-point	Open/OFF	Open/OFF

*Factory setting for all switches

Example:

Slot C with a 16-point module

SW5, position 5 - ON

SW5, position 6 - OFF

Slot G with an 8-point module

SW4, position 5 - OFF

SW4, position 6 - ON

Figure 7-7 — Parallel I/O Module SW3, SW4 And SW5 DIP Switch Settings For Slot Configuration

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7.3.8 Serial I/O Module

The Serial I/O Module (SIOM), models 621-9938R or 621-9940, is installed in slot N in the remote Serial I/O card file. It controls the Serial I/O card file and interfaces with the model 621-9939 SLM in the processor card file.

The model 621-9938R SIOM is used in redundant serial I/O configurations, but can be used in nonredundant applications.

Terminating resistors are required at each extreme end of the serial channel to match the cable impedance. Use the 200 ohm resistors supplied in the SLM resistor kits for installation using Belden 9729 cable and 300 ohm resistors also supplied in the SLM resistor kits for installations using Belden 9182 cable.

Two terminating resistors are required in each SIOM that is at the extreme end of the link. Refer to subsection 5.6.2 for additional information, and to Figures 5-4 and 5-5 in Section 5 for examples. Install terminating resistors in SIOMs that are at the end of communications links according to the following procedure:

1. Remove the four screws that secure the module side cover and remove the cover.
2. Locate the R36 and R38 lead receptacles on the SIOM circuit board per Figure 7-8.
3. Insert terminating resistors into both the R36 and R38 lead receptacles.
4. Replace the module side cover.

Configure the SIOM by using the four 8-position and one 4-position DIP switch modules located near the top edge of the SIOM circuit board. These DIP switches select the module starting address, card fault recognition, communication fault responses, normal or test mode operation, and the number of I/O points located in each I/O card file.

DIP switch module SW1 selects the absolute starting address for each card file. The starting address is selected from a range of 0 through 2,032, in increments of eight. On startup, the SLM queries all connected SIOMs for their starting addresses and the number of allocated addresses in the card file. The SLM also determines the sequence in which the I/O card files on the channel are accessed, based on the absolute starting address of each SIOM. Overlapping SIOM starting or I/O addresses and addresses higher than 2,032 are prohibited. Also, be sure to never install an 8- or 16-point I/O module in a 32-point slot.

DIP switch module SW2 determines rack output states in case of a fault and allows for the option of auto-test or normal operation. DIP switch modules SW3, SW4, and SW5 determine the number of I/O points per card file slot.

Refer to Figure 7-8 and Table 7-12 for the SIOM DIP switch module locations and selections.

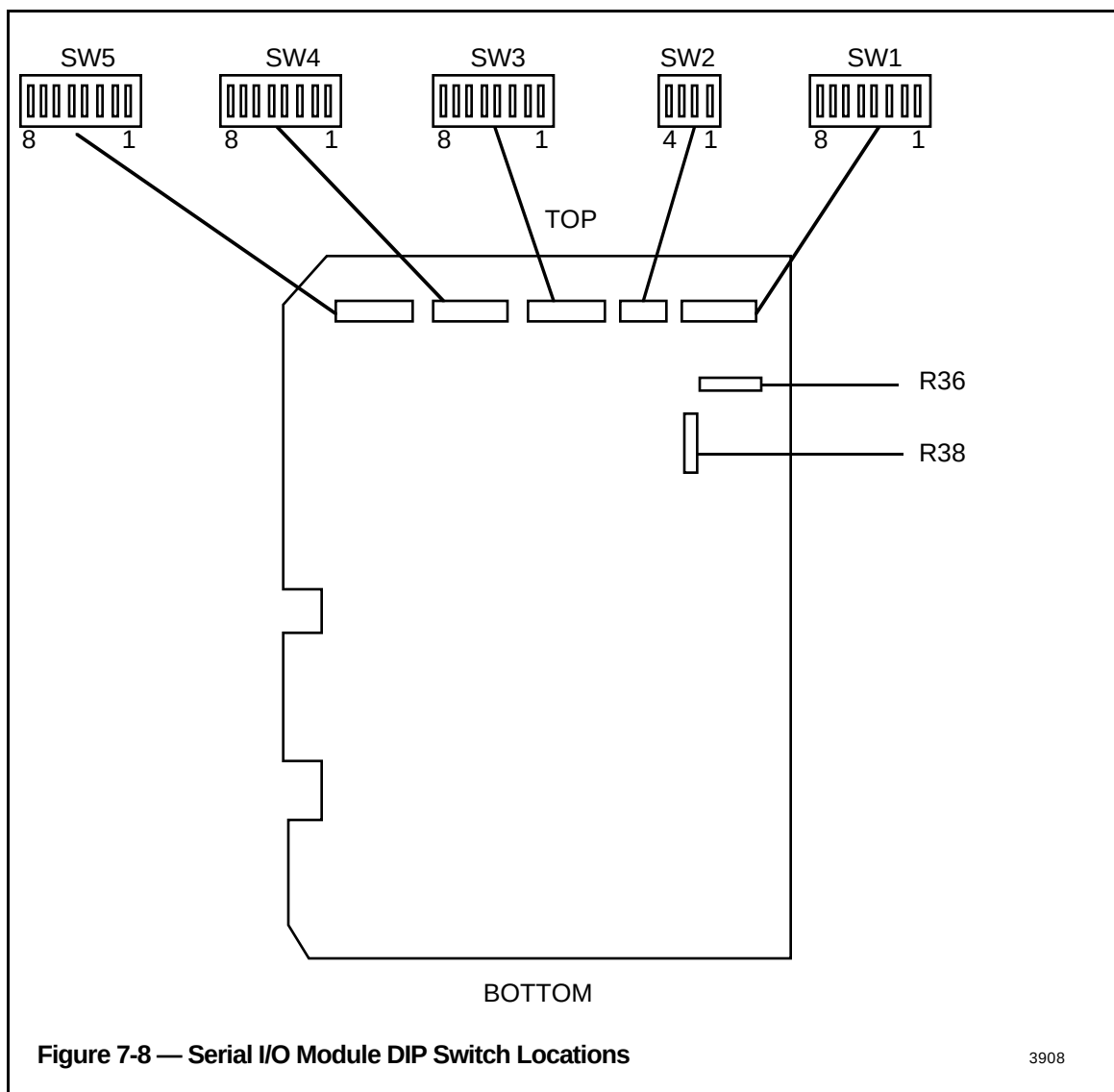


Table 7-12 — Serial I/O Module DIP Switch Settings

SW1 STATE	SWITCH POSITION ADDRESS VALUE							
	1	2	3	4	5	6	7	8
Open/Off*	0	0	0	0	0	0	0	0
Closed/On	8	16	32	64	128	256	512	1024

SW2 POSITION	STATE	FUNCTION
1	Closed/On	Clear outputs and PUSH data with: 1) SW2 position 2 closed and an I/O fault occurrence 2) Loss of SLM communications 3) External power failure 4) Processor in Program or Disable mode (CLROL low)
	Open/Off *	Freeze outputs and Push data with conditions above
2	Closed/On	Recognize I/O card fault and respond according to SW2 position 1 setting
	Open/Off *	Ignore an output module fault
3		Not used
4	Closed On	Auto-test
	Open/Off*	Normal operation

CONTROLLING SWITCH			SELECTION	NUMBER OF I/O POINTS PER SLOT			
SW5	SW4	SW3	SWITCH**	0	8	16*	32
I/O SLOT				SWITCH STATE			
D	H	L	1 (2)	Closed/On	Closed/On	Open/Off	Open/Off
			2 (1)	Closed/On	Open/Off	Closed/On	Open/Off
C	G	K	3 (2)	Closed/On	Closed/On	Open/Off	Open/Off
			4 (1)	Closed/On	Open/Off	Closed/On	Open/Off
B	F	J	5 (2)	Closed/On	Closed/On	Open/Off	Open/Off
			6 (1)	Closed/On	Open/Off	Closed/On	Open/Off
A	E	I	7 (2)	Closed/On	Closed/On	Open/Off	Open/Off
			8 (1)	Closed/On	Open/Off	Closed/On	Open/Off

* Factory setting

** Switch selection in parenthesis denotes the silkscreen labeling on the SIOM printed circuit board.

Note: No two I/O card files are allowed to have overlapping addresses.

7.3.9 Enhanced Diagnostic Module

The Enhanced Diagnostic Module (EDM), model 621-0021, provides the Logic Manager System with additional system monitoring capability such as detection of problems with data and control lines on the I/O bus. It should be assigned a quantity of eight I/O addresses and should always be placed in the leftmost slot (A) of a Parallel or Serial I/O card file.

A four-position DIP switch module on the top left edge of the EDM board sets operating parameters. Switches 1 and 2 select the Watchdog Timer, and switch 3 selects Parallel or Serial I/O operation. Switch 4 selects the way the EDM reacts to a fault.

See Table 7-13 for EDM DIP switch selections.

Table 7-13 — Enhanced Diagnostic Module DIP Switch Module Settings

SWITCH AND STATE		FUNCTION
SW1	SW2	
Open/Off	Open/Off	1.74 sec. timeout
Closed/On *	Open/Off *	200 ms timeout
Open/Off**	Closed/On**	4.5 sec. timeout used for redundant LMs
Closed/On	Closed/On	490 ms timeout

SWITCH	STATE	FUNCTION
SW3	Closed/On*	Identifies serial I/O operation
	Open/Off	Identifies parallel I/O operation

SWITCH	STATE	FUNCTION
SW4	Closed/On *	A data bus or control line fault de-energizes relay immediately.
	Open/Off	Signal from SIOM, PIOM or LM de-energizes relay.

* Factory setting for nonredundant systems

** Setting required for redundant systems (4.5 sec.)

Note: Set EDM SW3 to On or Off depending upon the type of I/O card file, parallel or serial.

7.3.10 Parallel Bus Selector Module

The Parallel Bus Selector Module (PBS), model 621-9929, is a double wide module installed in slots B and C of one I/O card file per Parallel I/O system in a redundant configuration only. The PBS connects the primary Logic Manager processor to the Parallel I/O system and switches the Parallel I/O system, upon request, from the primary to the secondary Logic Manager processor.

The PBS's only connections to the I/O card file backpanel are the +5 Vdc supply, ground, and the power fail signals. All other connections are made through the front panel connectors. The state of the PBS's bus selector circuitry is controlled by the Redundant Control Module (RCM) which sends a switch select output signal to the PBS through the two 3-position front panel connectors.

Two 50-pin D-type connectors, labeled A and B, interface with the Parallel Link Driver Modules (PLDM) in the Logic Manager Processor card files A and B, respectively. The third D connector, labeled Out, interfaces with the first Parallel I/O Module (PIOM) in the parallel I/O system.

Two on-line LEDs, labeled A and B, indicate which Logic Manager processor, A or B, has control of the I/O system.

A two-position DIP switch module on the PBS circuit board sets operating parameters. See Table 7-14 for PBS DIP switch module selections.

Table 7-14 — Parallel Bus Selector Module DIP Switch Settings

SW1 POSITION	STATE	FUNCTION
1	Closed/On	A 100 ohm resistor is placed across the A switch control input.
	Open/Off*	No resistor is placed across the A switch control input.
2	Closed/On	A 100 ohm resistor is placed across the B switch control input.
	Open/Off*	No resistor is placed across the B switch control input.

* Factory setting

7.3.11 Serial Link Selector Module

The Serial Link Selector Module (SLS), model 621-9928R, is a double wide module installed in slots B and C of one I/O card file per Serial I/O System in a redundant configuration. The SLS serves as a general purpose switching device and switches serial communication devices shared by the redundant Logic Managers processors.

The SLS's only connections to the I/O card file backpanel are the +5 Vdc supply, ground, and the power fail signal. All other are made through the front panel connectors. The state of the SLS's bus selector circuitry is controlled by the Redundancy Control Module (RCM), that sends a switch select output signal to the SLS through the two 3-position front panel connectors.

Two 12-position detachable screw terminals, labeled A and B, connect to the ports in Logic Managers card files A and B, respectively. Terminals 1 through 8 of the connector are communication signal terminals, and terminals 9 through 12 are dedicated for cable shield connections. A third 12-position terminal, labeled Out, connects to a serial device such as a host computer or Loader/Terminal.

Two LEDs indicate which Logic Manager processor, A or B, is connected to the serial device.

A 2-position DIP switch module on the SLS circuit board sets operating parameters. See Table 7-15 for SLS DIP switch module selections.

Table 7-15 — Serial Link Selector Module DIP Switch Settings

SW1 POSITION	STATE	FUNCTION
1	Closed/On	A 100 ohm resistor is placed across the A switchover control wiring connection.
	Open/Off*	No resistor is placed across the A switchover control wiring connection.
2	Closed/On	A 100 ohm resistor is placed across the B switchover control wiring connection.
	Open/Off*	No resistor is placed across the B switchover control wiring connection.

* Factory setting

Note: A terminating resistor is required if the SLS is at the end of the switchover control wiring link.

7.3.12 Power Supply Modules

The Power Supply Modules (PSM) provide power to operate the Logic Manager and I/O card files. The PSM is installed in slots J and K in the Logic Manager card file and either slot M or L and M in the I/O card module. Five models of power supply modules are available, two for use in the Logic Manager card file and three for use in the I/O card file. They are as follows:

Table 7-16 — Power Supply Modules

MODEL NUMBER	CARD FILE	INPUT VOLTAGE	AMPERAGE	SLOTS
620-0036	Logic Manager	115/230 Vac	15 A	2
620-0047	Logic Manager	24 Vdc	15 A	2
621-9932	I/O	24 Vdc	8 A	1
621-9933	I/O	115/230 Vac	15 A	2
621-9934	I/O	115/230 Vac	8 A	1

The system memory and register modules use battery backup power in the event of a power failure. The power supply module is shipped with an insulating wafer installed to isolate the battery in the battery holder. To activate the battery, remove the battery coverplate from the front of the module. Remove the battery and insulating wafer. Install the battery in the holder again and replace the coverplate on the front of the module.

The power supply front plates must be securely fastened to the chassis using the module thumbscrews in order to achieve proper grounding.

7.3.12.1 LM 115/230 Vac, 15 A Power Supply Module

The model 620-0036 provides a maximum of 15 A of +5 Vdc power for use by the Logic Manager card file modules. It also supplies 300 mA of ± 12 Vdc for optional modules requiring those voltages.

Terminals on the front panel of the module are labeled A (L1 or Line), B (L2 or Common), and GND (Ground) for connecting ac power wiring. The module operates on user selected voltages of 115 Vac or 230 Vac. The shorting board located inside the side cover at the top rear edge selects the proper ac voltage.

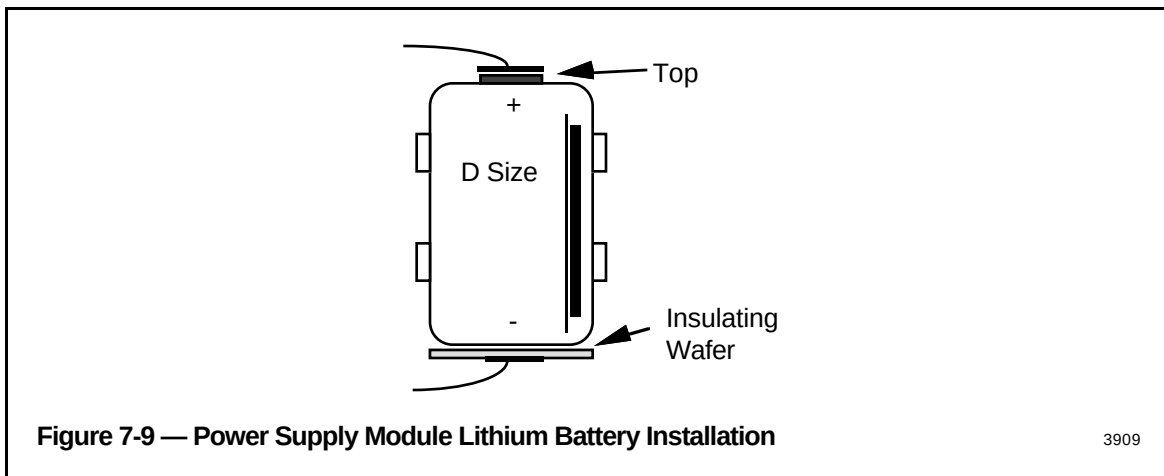
A battery compartment in the PSM contains a D size lithium battery which provides backup power to the memory and register modules in the Logic Manager Processor card file. The PSM is shipped with the battery installed.

WARNING

An insulating wafer prevents battery discharge during shipment and storage and must be removed before the system is operated. See Figure 7-9.

Two status indicators are located on the front of the PSM. A green LED labeled POWER indicates that power is applied to the module. A green LED labeled BATTERY PASS remains energized as long as the battery is in good condition and power is applied.

The module contains a front accessible fuse holder. When this power supply is shipped from the factory as part of a Logic Manager system the power supply is configured for the line voltage requested by the user. When it is shipped as a part, it is shipped with settings for 115 Vac operation. A fuse (1 amp slo-blo) for 230 Vac is shipped with the part, if the user wants to operate the supply at 230 Vac. It is the users responsibility to replace the 115 Vac fuse with the 230 Vac fuse.



7.3.12.2 LM 24 Vdc, 15 A Power Supply Module

The model 620-0047 provides a maximum of 15 A of +5 Vdc power for use by the Logic Manager card file modules. It also supplies 300 mA of ± 12 Vdc for optional modules requiring those voltages.

Terminals on the front of the module are labeled +, -, and GND. The module operates with a nominal input voltage of 20 to 28 Vdc.

A battery compartment in the PSM contains a D size lithium battery which provides backup power to the memory and register modules in the Logic Manager Processor card file. The PSM is shipped with the battery installed.

WARNING

An insulating wafer prevents battery discharge during shipment and storage and must be removed before the system is operated. See Figure 7-9.

Two status indicators are located on the front of the PSM. A green LED labeled POWER indicates that power is applied to the module. A green LED labeled BATTERY PASS remains energized as long as the battery is in good condition and power is applied.

The module has a front accessible fuse holder and is shipped with a 8 A slow blow fuse.

7.3.12.3 I/O 24 Vdc, 8 A Power Supply Module

The model 621-9932 I/O Power Supply Module provides 8 A of +5 Vdc for the I/O logic circuitry in the card file. It also supplies 600 mA of ± 15 Vdc power.

The maximum power consumption of the module is 96 VA. A cold start surge of this power supply module requires a maximum of 40 A.

Terminals on the front of the module are labeled +, -, and GND. The module operates with a nominal input voltage of 20 to 28 Vdc.

The module has a front accessible fuse holder and is shipped with a 8 A fast blow fuse.

7.3.12.4 I/O 115/230 Vac, 15 A Power Supply Module

The model 621-9933 I/O Power Supply Module is a double wide module that provides up to 15 A of +5 Vdc for the logic circuitry in the I/O card file. It also supplies up to 2 A of ± 15 Vdc. The total power output is limited to 110 Watts. The graph in Figure 7-10 shows the relationship of current loads between the ± 15 Vdc source and the +5 Vdc source. If, for example, the ± 15 Vdc requires 2 A (2 A of +15 Vdc and 2 A of -15 Vdc), the +5 Vdc can draw a maximum of 10 A. If the ± 15 Vdc requires 1.2 A (or less), the +5 Vdc can draw the maximum of 15 A.

The maximum power consumption of the module is 110 VA. A cold start surge requires a maximum of 20 A for one cycle.

The input is selectable for 115 or 230 Vac operation by the position of the toggle switch located under the component cover of the module. The 115 Vac selection allows a voltage range of 85 to 132 Vac. The 230 Vac selection allows a voltage range of 170 to 250 Vac operation.

The module contains a front accessible fuse holder. When this power supply is shipped from the factory as part of a Logic Manager system the power supply is configured for the line voltage requested by the user. When it is shipped as a part, it is shipped with settings for 115 Vac operation. A fuse (1 amp slo-blo) for 230 Vac is shipped with the part, if the user wants to operate the supply at 230 Vac. It is the users responsibility to replace the 115 Vac fuse with the 230 Vac fuse.

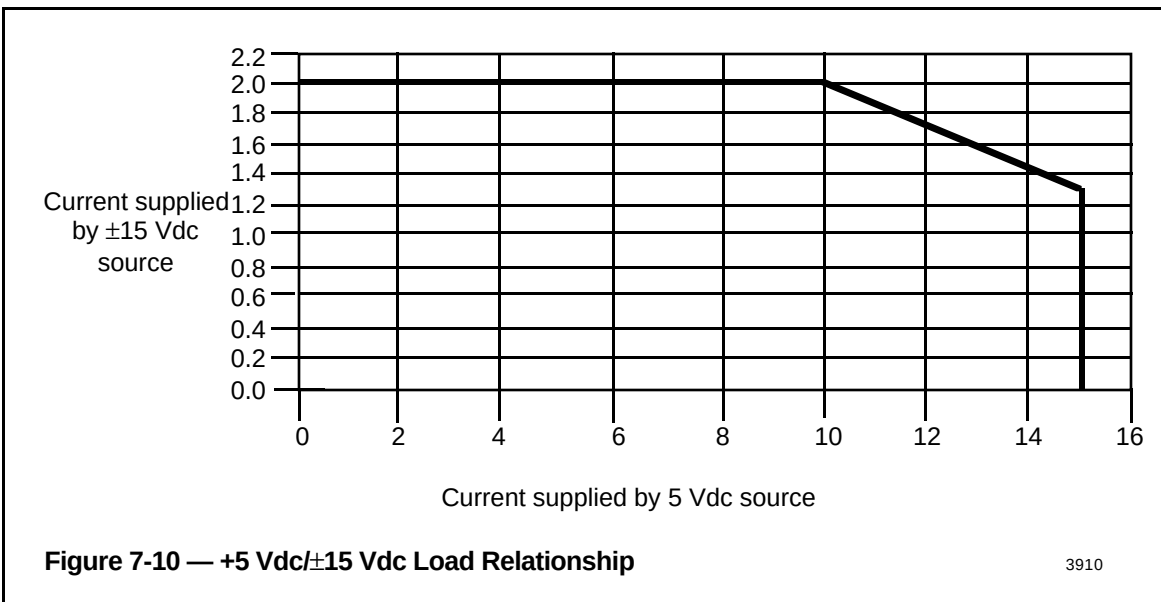
WARNING

Be sure to match the 115/230 Vac toggle switch position with the proper fuse selection.

7.3.12.5 I/O 115/230 Vac, 8 A Power Supply Module

The model 621-9934 I/O Power Supply Module provides 8 A of +5 Vdc power. It also provides 600 mA of ± 15 Vdc power for operation of some special function I/O and analog I/O modules. The module is selectable for 115/230 Vac operation, 46 to 63 Hz.

A terminal block at the top of the module front coverplate is labeled with an A (L1 or Line), a B (L2 or Common), and a ground symbol for AC input wiring connection.



PROCESS I/O WIRING INSTALLATION

Section 8

This section details the specification, configuration, and installation wiring information for the Logic Manager I/O modules.

8.1 INTRODUCTION

This section provides information to be used for the installation of **TotalPlant** Solution (TPS) systems with Logic Managers. It is intended to provide information necessary to select and connect the input and output signal wiring to the Logic Manager's I/O subsystem.

8.2 REFERENCES

Additional information concerning the I/O modules documented in this section may be referenced in the following Honeywell 620 Logic Controller System publications, which are available from the IACD Technical Resources Distribution Center (internal Honeywell address: HED AZ15/E1):

- Honeywell 621 Logic Controller System I/O Specification, 620-8995
- Honeywell 620 Logic Controller System Installation Manual, 620-8996

8.3 I/O POINT MODULES

Table 8-1 lists the I/O point input and output modules, both discrete and analog, that are available for installation in Logic Manager I/O card files.

Table 8-1 — Logic Manager I/O Point Modules

MODULE NUMBER	DESCRIPTION	POINTS
621-0010-AR	4-20 mA Analog Output	4
621-0010-VR	0-10 V Analog Output	4
621-1101R	115 Vac/Vdc Isolated Discrete Input	6
621-1201R	230 Vac/Vdc Isolated Discrete Input	6
621-0007R	Reed Relay Output	6
621-2101R	115 Vac Isolated Discrete Output	6
621-2201R	230 Vac Isolated Discrete Output	6
621-0022-AR	0-20 mA Isolated Analog Input	8
621-0022-VR	0-10 V Isolated Analog Input	8
621-1100R	115 Vac/Vdc Discrete Input	8
621-1200R	230 Vac/Vdc Discrete Input	8
621-1500R	24 Vac/Vdc Discrete Input	8
621-3300R	5 Vdc Sink Discrete Input	8
621-3500R	12-24 Vdc Sink Discrete Input	8
621-3502R	12-24 Vdc Sink Fast Response Discrete Input	8
621-3600R	48 Vdc Sink Discrete Input	8
621-4300R	5 Vdc Source Discrete Input	8
621-4500R	12-24 Vdc Source Discrete Input	8
621-4502R	12-24 Vdc Source Fast Response Discrete Input	8
621-2100R	115 Vac Discrete Output	8
621-2102R	115 Vac Self-Protected Discrete Output	8
621-2200R	230 Vac Discrete Output	8
621-2500R	24 Vac Discrete Output	8
621-6300R	5 Vdc Source Discrete Output	8
621-6500R	12-24 Vdc Source Discrete Output	8
621-6503R	12-24 Vdc Source Self-Protected Discrete Output	8
621-6600R	48 Vdc Source Discrete Output	8
621-6700R	120 Vdc Source 0.5 A Discrete Output	8
621-6701R	120 Vdc Source 2 A Discrete Output	8
621-0020R	Universal Analog Input	16
621-1160R	115 Vac Discrete Input	16
621-1250R	240 Vac Discrete Input	16
621-1251R	240 Vac Isolated Discrete Input	16
621-1550R	24 Vac/Vdc Discrete Input	16
621-3450R	12 Vdc Sink Discrete Input	16
621-3552R	24 Vdc Sink Fast Response Discrete Input	16
621-3560R	24 Vdc Sink Discrete Input	16
621-3660R	48 Vdc Sink Discrete Input	16
621-4350R	5 Vdc TTL Discrete Input	16
621-4550R	24 Vdc Source Discrete Input	16
621-2150R	115 Vac Discrete Output	16
621-2550R	24 Vac Discrete Output	16
621-6350R	5 Vdc TTL Source Discrete Output	16
621-6450R	12 Vdc Source Discrete Output	16
621-6550R	24 Vdc Source Discrete Output	16
621-6551R	24 Vdc Source Low Power Discrete Output	16
621-6650R	48 Vdc Source Discrete Output	16
621-1180R	115 Vac Discrete Input	32
621-3580R	24 Vdc Sink Discrete Input	32
621-2175R	115 Vac Discrete Output	32
621-6575R	24 Vdc Source Discrete Output	32

8.4 I/O TERMINAL BLOCK SETS (Terminal Blocks are included with the I/O Modules)

I/O terminal block sets provide the field wiring termination for the I/O modules. Terminal blocks are required for all I/O modules (including discrete, analog, and certain special function I/O modules) unless otherwise stated in the module specifications. Included is one 8 point I/O terminal block set for each 6- or 8- point module, one 16-point I/O terminal block set for each 16-point module, and one 32- point I/O terminal block set for each 32-point module.

8.4.1 8-Point I/O Terminal Block Sets

The 8-point terminal block set provides the field wiring termination for any 6- point or 8-point I/O module. Each set, consisting of a clearly labeled TOP and BOTTOM half, allows connection of a maximum of two #14 MTW wires per terminal.

The 8-point terminal blocks are ordered as single sets (model 621-9949) or as 10 pack sets (model 621-9947). These sets can be ordered separately for spare parts.

Each terminal block consists of a thumbscrew, two locating pins, and a cover. The thumbscrew secures the wiring arm to the I/O module, and the locating pins align the wiring arm with the I/O card file. A hinged terminal block cover allows placement of I/O address labels.

Wiring details for each module are found in the respective module description sections of this manual. Two pairs of terminals on the blocks, T1 and T2, B1 and B2, provide field power to the module. Each pair has a removable jumper, which distributes module load current between the two terminals of the pair. Each terminal block set is rated for 12 amp total load current. Do not remove the T1/T2 or B1/B2 jumpers unless instructed to do so for the module installed.

Four keying pins are included with each 8-point terminal block set. They are inserted into the terminal block according to the type of I/O module used.

8.4.2 16-Point I/O Terminal Block Sets

The 16-point terminal block set provides field wiring termination for all 16-point I/O modules. Each set, consisting of a clearly labeled TOP and BOTTOM half, allows connection of a maximum of one #14 MTW wire per terminal. A 10 pack consists of ten TOP halves and ten BOTTOM halves.

The 16-point terminal blocks are ordered as single sets (model 621-9950) or as 10 pack sets (model 621-9952). These sets can be ordered separately as spare parts.

Each of the 22 terminals in the 16-point set is a screw terminal with a pressure pad. Each terminal block has a thumbscrew and two locating pins. The thumbscrew secures the I/O wiring arm to the I/O module, and the locating pins align the I/O arm with the I/O card file. A hinged terminal block cover allows placement of I/O address labels.

Wiring details are found in the respective module description sections of this manual. Two groups of three terminals on the terminal blocks, T1, T2, T3, and B1, B2, B3, provide field power to the I/O module. Each group has removable jumpers, which distribute module load current between the terminals. The terminal block set is rated for 18 A total load current. Do not remove T1/T2/T3 or B1/B2/B3 jumpers unless instructed to do so.

Four keying pins are included with each 16-point terminal block set. They are inserted in the terminal block according to the type of I/O module used.

8.4.3 32-Point I/O Terminal Block Set

The 32-point I/O terminal block set (model 621-9977) provides the field wiring termination for all 32-point I/O modules. Each set contains two interchangeable terminal blocks and allows connection of one #14 AWG wire per terminal.

The terminal block is inserted into a block header located on the front of the 32-point I/O module. The field wiring is installed on the front of the connector using clamping screws that can be accessed without removing the connector from the module. If a connector must be removed, it can be ejected by using the terminal block ejector lever.

Additional wiring details are found in the respective I/O module description sections in this manual.

8.4.4 32-Point I/O Terminal Block Set With LEDs

This I/O terminal block set (model 621-9976) is used with the 24 Vdc 32-point I/O module. The terminal blocks provide field side LED indication for each point. The LEDs are located inside the terminal block, next to the wiring terminals.

8.4.5 I/O Address Labels (Included with LM Processor Card File)

The I/O address paper labels are available for use with I/O module terminal blocks.

The paper labels slide behind the clear I/O terminal block cover. The paper labels cannot be used with the old terminal blocks. These labels are numbered sequentially on the front and provide space for notes on the back. The paper labels are available for 6-, 8-, 16- and 32-point I/O modules.

All address labels can be ordered as spares by model number, according to the types of I/O modules used in the system. The model numbers are listed in the following tables for reference.

Table 8-2 — Paper I/O Address Labels

MODEL NUMBER	POINT ADDRESS
629-9980	8-Point Address 0-511
629-9981	8-Point Address 0-2047
629-9982	16-Point Address 0-511
629-9983	16-Point Address 0-2047
629-9988	6-Point Address 0-511
629-9997	32-Point Address 0-2047

8.5 I/O MODULE INSTALLATION

8.5.1 Pre-Installation Considerations

This section provides information that applies generally to the installation of all I/O modules. Specific Logic Manager System installation procedures are provided in previous sections of this manual. Review these sections in this manual before attempting any installation procedures:

- I/O Module Addressing
- General Guidelines for Safe and Proper I/O Module Installation

These guidelines are designed to aid system configurations and to ensure optimal system performance and operator safety.

8.5.1.1 I/O Module Addressing

Keep the following factors in mind when assigning I/O addresses:

- Parallel I/O is addressed first (beginning with address of 0), followed by the various serial I/O channels.
- Card files with 32-point I/O modules require one of the following, depending on the card file type: 621-9937 PIOM or 621-9940/-9938R SIOM. Note that these modules also accommodate any combination of 4-, 6-, 8-, 16-, and/or 32-point I/O modules.
- Each serial channel accommodates a maximum of 2040 I/O points, provided the total does not exceed the processor maximum of 2048.
- Overlapping I/O addresses and identical starting addresses within a system are prohibited.
- Do not duplicate starting addresses.
- Skipping addresses is permitted.
- Note that the SIOMs (models 621-9938R and 621-9940) determine the absolute starting address and the number of I/O for each I/O card file. These SIOMs can interface only with a model 621-9939 Serial Link Module (SLM).

Input Module Addressing

Input circuit number one is located at the top terminal of the module. This input's address is always the least significant address (LSA) of the I/O card slot in which the module is installed. Other input terminals on the module assume higher addresses according to their position on the module.

Output Module Addressing

Output circuit number one is located at the top terminal of the module. This output's address is always the least significant address (LSA) of the I/O card slot in which the module is installed. Other outputs on the module assume higher addresses according to their position on the module.

8.5.1.2 Precautions for Proper Installation

Ensure optimal system performance and operator safety by adhering to the following precautions for safe and proper installation.

- Carefully review all literature pertaining to the Logic Manager System, paying close attention to cautionary notes.
- The installation of Logic Manager System equipment must be in accordance with the current edition of the National Electric Code ANSI/NFPA 70.
- Some I/O power supply modules allow voltage selection. Ensure that the proper voltage selection, 115 Vac or 230 Vac, is made before supplying power to the system.
- I/O modules should be installed and removed only when power is removed from the system, if you are using a parallel I/O system. Do not attempt to install or remove modules from the card file while the card file is powered. This could damage the module, possibly affect the backpanel communications, and turn on outputs unintentionally for one program scan.
- I/O modules may be inserted or removed from a card file while power is applied IF a 621-9938R SIOM is located in the I/O card file.
- The coverplates supplied with Logic Manager System card files and modules shield the Logic Manager electronics from electromagnetic noise. Ensure that coverplates are installed over all blank slots.
- Ensure that all thumbscrews on module frontplates and coverplates are firmly tightened.
- All control wiring should be installed according to IEEE Standard 518-1977 and the National Electric Code. Also review "Cabling Considerations," subsection 5.5.1, in this manual before wiring any Logic Manager System components.

8.5.2 Installing I/O Modules

The following procedure provides general guidelines for installing I/O modules. Be sure to review the instructions for each type of module, because certain I/O modules may require special installation steps not mentioned here. This procedure assumes that the appropriate DIP switch/jumper settings and address assignments have been made according to the module specifications and system requirements.

WARNING

Do not install or remove I/O modules while the card file is powered on.

1. Review the previous section.
2. Review the system configuration (card file layout and module positions within each card file). Refer to the Logic Manager Configuration Guide and prior sections in this manual for examples.
3. Confirm all jumper and DIP switch settings, module by module.
4. Review the selected addressing for all modules in all card files (processor and I/O) to ensure proper and sequential addressing throughout the system.
5. If using an augmented I/O card file, ensure that the card file backpanel jumpers are positioned properly. In a motion control configuration, more than one PTP Control Module may reside in the augmented card file, depending on the jumper placement. Refer to the 622 Motion Control System User Manual, Form No. 622-8997, for more information.
6. Carefully slide each I/O module into the I/O card file slot according to the selected system configuration. Modules are installed with the component side facing left. Seat each module firmly in the backpanel.
7. Install module frontplates, ensuring that each is securely fastened. Install blank coverplates over any empty module slots, and ensure that all thumbscrews (on coverplates and module frontplates) are firmly tightened.
8. After all modules are configured and properly installed in their designated rack slots, proceed with cable routing and wiring. Refer to previous sections of this manual for recommended cable selection and routing.

8.6 I/O Input Module Descriptions

The I/O system includes a variety of input modules that can be classified as either discrete or analog. As shown in Tables 8-4 and 8-5, the model 621 input modules are available in different voltage levels to match the voltage requirements of the field devices to which the modules are connected. Each module provides a standard number of inputs, either 4, 6, 8, 16, or 32.

This section provides individual input module specifications, references to wiring diagrams, and in certain cases, special instructions for installation and operation. The modules are divided into discrete and analog sections and are listed in numerical model number order as shown in Tables 8-4 and 8-5.

Table 8-4 — Discrete Input Modules

MODULE DESCRIPTION	NUMBER OF INPUTS	CABLE CLASS	MODEL NUMBER
115 Vac/Vdc Discrete Input	8	DATA 3	621-1100R
115 Vac/Vdc Isolated Discrete Input	6	DATA 3	621-1101R
115 Vac Discrete Input	16	DATA 3	621-1160R
115 Vac Discrete Input	32	DATA 3	621-1180R
230 Vac/Vdc Discrete Input	8	DATA 3	621-1200R
230 Vac/Vdc Isolated Discrete Input	6	DATA 3	621-1201R
240 Vac Discrete Input	16	DATA 3	621-1250R
240 Vac Isolated Discrete Input	16	DATA 3	621-1251R
24 Vac/Vdc Discrete Input	8	DATA 2	621-1500R
24 Vac/Vdc Discrete Input	16	DATA 2	621-1550R
5 Vdc Sink Discrete Input	8	DATA 1	621-3300R
12 Vdc Sink Discrete Input	16	DATA 1	621-3450R
12-24 Vdc Sink Discrete Input	8	DATA 1, 2 *	621-3500R
12-24 Vdc Sink Fast Response Discrete Input	8	DATA 1, 2 *	621-3502R
24 Vdc Sink Fast Response Discrete Input	16	DATA 2	621-3552R
24 Vdc Sink Discrete Input	16	DATA 2	621-3560R
24 Vdc Sink Discrete Input	32	DATA 2	621-3580R
48 Vdc Sink Discrete Input	8	DATA 2	621-3600R
48 Vdc Sink Discrete Input	16	DATA 2	621-3660R
5 Vdc Source Discrete Input	8	DATA 1	621-4300R
5 Vdc TTL Discrete Input	16	DATA 1	621-4350R
12-24 Vdc Source Discrete Input	8	DATA 1, 2 *	621-4500R
12-24 Vdc Source Fast Response Input	8	DATA 1, 2 *	621-4502R
24 Vdc Source Discrete Input	16	DATA 2	621-4550R

* The IEEE 518 wiring class is application dependent and can vary with the voltage and current present in the wiring.

Table 8-5 — Analog Input Modules

MODULE DESCRIPTION	NUMBER OF INPUTS	CABLE CLASS	MODEL NUMBER
Universal Analog Input	16	DATA 1	621-0020R
Isolated 0-20 mA Analog Input	8	DATA 1	621-0022-AR
Isolated 0-10 V Analog Input	8	DATA 1	621-0022-VR

8.6.1 Discrete Input Modules

Discrete Input Modules sense ON or OFF voltage levels of various input devices. Among the more common devices are limit switches, proximity sensors, and pushbuttons. Once this ON/OFF status is obtained, the input modules convert it into logic-level data to be used by the processor module.

LED indication for each input circuit on 6-, 8-, and 16-point modules is on the field device side of the input module circuitry. The 32-point input modules have the LED indication for each circuit on the 5 V logic side of the module. All discrete inputs are optically isolated to 2500 Vdc.

These modules are two piece units. A printed circuit board with a permanent front cover and label to identify each particular module, and a set of 8-, 16-, or 32-point terminal blocks make up each complete I/O module. The printed circuit board is inserted into the desired slot in the I/O card file. An offset backpanel connector makes it impossible to install a module upside-down.

The removable 8- and 16-point double swing terminal blocks allow for easy module insertion, removal, and wiring. These are installed by firmly seating the edge connector of each terminal block onto the front edge of the circuit board and securing the thumbscrew fastener. These terminal blocks are user-keyed to the circuit board. The 32-point terminal blocks are inserted into block headers located on the front of the module. Field wiring connects to the terminal block from the front of the module and can be accessed without removing the terminal block.

Each dc input module is either a sink or a source module. A sink input module provides the ground potential for the input device when the input device turns on. Current flows from the power supply through the field sensor input device (when it turns on) to the sink input module (ground). A source input module supplies the +Vdc potential to the field sensor input device when it turns on. Current flows from the power supply to the source input module to the field input sensor device to ground.

Allowable Leakage Current

Each discrete input module has a specification called “Allowable Leakage Current.” This specification is for the user who wants to connect a discrete output or an external sensor (such as a proximity switch) to a discrete input.

A small amount of current usually leaks from any discrete output or external sensor before the output energizes. Although this current, called “Off-State Leakage Current,” is usually less than 5 milliamps, it can cause an input to recognize an ON condition before the output actually turns on.

To prevent such a situation, drain the leakage current by connecting a resistor from the discrete input to ground. Use the following procedure to determine the value and wattage of the resistor:

1. Subtract the Allowable Leakage Current value for the discrete input module from the Off-State Leakage Current value of the discrete output module (or the external sensor).
2. Divide the Logic 0 voltage of the discrete input by the value determined in step 1. This will give the value for the resistor to be used.

$$R_v = \frac{LOV}{OSLC - ALC}$$

where: R_v = resistor value
 LOV = Logic 0 value
 $OSLC$ = Off State Leakage Current
 ALC = Allowable Leakage Current

3. Determine the wattage by dividing the squared maximum output voltage by the value determined in step 2.

$$Watt = \frac{(VO_{max})^2}{RV}$$

where: $Watt$ = wattage of R_v
 $(VO_{max})^2$ = maximum output voltage squared

Typically, the wattage should be 200 percent of the calculated value so that the resistor does not degrade over time.

8.6.1.1 6-Point Vac/Vdc Isolated Discrete Input Modules

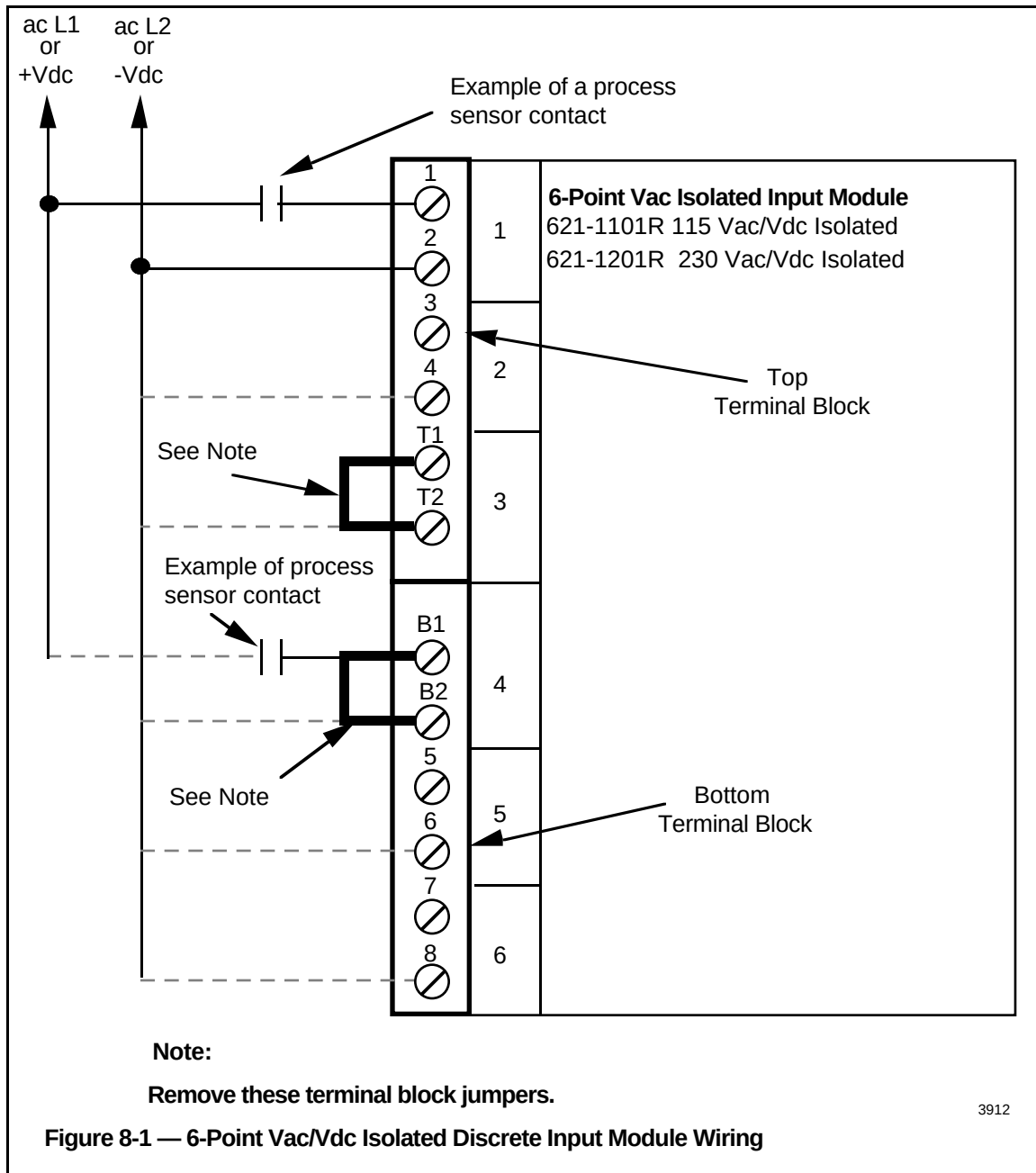
The 6-Point Vac/Vdc Isolated Discrete Input modules provide voltage source selectability for each of the six termination points. See Table 8-6 and Figure 8-1.

WARNING

The terminal block jumpers between terminals T1 and T2 and terminals B1 and B2 must be removed.

Table 8-6— 6-Point Isolated Discrete Input Module Specifications

	MODEL NUMBER	
	621-1101R	621-1201R
POINT INPUTS	6	6
SPECIAL FEATURES	Isolated inputs	Isolated inputs
NOMINAL VOLTAGE	115 V	230 V
VOLTAGE RANGE	90-140 Vac/Vdc	195-250 Vac/Vdc
CURRENT RANGE	6-21 mA (12 mA at 115 Vac/Vdc typical)	4-10 mA (8.5 mA at 230 Vac/Vdc typical)
SWITCHING LEVEL		
Logic 1	75 Vac/85 Vdc	140 Vac/155 Vdc
Logic 0	40 Vac/60 Vdc	63 Vac/90 Vdc
ALLOWABLE LEAKAGE CURRENT	2 mA ac 3 mA dc	1 mA ac 1.5 mA dc
INPUT DELAY		
Off to On	2.4 ms $\pm$ 20% plus 0 to 0.5 cycles	2.4 ms $\pm$ 20% plus 0 to 0.5 cycles
On to Off	17 ms $\pm$ 20%	17 ms $\pm$ 20%
OPERATING CURRENT + 5 Vdc	30 mA max.	30 mA max.
TERMINAL BLOCK REQUIRED	621-9949	621-9949



8.6.1.2 8-Point Vac/Vdc Discrete Input Modules

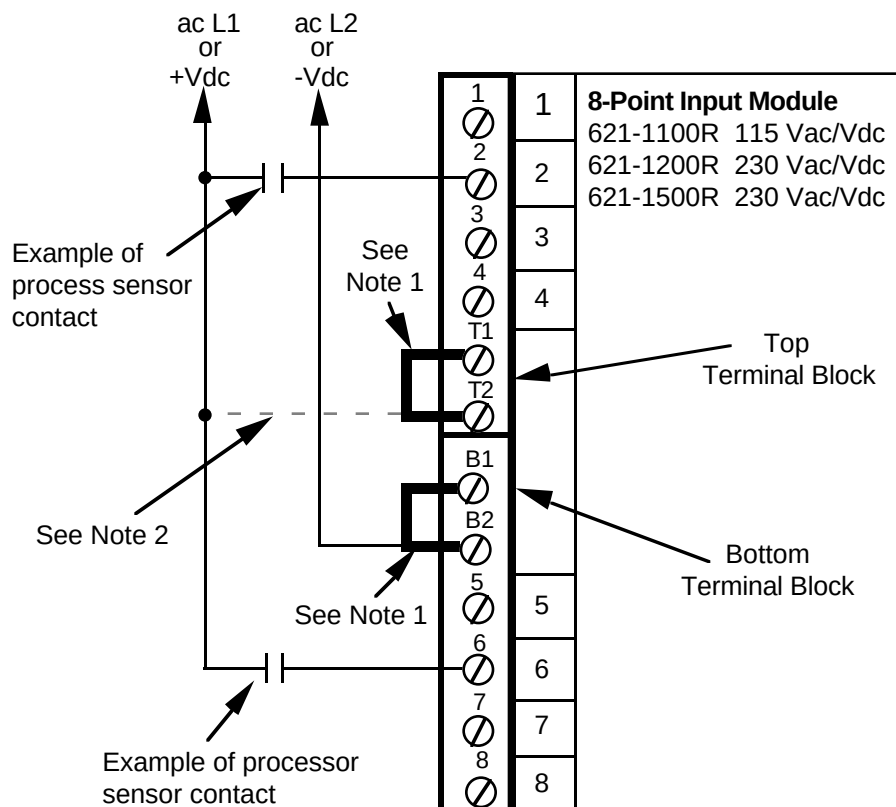
See Table 8-7 and Figure 8-2 for specifications and wiring information for the 8-Point Vac/Vdc Discrete Input modules. A separate connection to L1/+Vdc is needed when the module is used in an ungrounded ac control circuit. Input operation is unaffected by this connection. However, for ungrounded ac systems using ground fault detection circuitry, this connection balances the system and prevents erroneous ground fault indications.

NOTE

Terminals T1 and B1 provide voltage connections for circuits 1 through 4. Terminals T2 and B2 provide voltage connections for terminals 5 through 8. Separate voltage sources will power circuits 1 through 4 and 5 through 8 if the terminal block jumpers are removed.

Table 8-7 — 8-Point Vac/Vdc Discrete Input Module Specifications

	MODEL NUMBERS		
	621-1100R	621-1200R	621-1500R
POINT INPUTS	8	8	8
POINTS PER COMMON	4	4	4
SPECIAL FEATURE	-	-	-
NOMINAL VOLTAGE	115 V	230 V	24 V
VOLTAGE RANGE	90-140 V	195-250 V	18-28 V
CURRENT RANGE	6-21 mA (12 mA at 115 V typical)	4.5-10 mA (8.5 mA at 230 V typical)	4-15 mA (10 mA at 24 V typical)
SWITCHING LEVEL			
Logic 1	75 Vac/85 Vdc	140 Vac/155 Vdc	18 Vac/19 Vdc
Logic 0	40 Vac/60 Vdc	63 Vac/90 Vdc	9 Vac/11 Vdc
ALLOWABLE LEAKAGE CURRENT	2 mA ac 3 mA dc	1 mA ac 1.5 mA dc	1.3 mA ac 1.7 mA dc
INPUT DELAY			
Off to On	2.4 ms $\pm$ 20% plus 0 to 0.5 cycles	2.4 ms $\pm$ 20% plus 0 to 0.5 cycles	2.4 ms $\pm$ 20% plus 0 to 0.5 cycles
On to Off	17 ms $\pm$ 20%	17 ms $\pm$ 20%	17 ms $\pm$ 20%
OPERATING CURRENT + 5 Vdc	30 mA max.	30 mA max.	30 mA max.
TERMINAL BLOCK REQUIRED	621-9949	621-9949	621-9949



Notes:

1. To use two different power supplies, these terminal block jumpers are removed. Without the jumpers, T1 and B1 are the voltage references for terminals 1 through 4, and T2 and B2 are the voltage references for terminals 5 through 8.
2. This connection is needed only when this module is used in an ungrounded ac control circuit. Input operation is unaffected by this connection. However, for ungrounded ac systems using ground fault detection circuitry, this connection balances the system and prevents erroneous ground fault indications.

3913

Figure 8-2 — 8-Point Vac/Vdc Discrete Input Module Wiring

8.6.1.3 8-Point Vdc Sink Discrete Input Modules

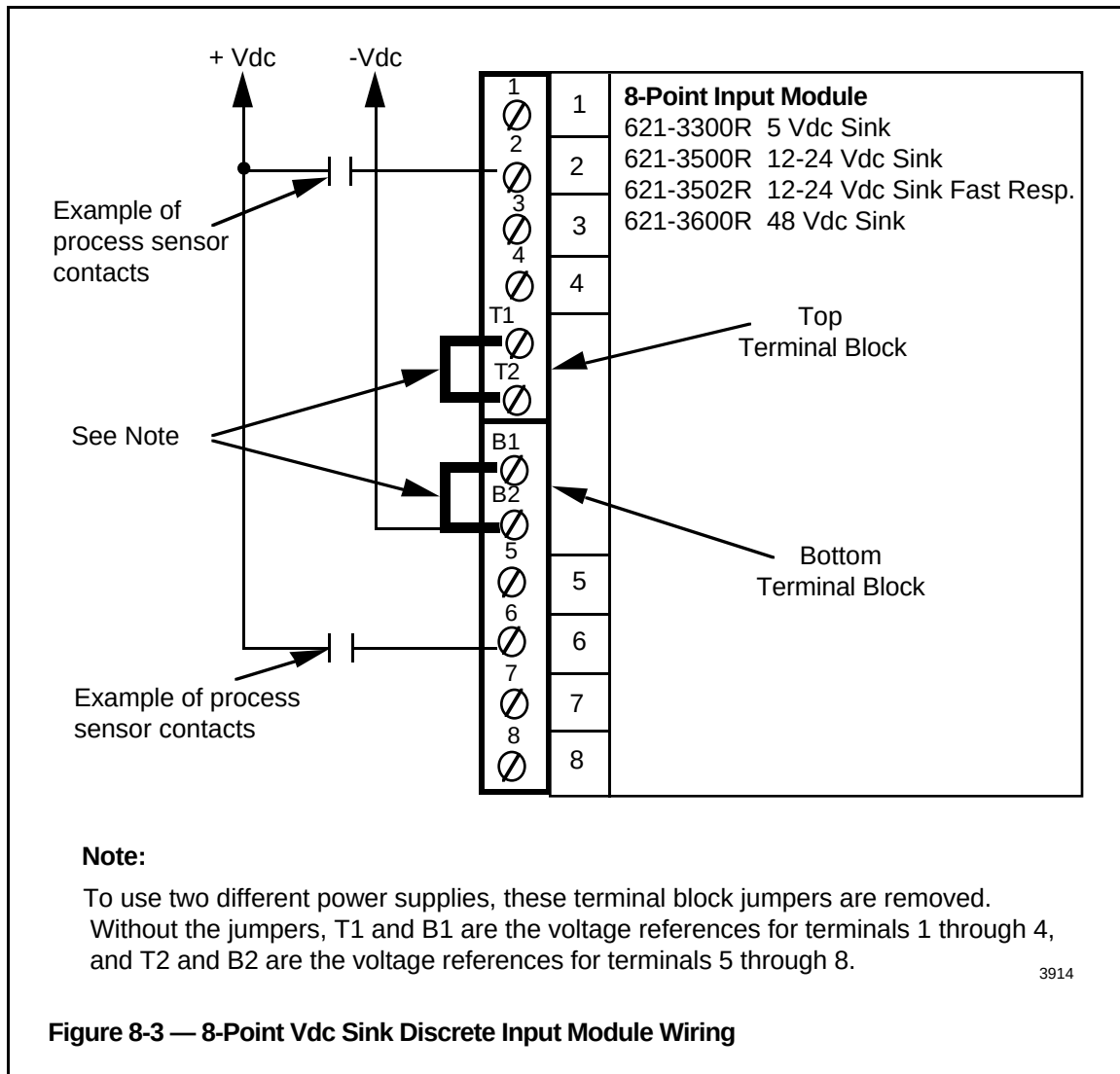
See Table 8-8 and Figure 8-3 for specifications and wiring information for the 8-Point Vdc Sink Discrete Input modules.

NOTE

Terminals T1 and B1 provide voltage connections for circuits 1 through 4. Terminals T2 and B2 provide voltage connections for terminals 5 through 8. Separate voltage sources will power circuits 1 through 4 and 5 through 8 if the terminal block jumpers are removed.

Table 8-8 — 8-Point Vdc Sink Discrete Input Module Specifications

	MODEL NUMBER			
	621-3300R	621-3500R	621-3502R	621-3600R
POINT INPUTS	8	8	8	8
POINTS PER COMMON	4	4	4	4
SPECIAL FEATURE	Sink input	Sink input	Sink input fast response	Sink input
NOMINAL VOLTAGE	5 Vdc	12-24 Vdc	12-24 Vdc	48 Vdc
VOLTAGE RANGE	4-6 Vdc	10-32 Vdc	10-32 Vdc	38-66 Vdc
CURRENT RANGE	3-12 mA (7.5 mA at 5 V typical)	4.5-28 mA (18 mA at 24 V typical)	4.5-28 mA (18 mA at 24 V typical)	8.5-20 mA (12 mA at 48 V typical)
SWITCHING LEVEL				
Logic 1	4 Vdc	10 Vdc	10 Vdc	22 Vdc
Logic 0	1.5 Vdc	3 Vdc	3 Vdc	12 Vdc
ALLOWABLE LEAKAGE CURRENT	1 mA	1.2 mA	1.2 mA	1.4 mA
INPUT DELAY				
Off to On	2.4 ms ± 20%	2.4 ms ± 20%	0.26 ms max.	2.4 ms ± 20%
On to Off	17 ms ± 20%	17 ms ± 20%	1.4 ms max.	17 ms ± 20%
OPERATING CURRENT +5 Vdc	30 mA max.	30 mA max.	30 mA max.	30 mA max.
TERMINAL BLOCK REQUIRED	621-9949	621-9949	621-9949	621-9949



8.6.1.4 8-Point Vdc Source Discrete Input Modules

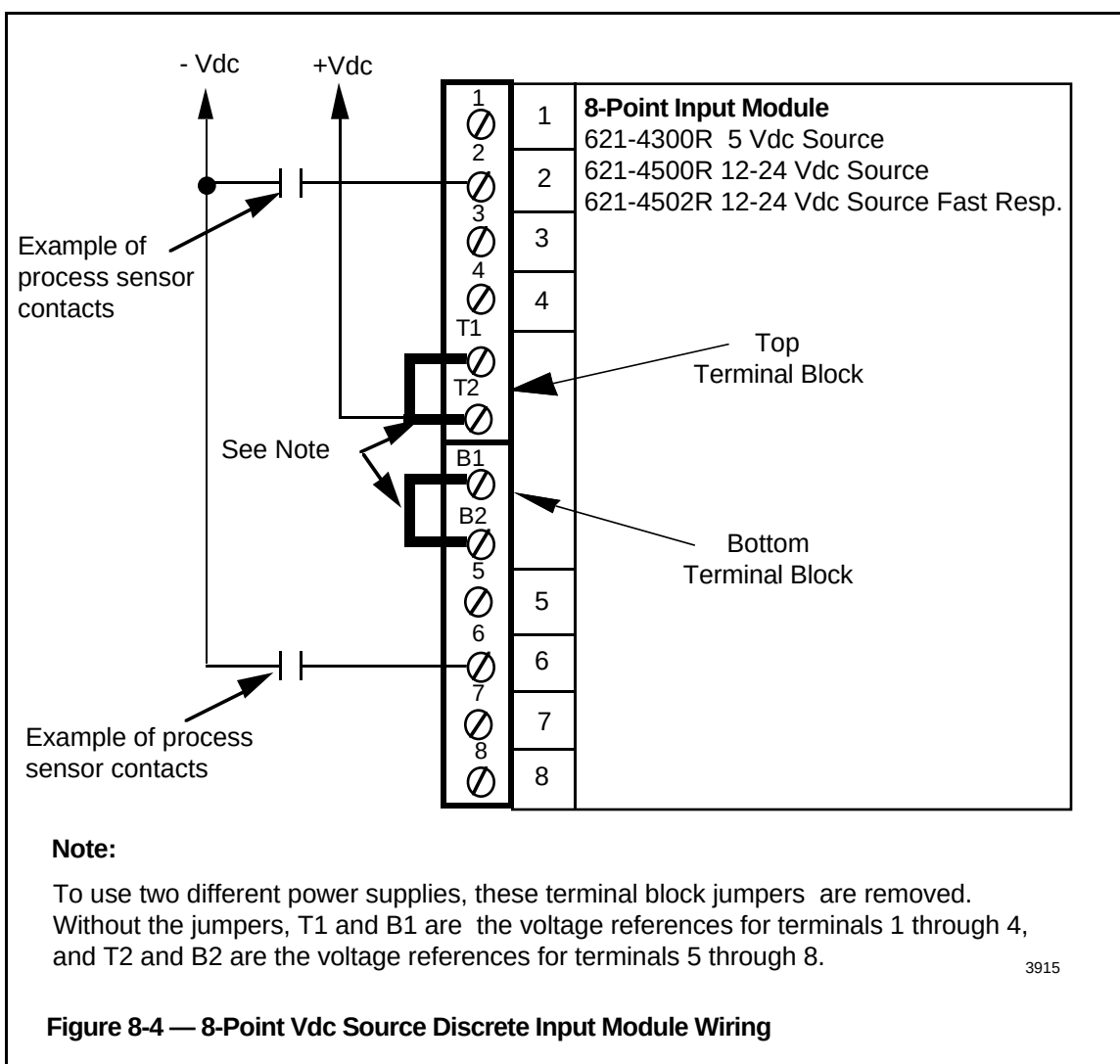
See Table 8-9 and Figure 8-4 for specifications and wiring information for the 8-Point Vdc Source Discrete Input modules.

NOTE

Terminals T1 and B1 provide voltage connections for circuits 1 through 4. Terminals T2 and B2 provide voltage connections for terminals 5 through 8. Separate voltage sources will power circuits 1 through 4 and 5 through 8 if the terminal block jumpers are removed.

Table 8-9 — 8-Point Vdc Source Discrete Input Module Specifications

	MODEL NUMBER		
	621-4300R	621-4500R	621-4502R
POINT INPUTS	8	8	8
POINTS PER COMMON	4	4	4
SPECIAL FEATURE	Source input	Source input	Source input fast response
NOMINAL VOLTAGE	5 Vdc	12-24 Vdc	12-24 Vdc
VOLTAGE RANGE	4-6 Vdc	10-32 Vdc	10-32 Vdc
CURRENT RANGE	3-12 mA (7.5 mA at 5 V typical)	4.5-28 mA (18 mA at 24 V typical)	4.5-28 mA (18 mA at 24 V typical)
SWITCHING LEVEL			
Logic 1	1.5 Vdc	3 Vdc	3 Vdc
Logic 0	4 Vdc	10 Vdc	10 Vdc
ALLOWABLE LEAKAGE CURRENT	1.0 mA	1.2 mA	1.2 mA
INPUT DELAY			
Off to On	2.4 ms $\pm$ 20%	2.4 ms $\pm$ 20%	0.26 ms max.
On to Off	17 ms $\pm$ 20%	17 ms $\pm$ 20%	1.4 ms max.
OPERATING CURRENT +5 Vdc	30 mA max.	30 mA max.	30 mA max.
TERMINAL BLOCK REQUIRED	621-9949	621-9949	621-9949



8.6.1.5 16-Point Vac/Vdc Discrete Input Module

The model 621-1550R 24 Vac/Vdc Discrete Input module for 16-point discrete input capability. See Figure 8-5.

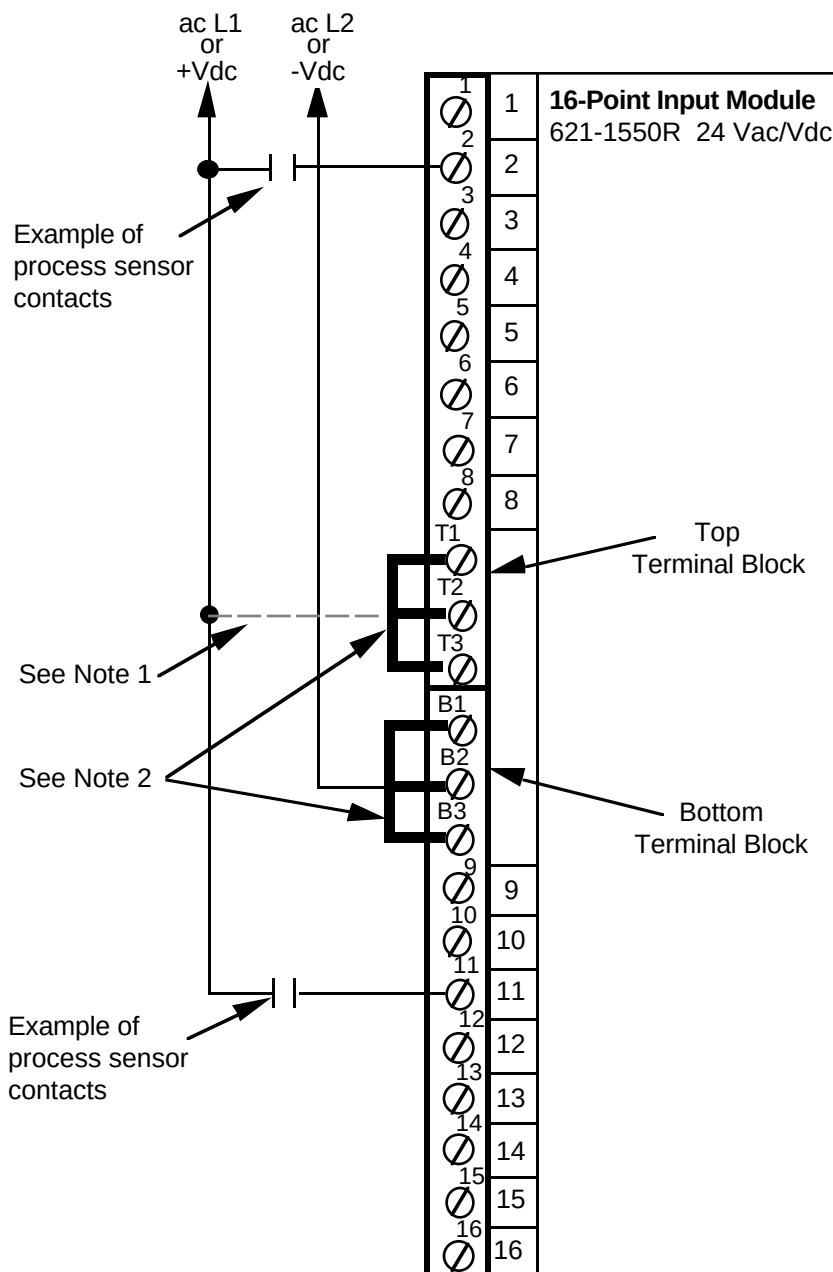
A separate connection to L1/+Vdc is needed when the module is used in an ungrounded ac control circuit. Input operation is unaffected by this connection. However, for ungrounded ac systems using ground fault detection circuitry, this connection balances the system and prevents erroneous ground fault indications.

Specifications:

INPUTS	16
POINTS PER COMMON	16
NOMINAL VOLTAGE	24 Vac/Vdc
VOLTAGE RANGE	18-28 Vac/Vdc
CURRENT RANGE	4-15 mA (10 mA at 24 Vac/Vdc typical)
SWITCHING LEVEL	
Logic 1	18 Vac/19 Vdc
Logic 0	9 Vac/11 Vdc
ALLOWABLE LEAKAGE CURRENT	1.3 mA ac 1.7 mA dc
INPUT DELAY	
Off to On	2.4 ms $\pm$ 20%
On to Off	17 ms $\pm$ 20%
MODULE OPERATING CURRENT	
+5 Vdc	54 mA max.
TERMINAL BLOCK REQUIRED	621-9950

WARNING

Do not remove the terminal block jumpers.



Notes:

1. This connection is needed only when this module is used in an ungrounded ac control circuit. Input operation is unaffected by this connection. However, for ungrounded ac systems using ground fault detection circuitry, this connection balances the system and prevents erroneous ground fault indications.
2. **Do not remove these terminal block jumpers.**

3916

Figure 8-5 — 16-Point Vac/Vdc Discrete Input Module Wiring

8.6.1.6 16-Point Vac Discrete Input Modules

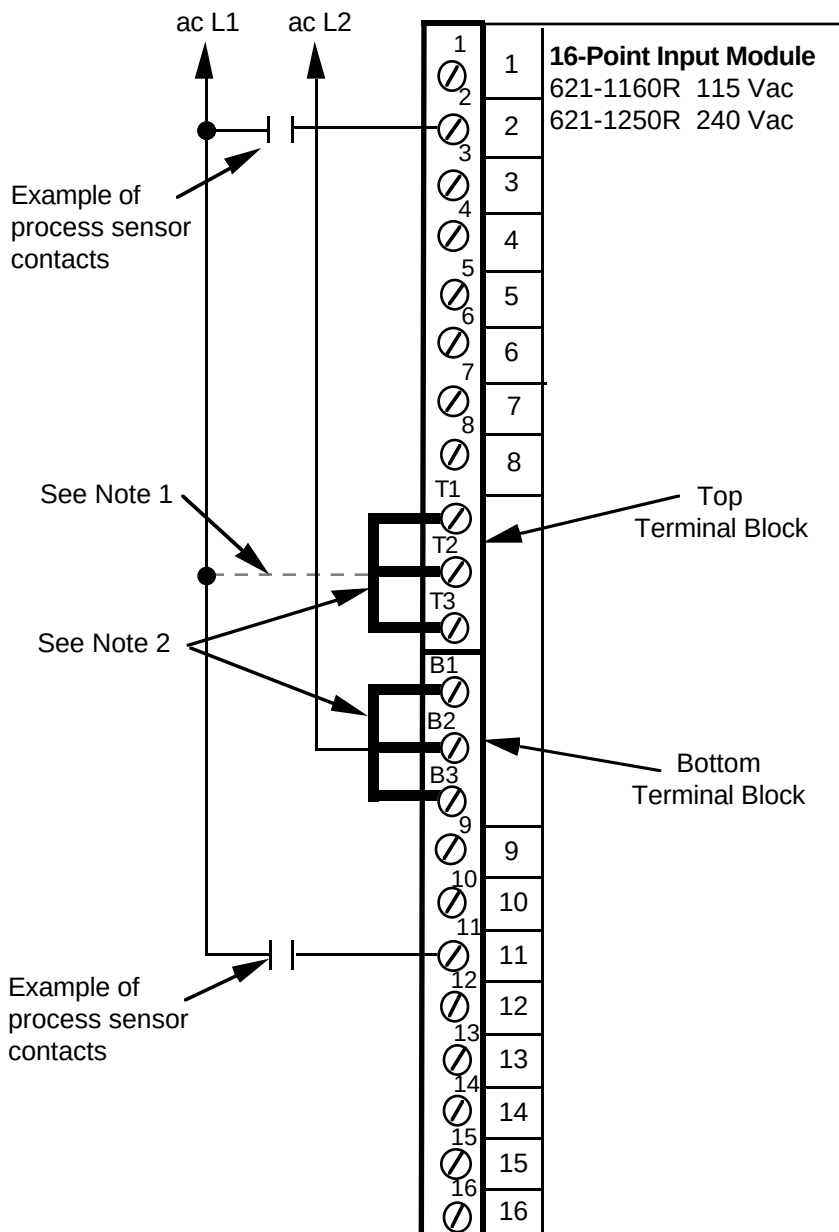
See Table 8-10 and Figure 8-6 for specifications and wiring information for the 16-point Vac Discrete Input modules. A separate connection to L1/+Vdc is needed when the module is used in an ungrounded ac control circuit. Input operation is unaffected by this connection. However, for ungrounded ac systems using ground fault detection circuitry, this connection balances the system and prevents erroneous ground fault indications.

Table 8-10 — 16-Point Vac Discrete Input Module Specifications

	MODEL NUMBER	
	621-1160R	621-1250R
POINT INPUTS	16	16
POINTS PER COMMON	16	16
NOMINAL VOLTAGE	115 Vac	240 Vac
VOLTAGE RANGE	80-140 Vac	180-276 Vac
CURRENT RANGE	4-15 mA (8 mA at 115 Vac typical)	4-13 mA (8 mA at 240 V typical)
SWITCHING LEVEL		
Logic 1	75 Vac	150 Vac
Logic 0	43 Vac	63 Vac
ALLOWABLE LEAKAGE CURRENT	1.5 mA ac	1.3 mA ac
WATTS DISSIPATED PER ON INPUT		0.75 at 276 Vac
INPUT DELAY		
Off to On	2.4 ms $\pm$ 25%	2.4 ms $\pm$ 25%
On to Off	30 ms $\pm$ 50%	24 ms $\pm$ 50%
OPERATING CURRENT + 5 Vdc	$\leq$ 50 mA max.	$\leq$ 195 mA max.
INPUT SURGE CURRENT LIMIT	500 mA per input	500 mA per input
TERMINAL BLOCK REQUIRED	621-9950	621-9950

WARNING

Do not remove the terminal block jumpers.



Notes:

1. This connection is needed only when this module is used in an ungrounded ac control circuit. Input operation is unaffected by this connection. However, for ungrounded ac systems using ground fault detection circuitry, this connection balances the system and prevents erroneous ground fault indications.
2. **Do not remove these terminal block jumpers.**

3917

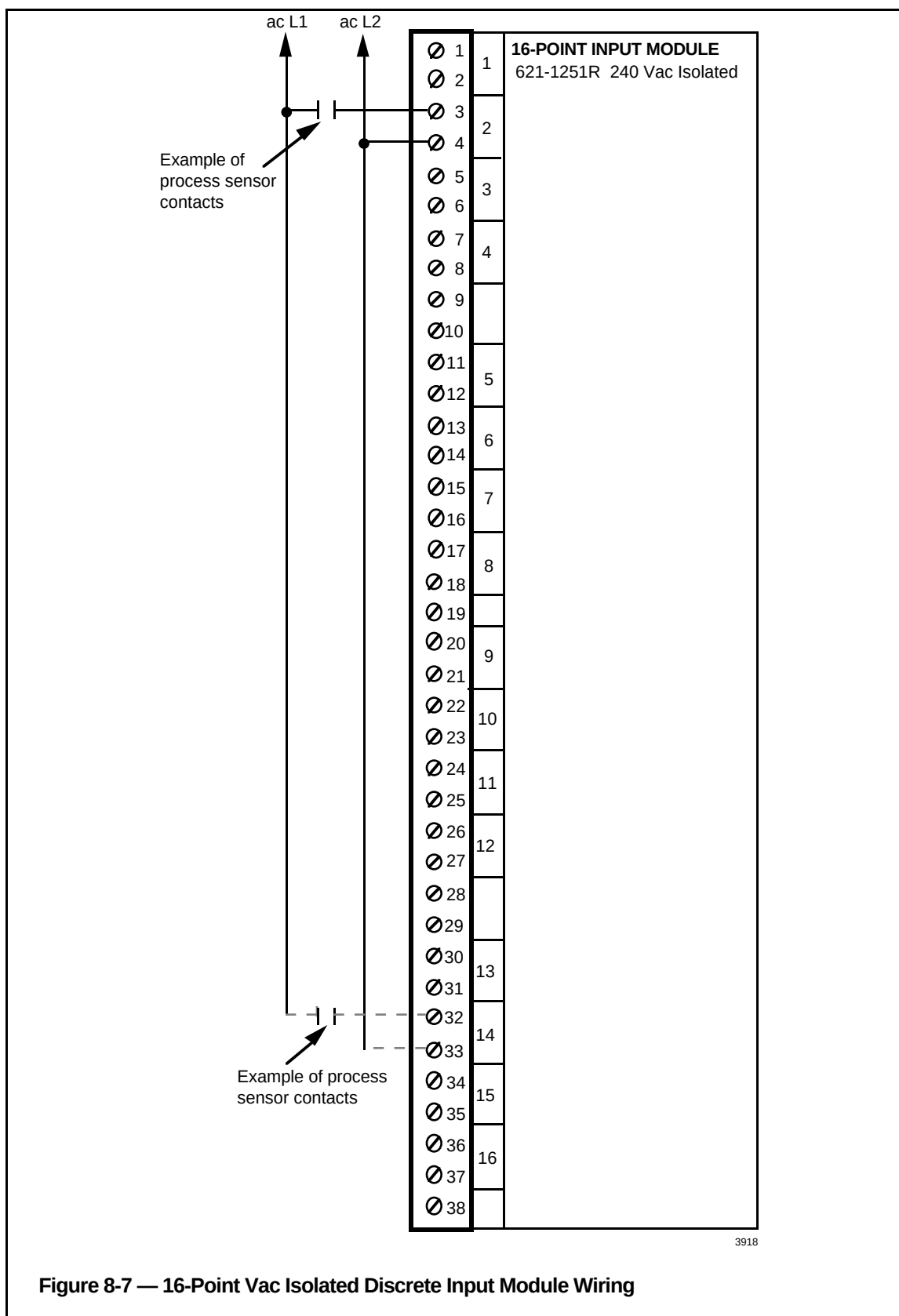
Figure 8-6 — 16-Point Vac Discrete Input Module Wiring

8.6.1.7 16-Point Vac Isolated Discrete Input Module

The model 621-1251R 240 Vac Isolated Discrete Input module provides 16-point discrete input capability. See Figure 8-7.

Specifications:

INPUTS	16
NOMINAL VOLTAGE	240 Vac
VOLTAGE RANGE	180-256 Vac
CURRENT RANGE	4-13 mA
INPUT SURGE	Limited to 500 mA per input
SWITCHING LEVEL	
Logic 1	150 Vac
Logic 0	63 Vac
ALLOWABLE LEAKAGE CURRENT	1.3 mA
WATTS DISSIPATED PER ON CIRCUIT	0.75 at 276 Vac
INPUT DELAY	
Off to On	2.4 ms $\pm$ 25% plus 0 to 0.5 cycles
On to Off	24 ms $\pm$ 50%
MODULE OPERATING CURRENT	
+5 Vdc	$\leq$ 195 mA max.
TERMINAL BLOCK REQUIRED	621-9977



8.6.1.8 16-Point Vdc Sink Discrete Input Modules

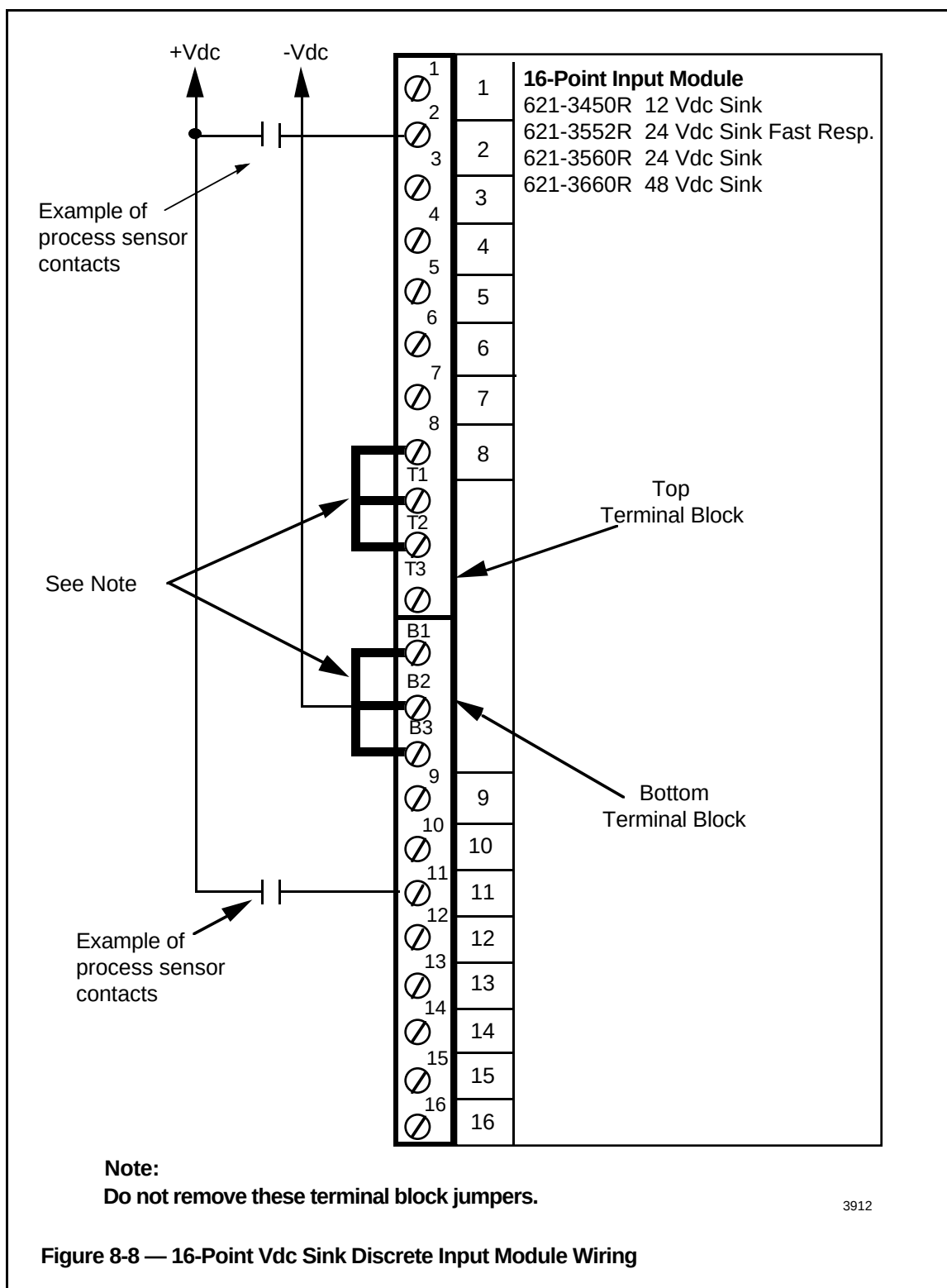
See Table 8-11 and Figure 8-8 for specifications and wiring information for the 16-point Vdc Sink Discrete Input modules.

CAUTION

Do not remove the terminal block jumpers.

Table 8-11 — 16-Point Vdc Sink Discrete Input Module Specifications

	MODEL NUMBER			
	621-3450R	621-3552R	621-3560R	621-3660R
POINT INPUTS	16	16	16	16
POINTS PER COMMON	16 (-Vdc)	16 (-Vdc)	16 (-Vdc)	16 (-Vdc)
SPECIAL FEATURE	Sink input	Sink input fast response	Sink input	Sink input
NOMINAL VOLTAGE	12 Vdc	24 Vdc	24 Vdc	48 Vdc
VOLTAGE RANGE	10-16 Vdc	18-28 Vdc	18-28 Vdc	38-66 Vdc
CURRENT RANGE	4.5-15 mA (7.5 mA at 12 V typical)	3-9 mA (6 mA at 24 V typical)	6-15 mA (10 mA at 24 V typical)	5.5-11.5 mA (8 mA at 48 V typical)
SWITCHING LEVEL				
Logic 1	10 Vdc	18 Vdc	18 Vdc	38 Vdc
Logic 0	3 Vdc	11 Vdc	11 Vdc	20 Vdc
ALLOWABLE LEAKAGE CURRENT	1.2 mA	1.3 mA	2.5 mA	2 mA
INPUT DELAY				
Off to On	2.4 ms ± 20%	0.26 ms (max.)	2.4 ms ± 25%	2.4 ms ± 20%
On to Off	17 ms ± 20%	1.4 ms (max.)	17 ms ± 25%	17 ms ± 20%
OPERATING CURRENT +5 Vdc	54 mA max.	54 mA max.	≤50 mA max.	54 mA max.
TERMINAL BLOCK REQUIRED	621-9950	621-9950	621-9950	621-9950



3912

8.6.1.9 16-Point 5 Vdc TTL Discrete Input Module

The model 621-4350R 5 Vdc TTL Discrete Input module is designed to interface with TTL output devices. The module provides 16 inputs that can be selected for high or low true logic.

Specifications:

INPUTS	16
POINTS PER COMMON	16
VOLTAGE	TTL (7 Vdc max.)
CURRENT	≤ 690 μ A at 0.4 Vdc, ≤ 280 μ A at 2.7 Vdc, ≤ 100 μ A at 7.0 Vdc
SWITCHING LEVELS	
High True:	
Logic 1	≥ 2 Vdc
Logic 0	≤ 0.8 Vdc
Low True:	
Logic 1	≤ 0.8 Vdc
Logic 0	≥ 2 Vdc
INPUT DELAY (High True)	
Off to On	0.26 ms
On to Off	1.4 ms
MODULE OPERATING CURRENT	107 mA (max.)
EXTERNAL POWER REQUIREMENTS	
Voltage	5 Vdc ± 5%
Current	248 mA
TERMINAL BLOCK REQUIRED	621-9950

Operation:

The model 621-4350R can send all 16 input status bits to the Logic Manager processor when the module is programmed with a PULL 1 instruction to any of the module's least significant addresses. This feature allows the 16-point module to be used in slots configured for 8-point operation. The PULL instruction also allows the control program to read the module status several times during one program scan.

When installed in a slot configured for 16 point operation, the module also responds to CONTACT and BRING IN instructions.

The 5 V TTL Discrete Input Module can be selected for either high true or low true logic operation. A 10 k ohm pull-up resistor is provided at each input, which ensures a logic 1 (high true) when a high impedance driving device is connected.

An amber LED, labeled **LO/TRU**, on the front of the module will be illuminated if the jumper selection is set for low true logic. One red LED for each input, at the front of the module, indicates when voltage is present (input high).

Wiring:

The 5 V TTL Discrete Input Module requires an external power source. The power must be wired to the swing arm terminal blocks. The positive (+) side of the supply should be connected to T1, T2, and T3. The negative (-) side of the supply should be connected to B1, B2, and B3. See Figure 8-10.

The module may also be used with two power sources, one supplying the top eight inputs, another supplying the bottom eight inputs. The top eight inputs use T1 and T2 as the positive voltage connections and B1 and B2 as negative voltage connections. The bottom eight inputs use T3 as a positive voltage connection and B3 as a negative voltage connection. Connection to two power sources requires the removal of the jumpers between T2/T3 and B2/B3 on the swing arm terminal blocks.

Unused inputs for a module configured for a high true logic must be wired to the negative side of the power source (B1, B2, or B3). This forces the unused inputs low. There are no special wiring requirements for a module configured for low true logic.

Wiring to this module must be separated from all AC wiring. If the module is used in a noisy environment, or has wire runs of 50 feet or longer, use a shielded cable. The suggested cable is Belden 8761, a two conductor, 22 gauge, twisted pair shielded cable with a drain wire. When a shielded cable is used, tie the shield to earth ground at the external field device. Connect together the module's 5 V return (B1, B2, B3), the field device return, and the field device power supply return.

Configuration:

Two jumper selections must be made before using the module. Figure 8-9 shows the location of the jumpers. Table 8-12 explains the jumper functions.

When the module is used in a 16-point serial I/O card file, and the PULL instruction will not be used, select J3. This selection will allow faster updates to other special function modules in the serial I/O system. If the PULL instruction will be used, select J4 instead.

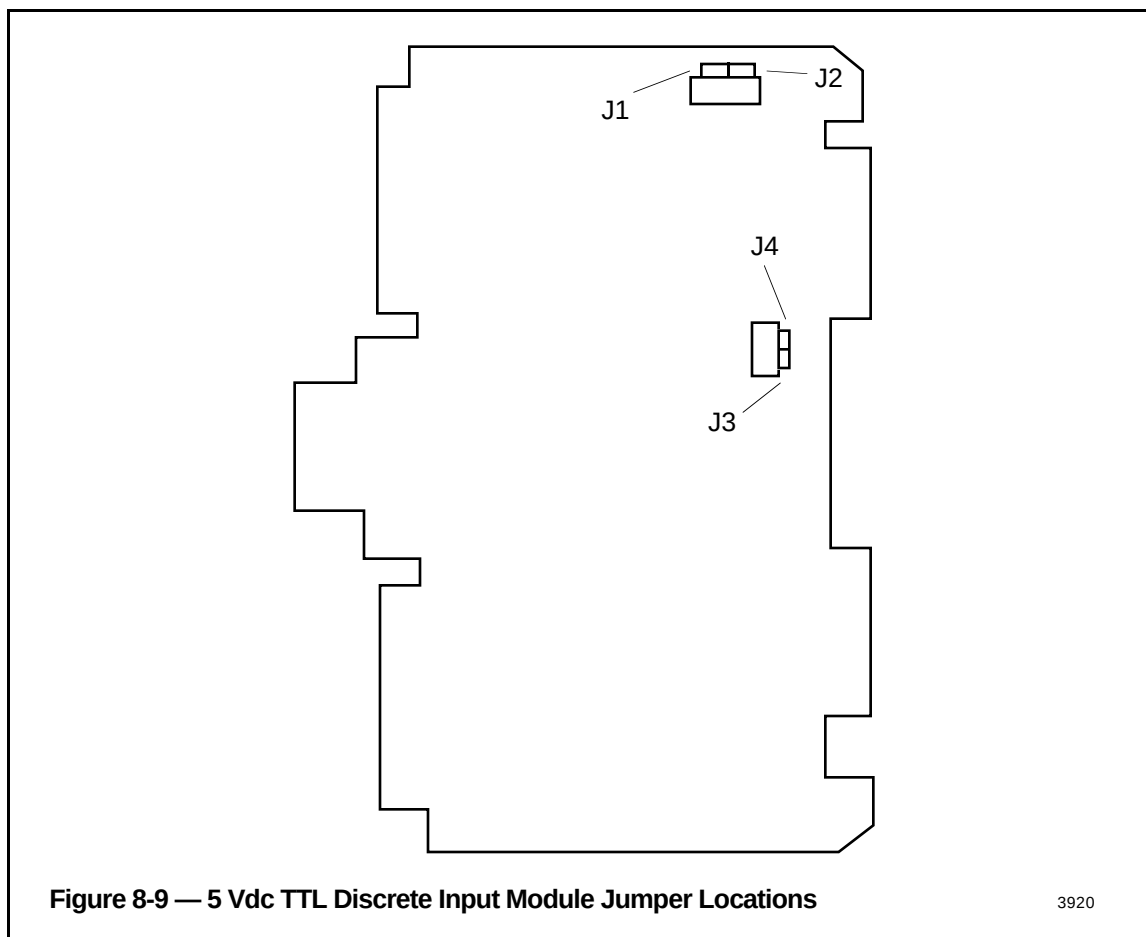
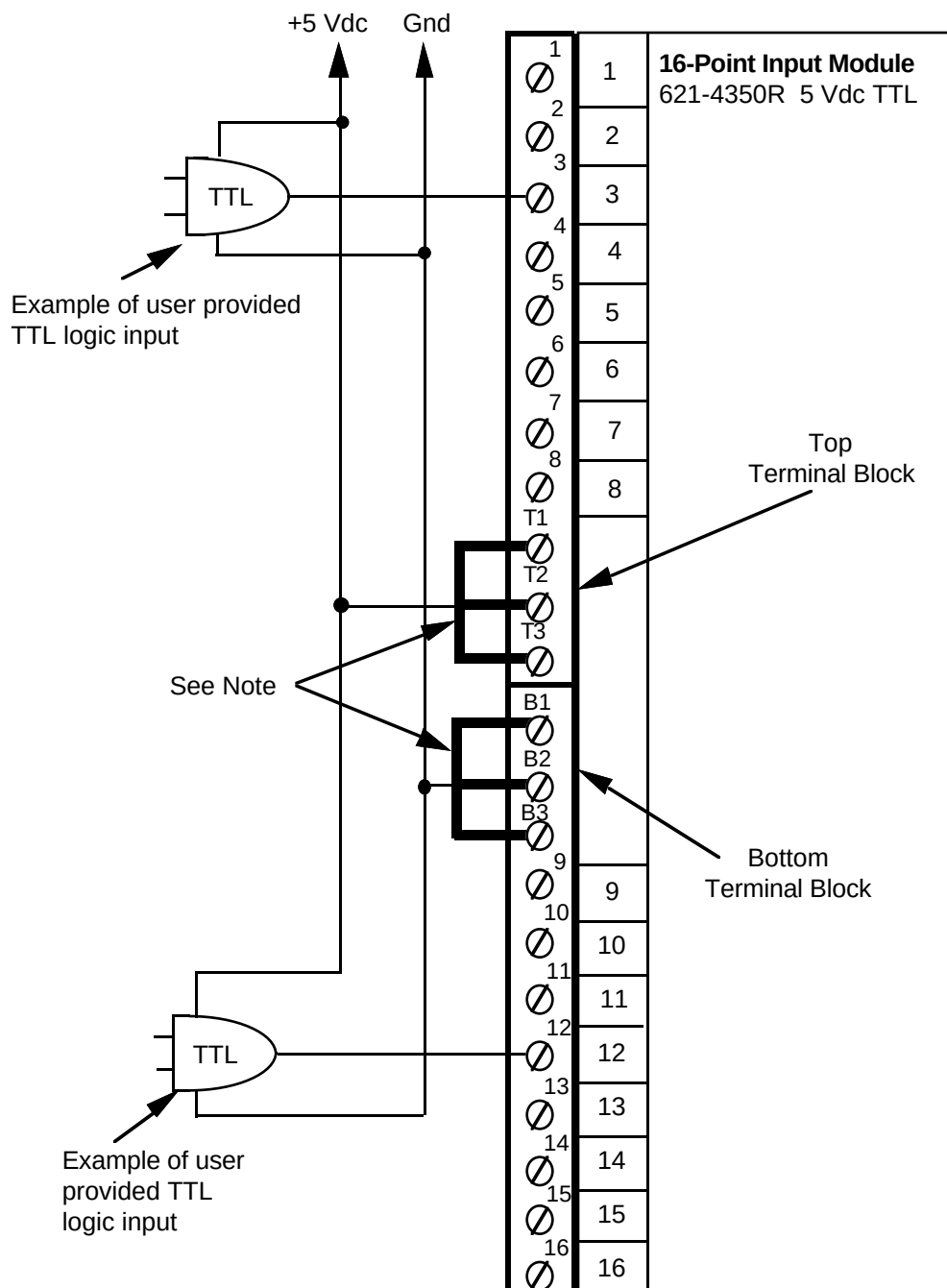


Table 8-12 — 5 Vdc TTL Discrete Input Module Jumper Selection

JUMPER	FUNCTION
J1*	High true logic
J2	Low true logic
J3	Disable serial PULL instruction
J4*	Enable serial PULL instruction

*Factory Setting

**Note:**

To provide separate power sources for the upper and lower sections of the terminal block, remove the terminal block jumpers between T2 and T3 and between B2 and B3. The top eight inputs use T1 and T2 as the positive connections and B1 and B2 as negative connections. The bottom eight inputs use T3 as a positive connection and B3 as a negative connection.

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Figure 8-10 — 16-Point TTL Discrete Input Module Wiring

8.6.1.10 16-Point Vdc Source Discrete Input Module

The model 621-4550R 24 Vdc Source Input Module provides 16-point source discrete input capability. See Figure 8-11.

Specifications:

INPUTS	16
POINTS PER COMMON	16
TYPE OF INPUT	Source
NOMINAL VOLTAGE	24 Vdc
VOLTAGE RANGE	20-28 Vdc
CURRENT RANGE	3-9 mA (6 mA at 24 Vdc typical)
SWITCHING LEVEL	
Logic 1	0.5 Vdc
Logic 0	18 Vdc
MAX. LEAKAGE CURRENT UNTIL INPUT TURNS ON	1.3 mA
INPUT DELAY	
Off to On	2.4 ms $\pm$ 20%
On to Off	17 ms $\pm$ 20%
MODULE OPERATING CURRENT +5 Vdc	54 mA maximum
TERMINAL BLOCK REQUIRED	621-9950

CAUTION

Do not remove the terminal block jumpers.

8.6.1.11 32-Point Discrete Input Modules

See Table 8-13 and Figure 8-12 for specifications and wiring information for the 32-point 115 Vac and 24 Vdc Sink Discrete Input modules.

A separate connection to L1 is needed when the model 621-1180R module is used in an ungrounded ac control circuit. Input operation is unaffected by this connection. However, for ungrounded ac systems using ground fault detection circuitry, this connection balances the system and prevents erroneous ground fault indications.

Table 8-13 — 32-Point Discrete Input Module Specifications

	MODEL NUMBER	
	621-1180R	621-3580R
POINT INPUTS	32	32
POINTS PER COMMON	16 (L2)	16 (-Vdc)
SPECIAL FEATURE		Sink Input
NOMINAL VOLTAGE	115 Vac	24 Vdc
VOLTAGE RANGE	80-140 Vac	18-28 Vdc
CURRENT RANGE	4-15 mA (8 mA at 115 Vac typical)	6-15 mA (10 mA at 24 Vdc typical)
INPUT SURGE CURRENT LIMIT	500 mA per input	
SWITCHING LEVEL		
Logic 1	75 Vac	18 Vdc
Logic 0	43 Vac	11 Vdc
ALLOWABLE LEAKAGE CURRENT	1.5 mA	2.5 mA
INPUT DELAY		
Off to On	2.4 ms to 3 ms, plus 0 to 0.5 cycles	2.4 ms $\pm$ 25%
On to Off	24 ms $\pm$ 50%	17 ms $\pm$ 25%
OPERATING CURRENT +5Vdc	$\leq$ 195 mA maximum	$\leq$ 195 mA maximum
TERMINAL BLOCK REQUIRED	621-9977	621-9976 or 621-9977

Terminal	621-1180R 115 Vac Sink Input	621-3580R 24 Vdc Sink Input
1	Input 1	Input 1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	Input 8	Input 8
9	* ac L1	Not Used
10	Not Used	Not Used
11	Input 9	Input 9
12	10	10
13	11	11
14	12	12
15	13	13
16	14	14
17	15	15
18	Input 16	Input 16
19	ac L2	** -Vdc
20	Input 17	Input 17
21	18	18
22	19	19
23	20	20
24	21	21
25	22	22
26	23	23
27	Input 24	Input 24
28	* ac L1	Not Used
29	Not Used	Not Used
30	Input 25	Input 25
31	26	26
32	27	27
33	28	28
34	29	29
35	30	30
36	31	31
37	Input. 32	Input. 32
38	ac L2	** -Vdc

* This connection is needed only when this module is used in an ungrounded ac control circuit. Input operation is unaffected by this connection. However, for ungrounded ac systems using ground fault detection circuitry, this connection balances the system and prevents erroneous ground fault indications.

** Inputs 1 through 16 and 17 through 32 can be powered by independent power sources. Terminal 19 provides the L2 reference for inputs 1 through 16, and terminal 38 provides the L2 reference for inputs 17 through 32.

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Figure 8-12 — 32-Point Discrete Input Module Wiring

8.6.2 Analog Input Modules

An analog input module accepts analog signals from field devices connected to the module's terminals and converts the signal data to 12-bit binary values (0 to 4095) via an Analog-to-Digital Converter contained in the module's circuitry. The digital values are stored in a holding register in the module and can be accessed by the Logic Manager control processor via a PULL instruction. The PULL instruction can read from one to eight 16-bit data words at a time.

Specifications, installation (programming and wiring) procedures, and operating instructions for each analog input module follow. Specific connectors or terminal block sets are required and must be ordered separately for each module, unless otherwise noted.

NOTE

The data from the analog input module must be configured for 12 bits. The Logic Manager module will convert the data to engineering units.

8.6.2.1 Universal Analog Input Module

The model 621-0020R Universal Analog Input Module (UAIM) allows you to obtain data from a maximum of 16 analog input sources. It contains an Analog-to-Digital Converter (ADC) that changes the analog signals into binary counts.

The model 623-0020 Configuration Utility is used with this module to customize the configuration for a specific application. This software is also used to calibrate the module. The 623-0020 is included with the 623-6010 or 623-6020 MS-DOS Loader Software.

Module Features:

- Compatibility with Parallel and 32-point Serial I/O Systems
- Access to data from the following types of analog inputs:

Linear (including square root):

0–1 Vdc	1–5 Vdc
0–5 Vdc	0–10 mVdc
10–50 mVdc	0–100 mVdc
4–20 mA (with shunt resistor)	

Direct Sensors:

Thermocouple types B, J, K, R, T, S, E, W5–W26, N**, and Ni-NiMoly
 Honeywell RH Radiamatic Pyrometer
 Carbon Probes *** (percent carbon, percent oxygen, dew point)

- Choice of output word sizes: 12, 13, 14, or 15 bits
- Installation in any one I/O slot configured for 32 points.
- Configurable fail safe output
- Selectable input scan time
- Configurable software filter
- Calibration software
- Manual bias
- Two LEDs that show module status
- Module fault reporting
- Configurable input ranging

** Nicrosil-Nisil or Omegalloy 14 AWG

*** One carbon probe per module. Probe can be manufactured by Marathon Monitors, Corning, AACC, Furnace Control Corp, MacDhui, Barber Coleman, or Bricesco.

NOTE

Refer to the 621-0020R Universal Analog Input Module and 623-0020 Configuration Utility User Manual (Form No. 621-8989) for more information concerning this module and its software.

Specifications:

NUMBER OF INPUTS	16 per module
RESOLUTION	12- to 15-bit configurable output
SPAN ACCURACY	0.05% full scale
TEMPERATURE COEFFICIENT	0.0042% per degree Celsius
INPUT IMPEDANCE	10 megohms minimum
CONTINUOUS OPERATING INPUT VOLTAGE	30 Vac/Vdc point to point or point to ground, Class 2 power limited
INPUT FILTER	1 ms RC nominal
REFERENCE JUNCTION ERROR	0.25° C (0.45° F)
CONVERSION RATE	20 conversions per second at 60 Hz, 16.7 conversions per second at 50 Hz
BURNOUT	1 k minimum, 4 k maximum loop resistance
MODULE OPERATING CURRENT	250 mA at 5 V, 70 mA at +15 V, 70 mA at -15 V
TERMINAL BLOCK	Supplied with module (621-9953)

8.6.2.2 8-Point 0-20 mA Isolated Analog Input Module

The model 621-0022-AR Isolated Analog Input Module provides eight isolated input channels and can be configured for various high level analog input signals. See Figure 8-13 for wiring information.

Module Features:

- Eight differential inputs per module
- Configurability for these ranges:

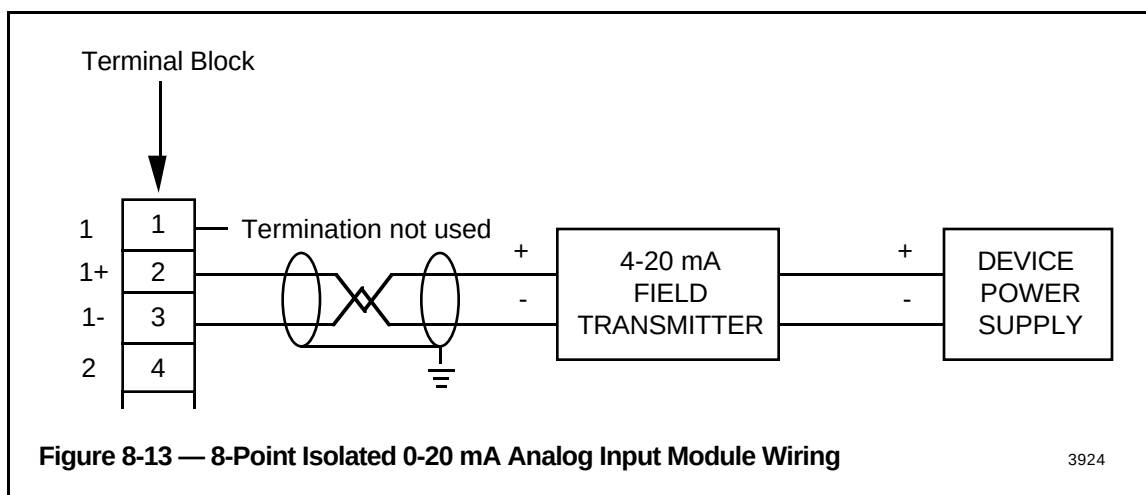
Unipolar

0 to 10 Vdc
 0 to 5 Vdc
 1 to 5 Vdc
 0 to 20 mA
 4 to 20 mA

Bipolar

-10 to +10 Vdc
 -5 to +5 Vdc

- Adjustable span and zero offset capability for each channel allows each input range to be calibrated exactly.
- Selectable input signal filtering
- Module fault reporting



Specifications:

NUMBER OF INPUTS	8 differential per module
RESOLUTION	1 part in 4096 (0–4095)
ACCURACY	±0.1% of span and ±1 LSB
NONLINEARITY	±1 LSB
TEMPERATURE COEFFICIENT	0.006% per °C
MAX. COMMON MODE VOLTAGE (Electrical Isolation)	
Input to Control Logic	1000 V peak ac or dc
Input to Input	2000 V peak ac or dc
MAX. NORMAL MODE INPUT	50 V RMS max. input voltage
NOISE REJECTION	
Common Mode (typical)	>88 dB (50–60 Hz) with up to 250 ohms unbalanced (typical)
Normal Mode (typical)	>38 dB (50–60 Hz) with up to 250 ohms source (typical) and no software filtering; >42 dB with 50% filtering; >48 dB with 75% filtering; >52 dB with 87.5% filtering
INPUT IMPEDANCE	
Current Mode	250 ohms
Voltage Mode	200k ohms minimum
MODULE UPDATE RATE	
DIP switch 5 OFF	25 ms
DIP switch 5 ON	33 ms
MODULE OPERATING CURRENT	
+5 Vdc	750 mA
±15 Vdc	75 mA maximum
ANALOG-TO-DIGITAL CONVERTER	12 bit
ENVIRONMENTAL CONDITIONS	
Operating Temperature	0 to 60° C (ambient)
Storage Temperature	-40 to 75° C
Humidity	5 to 95% noncondensing
TERMINAL BLOCK REQUIRED	Supplied with module (621-9954)

8.6.2.3 8-Point 0-10 V Isolated Analog Input Module

The model 621-0022VR Isolated Analog Input Module provides eight isolated input channels and can be configured for various high level analog input signals. See Figure 8-14 for wiring information.

Module Features:

- Eight differential inputs per module
- Configurability for these ranges:

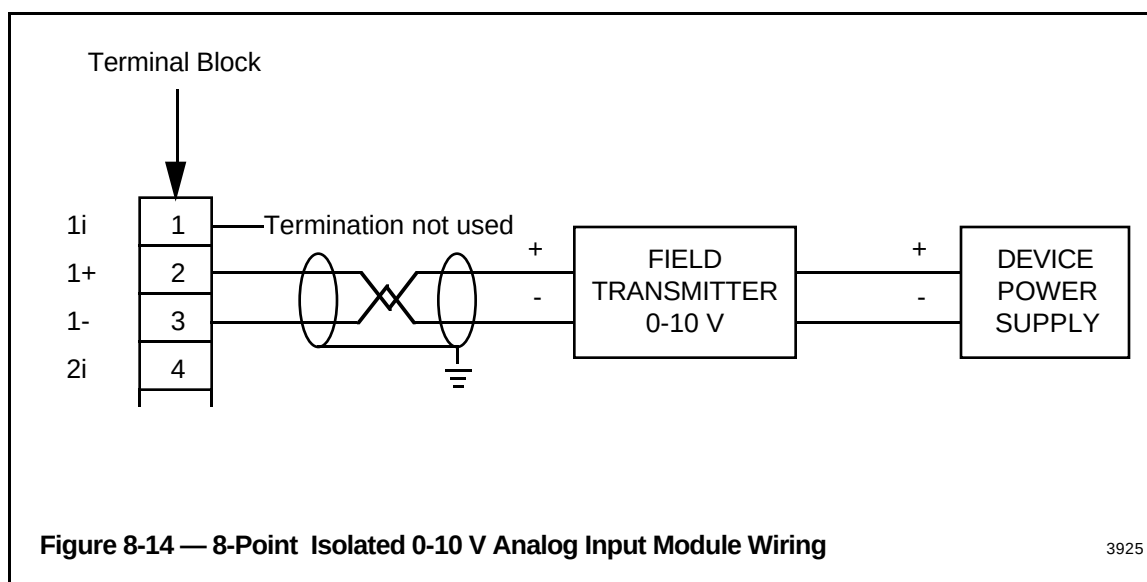
Unipolar

0 to 10 Vdc
0 to 5 Vdc
1 to 5 Vdc
0 to 20 mA
4 to 20 mA

Bipolar

-10 to +10 Vdc
-5 to +5 Vdc

- Adjustable span and zero offset capability for each channel allows each input range to be calibrated exactly.
- Selectable input signal filtering
- Module fault reporting



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Specifications:

NUMBER OF INPUTS	8 differential per module
RESOLUTION	1 part in 4096 (0–4095)
ACCURACY	±0.1% of span and ±1 LSB
NONLINEARITY	±1 LSB
TEMPERATURE COEFFICIENT	0.006% per °C
MAX. COMMON MODE VOLTAGE (Electrical Isolation)	
Input to Control Logic	1000 V peak ac or dc
Input to Input	2000 V peak ac or dc
MAX. NORMAL MODE INPUT	50 V RMS max. input voltage
NOISE REJECTION	
Common Mode (typical)	>88 dB (50–60 Hz) with up-to-250 ohms unbalanced (typical)
Normal Mode (typical)	>38 dB (50–60 Hz) with up-to-250 ohms source (typical) and no software filtering; >42 dB with 50% filtering; >48 dB with 75% filtering; >52 dB with 87.5% filtering
INPUT IMPEDANCE	
Current Mode	250 ohms
Voltage Mode	200 k ohms minimum
MODULE UPDATE RATE	
DIP switch 5 OFF	25 ms
DIP switch 5 ON	33 ms
MODULE OPERATING CURRENT	
+5 Vdc	750 mA
±15 Vdc	75 mA maximum
ANALOG-TO-DIGITAL CONVERTER	12 bit
ENVIRONMENTAL CONDITIONS	
Operating Temperature	0 to 60° C (ambient)
Storage Temperature	-40 to 75° C
Humidity	5 to 95% noncondensing
TERMINAL BLOCK REQUIRED	Supplied with module (621-9954)

8.7 I/O Output Module Descriptions

The Logic Manager I/O System uses a variety of output modules that can be classified as either discrete or analog. As shown in Tables 8-14 and 8-15, the I/O output modules are available in different voltage levels to match the voltage requirements of the field devices to which the modules are connected. Each module provides a standard number of outputs; 4, 6, 8, 16, or 32.

This section provides individual output module specifications, wiring diagrams, and in certain cases, special instructions for installation and operation. The modules are divided into Discrete and Analog sections and are listed in numerical order as shown in Tables 8-14 and 8-15.

Table 8-14 — Discrete Output Modules

MODULE DESCRIPTION	NUMBER OF OUTPUTS	CABLE CLASS	MODEL NUMBER
Reed Relay Discrete Output	6	DATA 1, 2, 3	621-0007R
115 Vac Discrete Output	8	DATA 3	621-2100R
115 Vac Isolated Discrete Output	6	DATA 3	621-2101R
115 Vac Self-Protected Discrete Output	8	DATA 3	621-2102R
115 Vac Discrete Output	16	DATA 3	621-2150R
115 Vac Discrete Output	32	DATA 3	621-2175R
230 Vac Discrete Output	8	DATA 3	621-2200R
230 Vac Isolated Discrete Output	6	DATA 3	621-2201R
24 Vac Discrete Output	8	DATA 2	621-2500R
24 Vac Discrete Output	16	DATA 2	621-2550R
5 Vdc Source Discrete Output	8	DATA 1	621-6300R
5 Vdc TTL Source Discrete Output	16	DATA 1	621-6350R
12 Vdc Source Discrete Output	16	DATA 1	621-6450R
12-24 Vdc Source Discrete Output	8	DATA 1, 2 *	621-6500R
12-24 Vdc Source Self-Protected Discrete Output	8	DATA 1, 2 *	621-6503R
24 Vdc Source Discrete Output	16	DATA 2	621-6550R
24 Vdc Source Low Power Discrete Output	16	DATA 2	621-6551R
24 Vdc Source Discrete Output	32	DATA 2	621-6575R
48 Vdc Source Discrete Output	8	DATA 2	621-6600R
48 Vdc Source Discrete Output	16	DATA 2	621-6650R
120 Vdc Source 0.5 A Discrete Output	8	DATA 3	621-6700R
120 Vdc Source 2 A Discrete Output	8	DATA 3	621-6701R

* The IEEE 518 wiring class is application dependent and can vary with the voltage and current present in the wiring.

Table 8-15 — Analog Output Modules

MODULE DESCRIPTION	NUMBER OF OUTPUTS	CABLE CLASS	MODEL NUMBER
Analog 4-20 mA Output	4	DATA 1	621-0010-A
Analog 0-10 V Output	4	DATA 1	621-0010-V

8.7.1 Discrete Output Modules

The typical discrete output module offers the ability to control the ON and OFF state of its end devices. Among the common output devices are motor starters, solenoid valves, and indicator lamps.

All dc output modules are source type modules. This means they provide the +V potential to the field output device when the circuit is turned on. Current flows from the external power supply to the source output module, to the field output device.

All AC output modules incorporate zero crossing. The output is turned ON at the zero crossing of the ac wave form.

LED indication for each output circuit on the 8- and 16-point modules is on the load side of the output (except isolated outputs and reed relay). The 32-point output modules have the LED indication for each circuit on the 5 Vdc logic side of the module. All discrete outputs are optically isolated to 2500 Vdc.

These output modules provide self-diagnostic assurance, since each module's coverplate displays an LED to signify a fault, thus reducing an otherwise lengthy troubleshooting procedure.

Each discrete output module consists of three components: a printed circuit board with a permanent front cover, address labels, and a set of 8-, 16-, or 32-point terminal blocks.

The printed circuit board is inserted into the desired slot in the Logic Manager I/O card file. An offset backpanel connector prevents installing a module upside down. The removable 8- and 16-point double swing terminal blocks allow easy module insertion, removal, and wiring. They are installed by firmly seating the edge connector of each terminal block onto the front edge of the circuit board and securing the thumbscrew fastener. The terminal blocks are keyed for the particular circuit board. The 32-point terminal blocks are inserted into block headers located on the front of the module. Field wiring connects to the 32-point terminal block from the front of the module and can be accessed without removing the terminal block.

Discrete 8-point modules use terminals T1 and B1 to supply voltage connections for circuits 1 through 4, and terminals T2 and B2 to supply voltage connections for circuits 5 through 8. Separate voltage sources may power circuits 1 through 4 and 5 through 8 by removing the factory installed jumpers across terminals T1 and T2 and terminals B1 and B2. The 8-point modules, therefore, have the potential for two common returns per module.

All discrete 16-point modules provide one common for all 16 outputs, requiring that all 16 circuits be powered from the same voltage source. The factory installed jumpers across terminals T1, T2, and T3 and terminals B1, B2, and B3 distribute the load current to all 16 outputs. For proper 16-point operation, do not remove these jumpers.

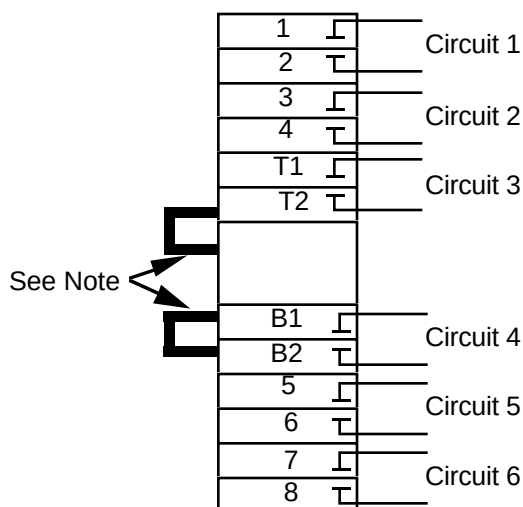
8.7.1.1 6-Point Reed Relay Output Module

The model 621-0007R Reed Relay Output Module provides six-isolated dry contact closures suitable for switching low level signals, as well as higher currents and voltages not exceeding the contact specifications.

A removable wire jumper is installed in series with each reed contact circuit. This jumper should be replaced with a properly sized resistor for protection of the reed contacts when a capacitive load is present (see Figure 8-15 and Table 8-16). Capacitive loading may be encountered when long runs of shielded cable are used. The load itself may also have some amount of capacitance, such as a model 621 input module.

Specifications:

OUTPUTS	6 normally open contacts
VOLTAGE RANGE	250 Vac/140 Vdc maximum
MAX. CONTACT RATING	0.5 A (not to exceed 30 VA)
CONTACT RESISTANCE	≤200 milliohms
MODULE OPERATING CURRENT	48 mA plus 40 mA for each energized output, 284 mA maximum
LED INDICATION	1 per circuit logic side
ISOLATION	1500 Vdc peak
ADDRESSING	Lower 6 addresses
TERMINAL BLOCK REQUIRED	621-9949



Note: Remove the terminal block jumpers.

Figure 8-15 — Reed Relay Output Module Terminal Block

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Table 8-16 — Reed Relay Jumper Replacement Resistors

VOLTAGE	RESISTOR VALUE (ohms) *	CURRENT LOAD MAX. AT 10% VOLTAGE DROP (milliamps)	CURRENT LOAD MAX. AT 5% VOLTAGE DROP (milliamps)
5	1**	500	250
10	2.7	370	185
20	11	180	90
30	24	125	62
40	43	90	45
50	68	72	36
60	100	60	30
70	130	50	25
80	180	45	23
90	220	40	20
100	270	36	18
110	330	33	16
120	390	30	15
130	470	27	14
140	560	25	12
150	620	24	12
160	680	23	11
170	820	21	11
180	910	20	10
190	1 k	19	10
200	1.1 k	18	9
210	1.2 k	18	9
220	1.3 k	17	8
230	1.5 k	15	8
240	1.6 k	15	8
250	1.8 k	14	7
260	2 k	13	7

* 1 watt unless otherwise noted

** 1/2 watt

8.7.1.2 6-Point Vac Isolated Discrete Output Modules

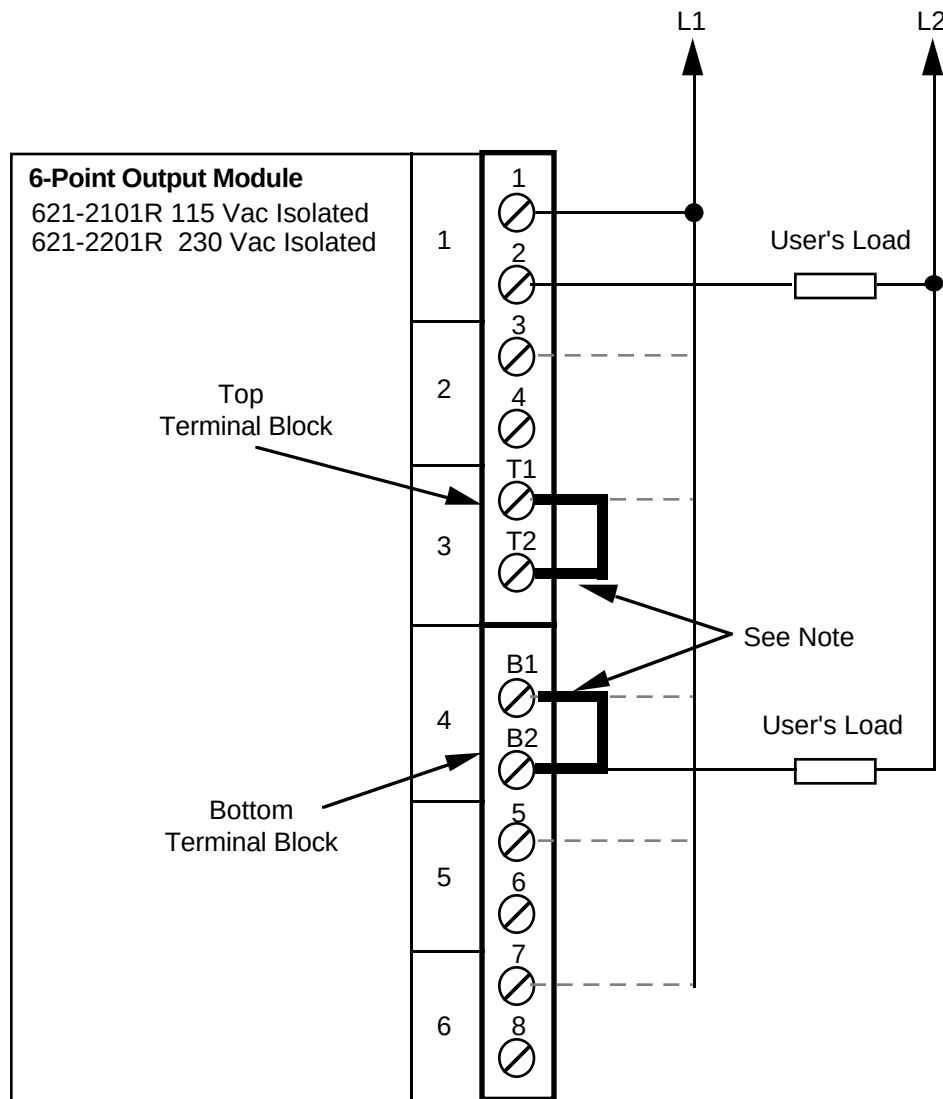
The Vac Isolated Discrete Output modules provide 6-point discrete output capability. See Table 8-17 and Figure 8-16 for specifications and wiring information.

WARNING

Remove the terminal block jumpers.

Table 8-17 — 6-Point Vac Isolated Discrete Output Module Specifications

	MODEL NUMBER	
	621-2101R	621-2201R
OUTPUTS	6 Output On at zero crossing	6 Output On at zero crossing
SPECIAL FEATURE	Isolated output	Isolated output
NOMINAL VOLTAGE	115 Vac	230 Vac
VOLTAGE RANGE	90-140 Vac	195-250 Vac
MAXIMUM CURRENT RANGE	2 A per circuit	2 A per circuit
SURGE CURRENT	15 A, 2 cycle (nonrepetitive)	10 A, 1 cycle (nonrepetitive)
OFF-STATE LEAKAGE CURRENT	≤5 mA (typical)	≤5 mA (typical)
ON -STATE VOLTAGE DROP	≤2 Vac at 2 A	≤2 Vac at 2 A
FUSING	1 per circuit, 3 A slo-blo	1 per circuit, 3 A fast blow
OPERATING CURRENT +5 Vdc	80 mA plus 24 mA for each energized output, 224 mA maximum	80 mA plus 24 mA for each energized output, 224 mA maximum
ADDRESSING	Lower 6 addresses	Lower 6 addresses
TERMINAL BLOCK REQUIRED	621-9949	621-9949



Note:
Remove these terminal block jumpers.

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Figure 8-16 — 6-Point Vac Isolated Discrete Output Module Wiring

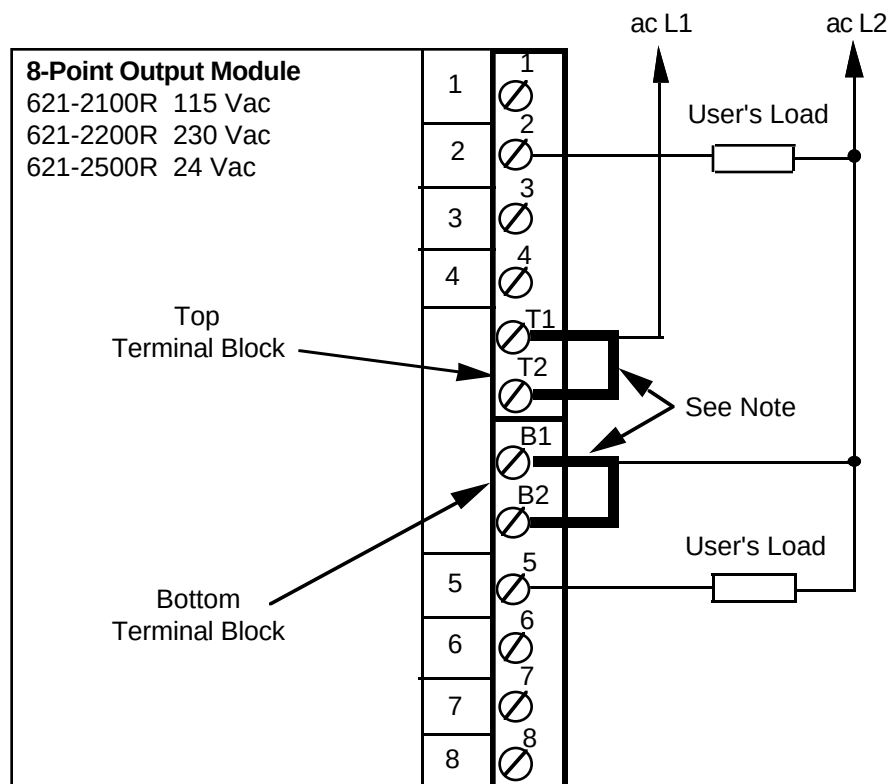
8.7.1.3 8-Point Vac Discrete Output Modules

Tables 8-18 and Figure 8-17 provide specifications and wiring information for the listed 8-point Vac Discrete Output modules.

To use two different power supplies, the terminal block jumpers are removed. Without the jumpers, terminals T1 and B1 are the voltage references for terminals 1 through 4, and terminals T2 and B2 are the voltage references for terminals 5 through 8.

Table 8-18 — 8-Point Vac Discrete Output Module Specifications

	MODEL NUMBER		
	621-2100R	621-2200R	621-2500R
OUTPUTS	8 Output On at zero crossing	8 Output On at zero crossing	8 Output On at zero crossing
POINTS PER COMMON	4	4	4
NOMINAL VOLTAGE	115 Vac	230 Vac	24 Vac
VOLTAGE RANGE	90-140 Vac	195-250 Vac	20-28 Vac
MAXIMUM CURRENT RANGE	2 A/circuit, 6 A/common, 8 A/module	2 A/circuit, 6 A/common, 8 A/module	2 A/circuit, 6 A/common, 8 A/module
SURGE CURRENT	15 A, 2 cycle (non-repetitive)	8 A, 1 cycle (non-repetitive)	8 A, 1 cycle (non-repetitive)
OFF-STATE LEAKAGE CURRENT	≤5 mA (typical)	≤5 mA (typical)	≤5 mA (typical)
ON-STATE VOLTAGE DROP	≤2 Vac at 2 A	≤2 Vac at 2 A	≤2 Vac at 2 A
FUSING	1/circuit, 3 A slo-blo	1/circuit, 3 A fast blow	1/circuit, 3 A fast blow
OPERATING CURRENT +5 Vdc	80 mA plus 24 mA for each energized output, 272 mA maximum	132 mA plus 24 mA for each energized output, 324 mA maximum	80 mA plus 24 mA for each energized output, 272 mA maximum
TERMINAL BLOCK REQUIRED	621-9949	621-9949	621-9949



Note:

To use two different power supplies, these terminal block jumpers are removed. Without the jumpers, terminals T1 and B1 are the voltage references for terminals 1 through 4, and terminals T2 and B2 are the voltage references for terminals 5 through 8.

Figure 8-17 — 8-Point Vac Discrete Output Module Wiring

8.7.1.4 8-Point 115 Vac Self-Protected Discrete Output Module

The specifications, operation, and wiring information for the 8-point 115 Vac Self-Protected Discrete Output Module follows. See Figure 8-18 for the wiring diagram.

Specifications:

NUMBER OF OUTPUTS	8 per module; outputs turned on at zero crossing
POINTS PER COMMON	8
VOLTAGE RANGE	90-140 Vac
SURGE CURRENT	15 A, 2 cycle nonrepetitive
FUSING	3 A slo-blo
OFF STATE LEAKAGE CURRENT	≤5 mA typical
ON STATE VOLTAGE DROP	≤2 Vac at 2 A
OPERATING CURRENT +5 VDC	130 mA plus 24 mA for each energized output, 322 mA maximum
TERMINAL BLOCK REQUIRED	621-9949

Operation:

The model 621-2102R compares the status of the output triac with the Control Processor Module (CPU) signal. When the two signals do not compare, the module indicates a malfunction.

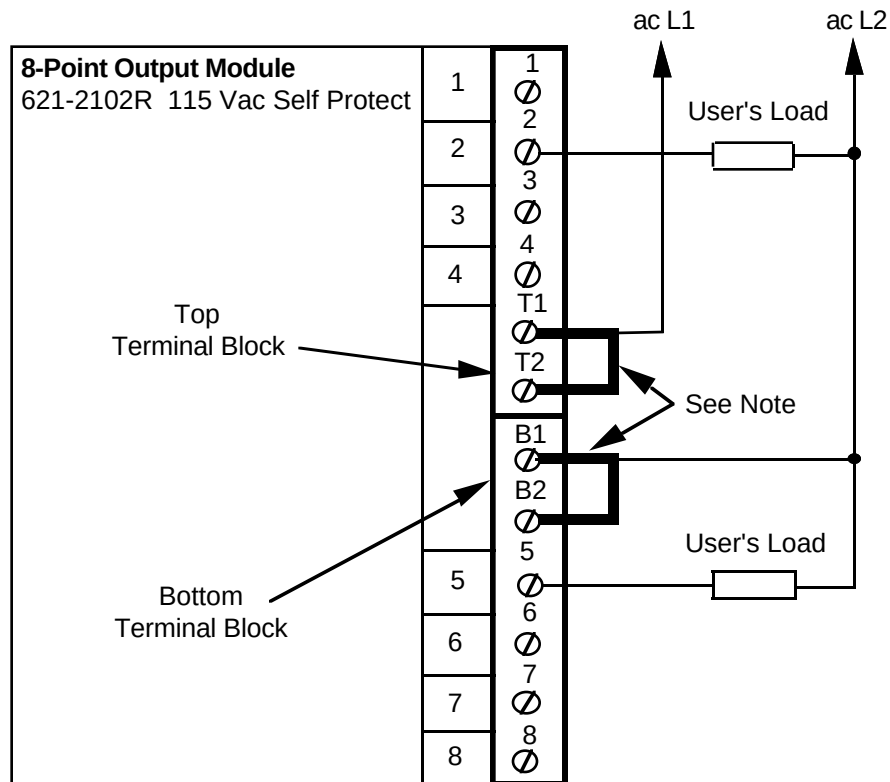
The model 621-2102R recognizes and responds to two types of malfunctions:

1. Each circuit has a 3 amp fuse. If the output triac is on (conducting current) and the CPM signal is off, the module blows the circuit's individual output fuse. The circuit is immediately disconnected from the load, and the **FUSE** LED illuminates. A card fault message is also reported to the CPM, and the module's **FAULT** LED illuminates.
2. If the output triac is off (not conducting current) while the CPM signal is on, a card message is sent to the CPM, and the module's **FAULT** LED illuminates. The card fault message is entered into the CPU's I/O Fault Table for the most significant address of the module.

An amber fault indicator, a fuse blown indicator, and individual red circuit indicators (field side) for output status are located at the front of the module.

Wiring:**WARNING**

Do not wire a switching device in parallel with a model 621-2102R output circuit. It will cause the fault detection circuitry to operate and the circuit fuse to blow.

**Note**

Terminals T1 and T2 are internally tied together. Terminals B1 and B2 are also internally tied together. Therefore, all eight circuits are referenced to T1 and B2. The external jumpers between T1 and T2 and between B1 and B2 should not be removed from the terminal blocks to distribute the load current.

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Figure 8-18— 8-Point 115 Vac Self-Protected Discrete Output Module Wiring

WARNING

Do not remove the terminal block jumpers.

8.7.1.5 8-Point Vdc Source Discrete Output Modules

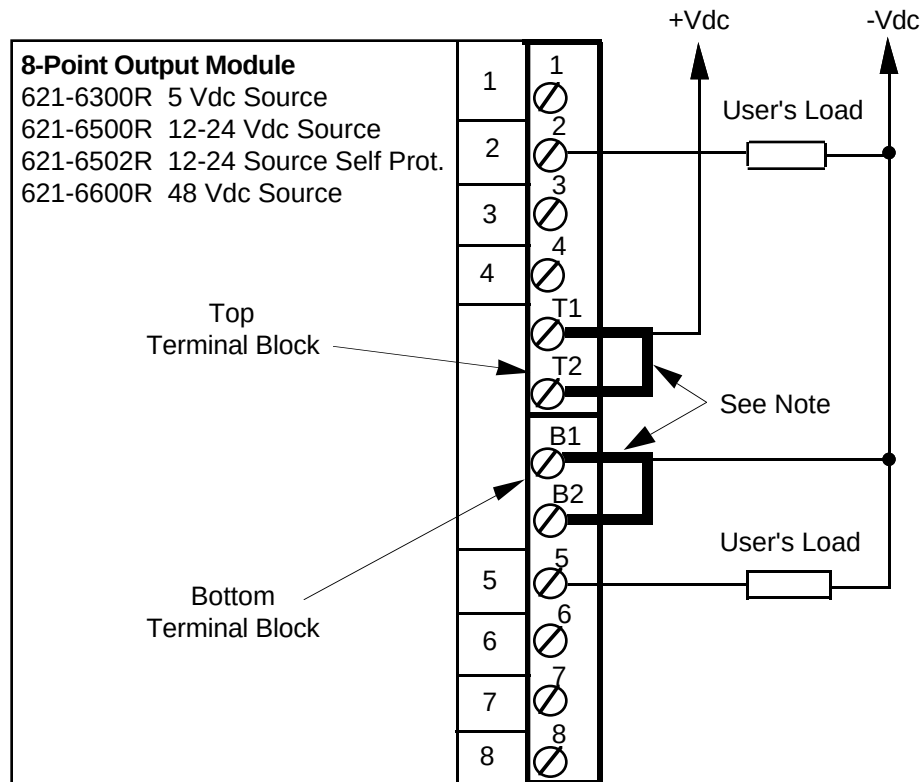
Tables 8-19 and 8-20 and Figure 8-19 provide specifications and wiring information for the 8-point Vdc Source Discrete Output modules.

Table 8-19 — 8-Point Vdc Source Discrete Output Module Specifications

	MODEL NUMBER			
	621-6300R	621-6500R	621-6503R*	621-6600R
OUTPUTS	8	8	8	8
POINTS PER COMMON	4	4	4	4
SPECIAL FEATURE	Source output	Source output	Source output self protected	Source output
NOMINAL VOLTAGE	5 Vdc	12-24 Vdc	12-24 Vdc	48 Vdc
VOLTAGE RANGE	4.8-5.2 Vdc	11-32 Vdc	11-28 Vdc	38-66 Vdc
Logic 0	0 < 0.4 Vdc			
Logic 1	1 > 2.4 Vdc			
MAXIMUM CURRENT RANGE	100 mA /output	2 A/circuit, 6 A/common, 8 A/module	2 A/circuit, 6 A/common, 8 A/module	2 A/circuit, 6 A/group of 4, 8 A/module
SURGE CURRENT	250 mA for 100 ms	8 A for 10 ms (nonrepetitive)	5 A for 10 ms (nonrepetitive)	8 A for 10 ms (nonrepetitive)
FIELD POWER REQUIREMENT	3 mA at 5 Vdc for each energized output	15 mA at 12 Vdc, 32 mA at 24 Vdc for each energized output	25 mA at 12 Vdc, ≤45 mA at 24 Vdc for each energized output	10 mA at 38 Vdc, 13 mA at 48 Vdc, 20 mA at 66 Vdc for each energized output
OFF-STATE LEAKAGE CURRENT	≤0.05 mA (typical)	≤5 mA (typical)	≤5 mA (typical)	≤5 mA (typical)
ON-STATE VOLTAGE DROP	≤1.2 Vdc at 100 mA	≤2 Vdc at 2 A	≤2 Vdc at 2 A	≤2 Vdc at 2 A
FUSING	0.8 A slow blow per group of 4	8 A fast blow per group of 4	3 A fast blow per circuit, 7 A fast blow/common	8 A fast blow per group of 4
OPERATING CURRENT +5 Vdc	70 mA plus 5 mA for each energized output, 110 mA maximum	70 mA plus 5 mA for each energized output, 110 mA maximum	70 mA plus 5 mA for each energized output, 110 mA maximum	70 mA plus 5 mA for each energized output, 110 mA maximum
TERMINAL BLOCK REQUIRED	621-9949	621-9949	621-9949	621-9949

* For the model 621-6503R module see the following operation and wiring guidelines.

NOTE: The 621-6503R replaces the 621-6502.

**Note:**

To use two different power supplies, these terminal block jumpers are removed. Without the jumpers, terminals T1 and B1 are the voltage references for terminals 1 through 4, and terminals T2 and B2 are the voltage references for terminals 5 through 8.

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Figure 8-19 — 8-Point Vdc Source Discrete Output Modules Wiring

Operation Guidelines:

The model 621-6503R (the 621-6503R replaces the 621-6502) module compares the status of the output transistor with the processor signal. When a noncompare between the two signals exists, the module indicates a malfunction.

The 621-6503R recognizes and responds to two types of malfunctions.

1. If the output transistor is actually ON (conducting current) and the processor signal is OFF, the module will blow the circuit's individual output fuse. The circuit is immediately disconnected from the load, and the FUSE LED is illuminated. A module fault message is also reported back to the processor, and the module's FAULT LED is illuminated.
2. If the output transistor is actually OFF (not conducting current) while the processor status is ON, a module fault message is sent to the processor, and the module's FAULT LED is illuminated. The card fault message is entered into the processor's I/O Fault Table for the most significant address (MSA) on the module.

Wiring Guidelines:

Each circuit has a 3 A fuse. An amber fault indicator, a fuse-blown indicator, and individual red circuit LEDs (field side) for output status are located on the front of the module.

WARNING

Do not wire any switching device in parallel with the model 621-6503R. This will cause the fault detection circuitry to operate and blow the fuse.

Any capacitive load that remains on in excess of 6 V after 8 ms of turning the output OFF may cause the module output fuse to be blown. Under these conditions, a diode may be installed, sized according to load in series with the output. The diode isolates the load from the module error detection circuitry to prevent blown fuses.

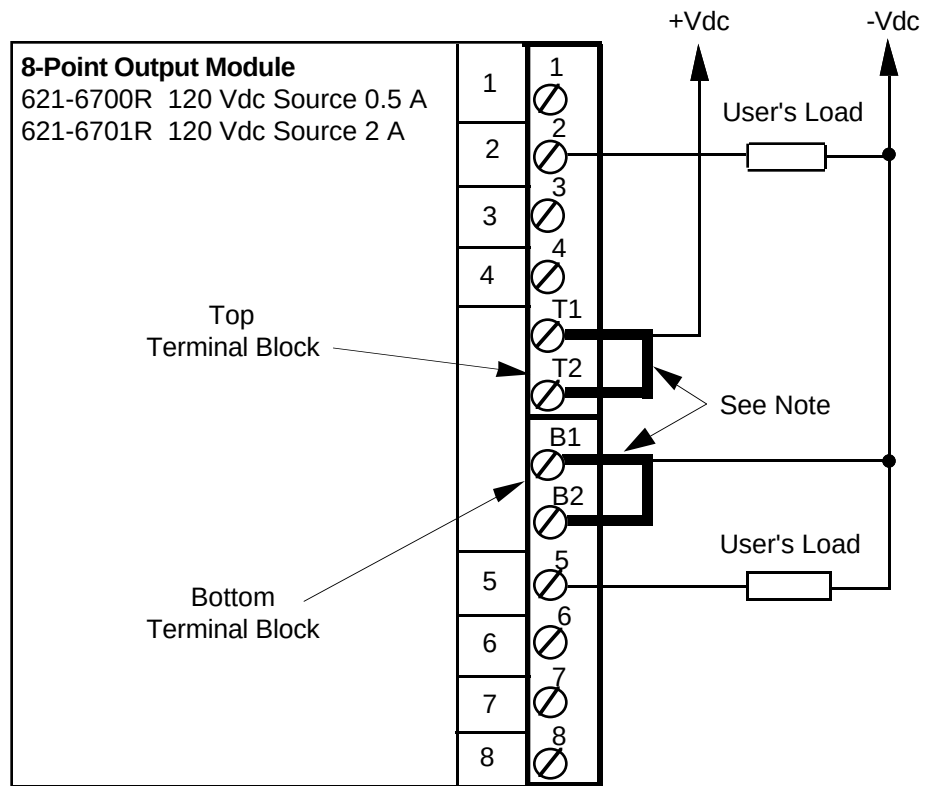
Circuits 1-4 may be powered separately from circuits 5-8 by removing the factory-installed jumpers across T1/T2 and B1/B2. Terminals T1 and B1 provide power to circuits 1-4 and terminals T2 and B2 provide power to circuits 5-8. All eight circuits can be powered from T1 and B1 by leaving the external jumpers on. See Figure 8-19.

8.7.1.5A 8-Point Vdc Source Discrete Output Modules

Table 8-20 and Figure 8-19A provides specification and wiring information for the 621-6700R and 621-6701R.

Table 8-20 — 8-Point 120 Vdc Source Discrete Output Modules Specifications

	MODEL NUMBER	
	621-6700R	621-6701R
OUTPUTS	8	8
POINTS PER BLOCK	4	4
SPECIAL FEATURE	Source output	2 amp source output
NOMINAL VOLTAGE	120 Vdc	120 Vdc
VOLTAGE RANGE	90-130 Vdc	90-130 Vdc (max. voltage of 130 Vdc includes ripple)
MAXIMUM CURRENT RANGE	0.5 mA /output	2 A/circuit, 6 A/group of 4, 12 A/module
SURGE CURRENT	4 A for 10 ms (nonrepetitive)	8 A for 10 ms (nonrepetitive)
FIELD POWER REQUIREMENT	7 mA at 120 Vdc for each energized output	30 mA/group of 4, 2.5 mA for each "On" circuit at 120 Vdc
OFF-STATE LEAKAGE CURRENT	≤5 mA (typical)	≤5 mA (typical)
ON-STATE VOLTAGE DROP	≤2 Vdc at .5 A	≤2 Vdc at 2 A
FUSING	1 per group of 4, 3 A fast blow	1 per group of 4, 8 A fast blow
OPERATING CURRENT +5 Vdc	70 mA plus 5 mA for each energized output, 110 mA (maximum)	70 mA plus 5 mA for each energized output, 110 mA (maximum)
TERMINAL BLOCK REQUIRED	621-9949	621-9949

**Note:**

To use two different power supplies, these terminal block jumpers are removed. Without the jumpers, terminals T1 and B1 are the voltage references for terminals 1 through 4, and terminals T2 and B2 are the voltage references for terminals 5 through 8.

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Figure 8-19A — 8 Point 120 Vdc Source Discrete Output Wiring

8.7.1.6 16-Point Vac Discrete Output Modules

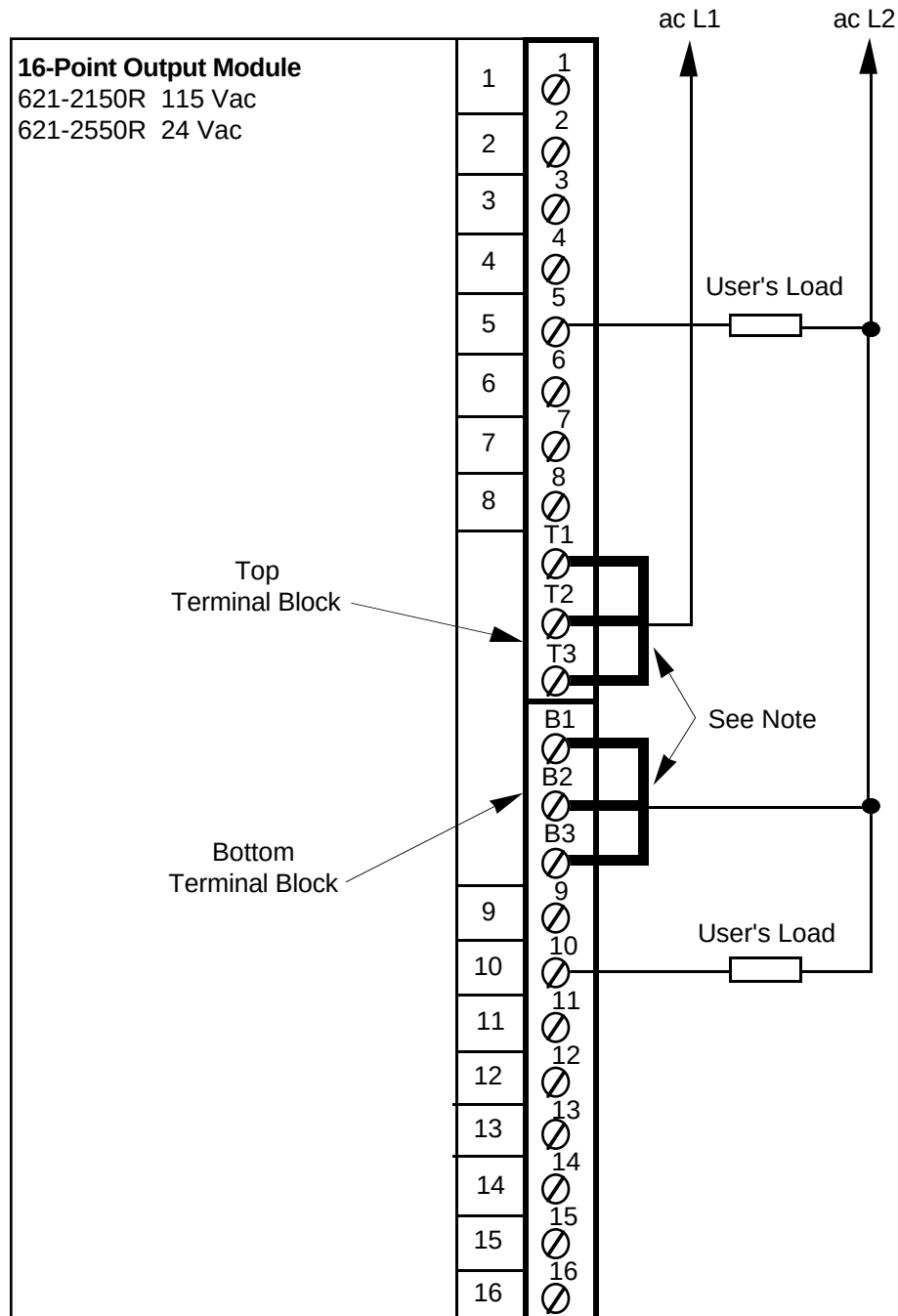
Table 8-21 and Figure 8-20 provide specification and wiring information for the listed 16-point Vac Discrete Output modules.

CAUTION

Do not remove the terminal block jumpers.

Table 8-21 — 16-Point Vac Discrete Output Module Specifications

	MODEL NUMBER	
	621-2150R	621-2550R
OUTPUTS	16 Output On at zero crossing	16 Output On at zero crossing
POINTS PER COMMON	16	16
NOMINAL VOLTAGE	115 Vac	24 Vac
VOLTAGE RANGE	90-140 Vac	20-28 Vac
MAXIMUM CURRENT RANGE	2 A per circuit, 12 A per module	2 A per circuit, 5 A per group of 4, 12 A per module
SURGE CURRENT	15 A, 2 cycle (nonrepetitive)	8 A, 1 cycle (nonrepetitive)
OFF-STATE LEAKAGE CURRENT	≤5 mA (typical)	≤5 mA (typical)
ON-STATE VOLTAGE DROP	≤2 Vac at 2 A	≤2 Vac at 2 A
FUSING	1 per circuit, 3 A slo-blo	1 per circuit, 3 A fast blow
OPERATING CURRENT +5 Vdc	145 mA plus 24 mA for each energized output, 529 mA maximum	145 mA plus 24 mA for each energized output, 529 mA maximum
TERMINAL BLOCK REQUIRED	621-9950	621-9950



Note:

Do not remove these terminal block jumpers.

Figure 8-20 —16-Point Vac Discrete Output Module Wiring

8.7.1.7 16-Point Vdc Source Discrete Output Modules

Table 8-22 and Figure 8-21 provide specification and wiring information for the 16-point Vdc Source Discrete Output modules.

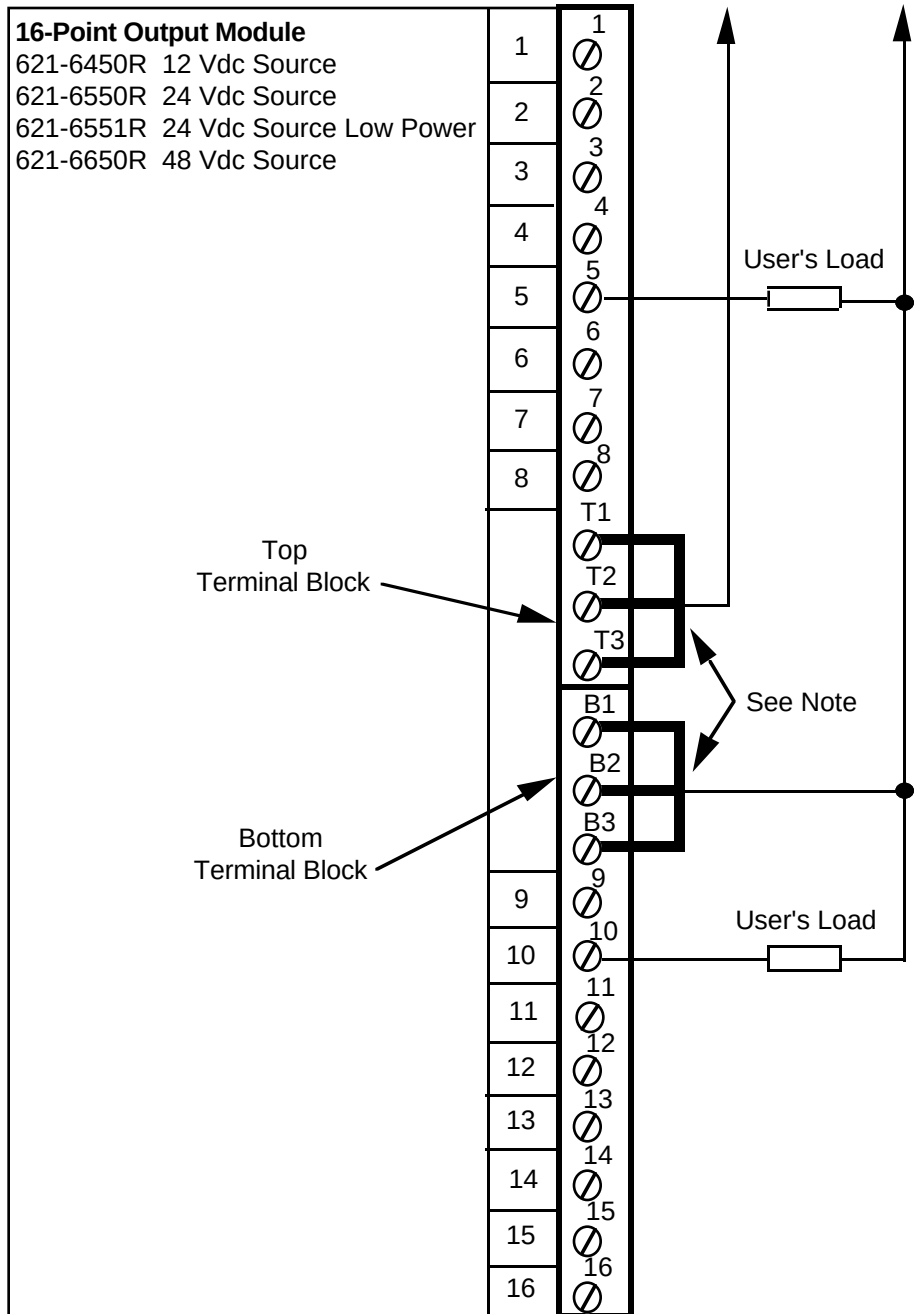
Do not remove the terminal block jumpers.

Table 8-22 — 16-Point Vdc Source Discrete Output Module Specifications

	MODEL NUMBER			
	621-6450R	621-6550R	621-6551R	621-6650R
OUTPUTS	16	16	16	16
POINTS PER COMMON	16	16	16	16
SPECIAL FEATURE	Source output	Source output	Source output low power	Source output
NOMINAL VOLTAGE	12 Vdc	24 Vdc	24 Vdc	48 Vdc
VOLTAGE RANGE	11-18 Vdc	18-28 Vdc	11-28 Vdc	38-66 Vdc **
MAXIMUM CURRENT RANGE	2 A per circuit, 12 A per module	2 A per circuit, 5 A per group of 4, 12 A/module	0.5 A per circuit, 4 A per module *	2 A per circuit, 5 A per group of 4, 12 A/module
SURGE CURRENT	8 A for 10 ms (nonrepetitive)	8 A for 10 ms (nonrepetitive)	5 A for 10 ms (nonrepetitive)	8 A for 10 ms (nonrepetitive)
FIELD POWER REQUIREMENT	25 mA for each energized output	25 mA for each energized output	25 mA for each energized output	25 mA for each energized output
OFF-STATE LEAKAGE CURRENT	≤5 mA (typical)	≤5 mA (typical)	≤5 mA (typical)	≤5 mA (typical)
ON-STATE VOLTAGE DROP	≤2 Vdc at 2 A	≤2 Vdc at 2 A	≤2 Vdc at .5 A	≤2 Vdc at 2 A
FUSING	1 per group of 4, 7 A fast blow	1 per group of 4, 7 A fast blow	1 per group of 4, 3 A fast blow	1 per group of 4, 7 A fast blow
OPERATING CURRENT +5 Vdc	124 mA plus 5 mA for each energized output, 204 mA maximum	124 mA plus 5 mA for each energized output, 204 mA maximum	124 mA plus 5 mA for each energized output, 204 mA maximum	123 mA plus 5 mA for each energized output, 203 mA maximum
TERMINAL BLOCK REQUIRED	621-9950	621-9950	621-9950	621-9950

* The model 621-6551R has 16 outputs capable of supplying 0.25 A concurrently, or 8 outputs capable of supplying 0.5 A concurrently.

** Operating voltages exceeding 48 Vdc for the model 621-6650R require power derating. Observe the following guidelines located on pages 8-65 & 8-66.



Note: Do not remove these terminal block jumpers.

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Figure 8-21 — 16-Point Vdc Source Discrete Output Module Wiring

Power Derating:

Voltages greater than 48 Vdc may require derating according to the ambient temperature (60° C maximum) of the card file. Figure 8-22 plots the derating required for a constant field load of 12 amps on the module. Given an ambient temperature and the dc voltage, the graph shows the maximum number of ON outputs that can be used at one time.

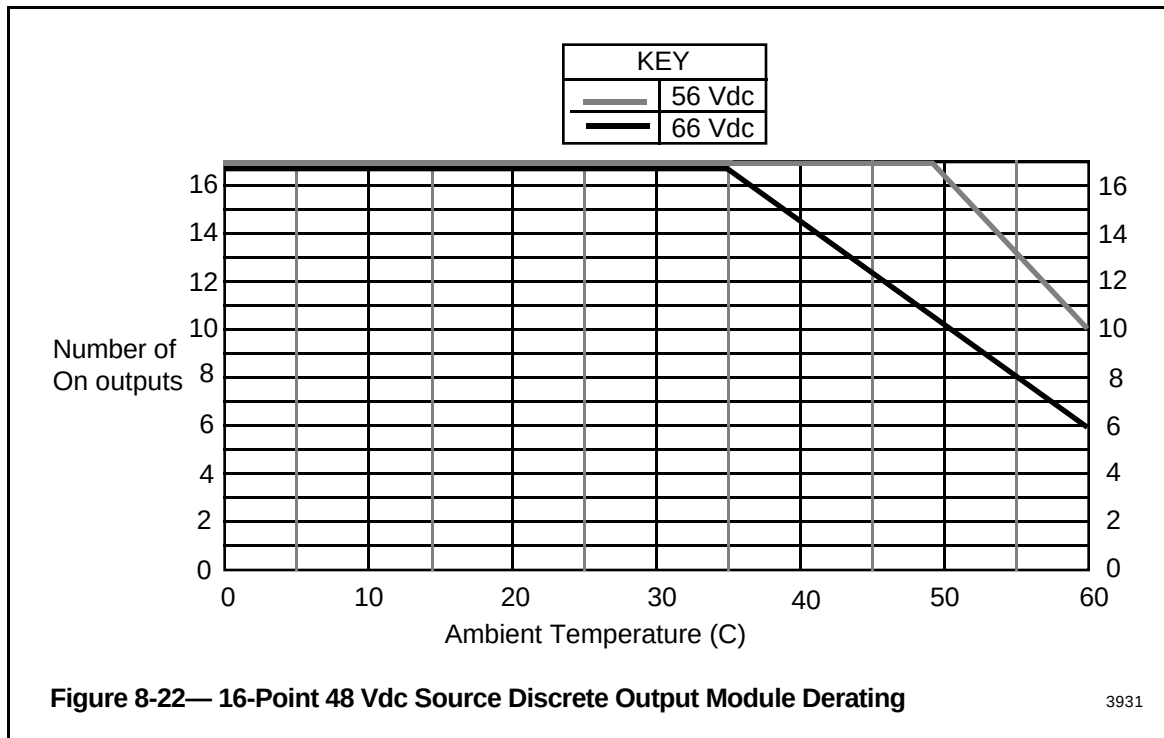
WARNING

Exceeding the recommended number of ON outputs may cause damage to the module.

Each 1 amp decrease in load permits an ambient temperature increase of 1° C for a given voltage. A 5 amp decrease, for example, increases the ambient temperature limit by 5° C. This maintains temperature stability on the module.

CAUTION

Voltages greater than 48 Vdc should not operate beyond 60° C.



8.7.1.8 16-Point 5 Vdc TTL Source Discrete Output Module

The model 621-6350R, 5 Vdc TTL Source Discrete Output module interfaces with devices using a TTL inputs. The module provides 16 TTL outputs. The outputs can be controlled as separate outputs or as one 16-bit register.

The operation of the module can be configured for high true or low true logic. On the front of the module, an amber LED, labeled **LO/TRU**, illuminates if the jumper selection is configured for low true logic.

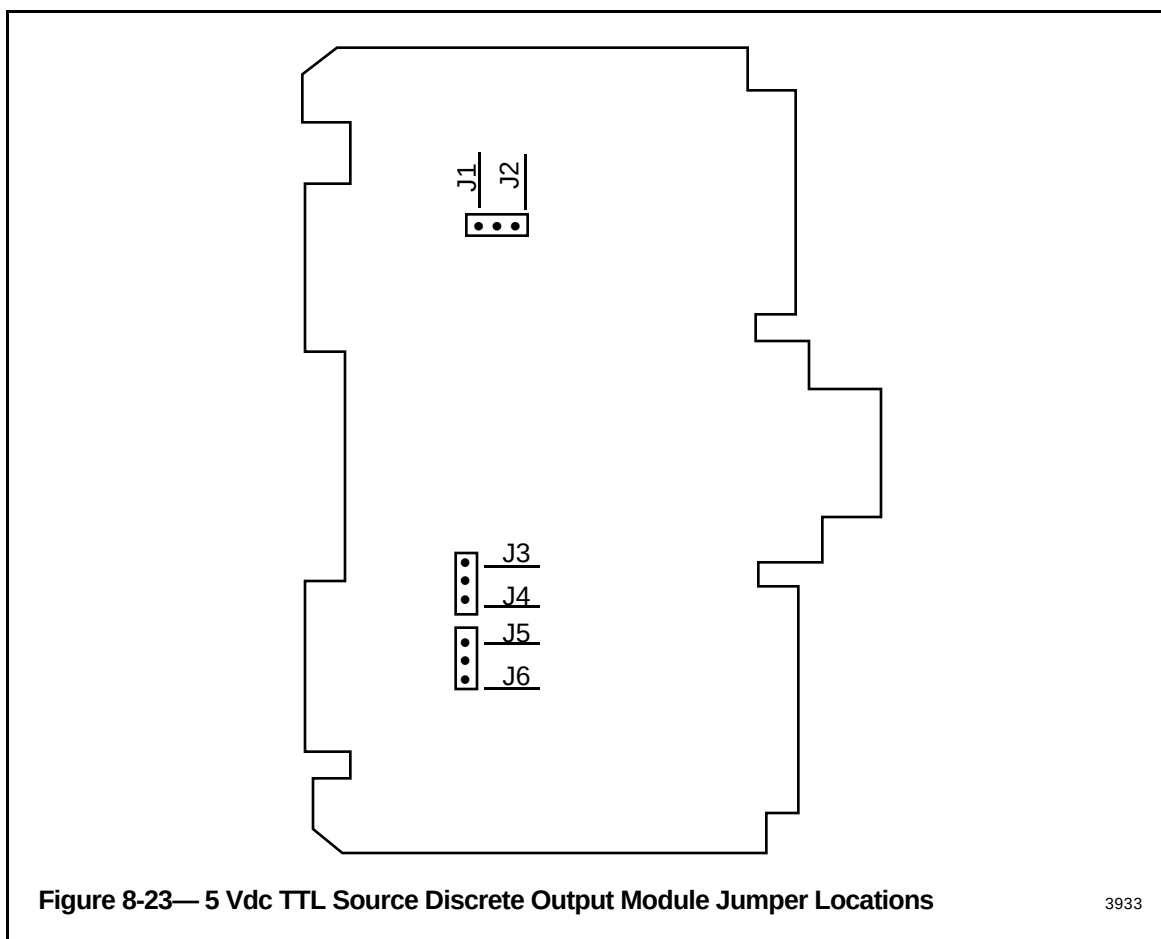
Operation:

The model 621-6350R TTL Output Module allows a PUSH 1 instruction to control all 16 outputs of the module. The module can be installed in an I/O card file slot configured for 8-point operation.

When the module is installed in a slot configured for 16-point operation, the module still responds to coil and SEND OUT instructions.

Configuration:

Three jumper selections must be made before using the module. Figure 8-23 shows the locations of the jumpers. Table 8-23 explains the jumper functions.



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Table 8-23 — 5 Vdc TTL Source Discrete Output Jumper Functions

JUMPER	FUNCTION
J1 *	High True Logic
J2	Low True Logic
J3	Enable SEND OUT to module
J4*	Enable PUSH instruction for module
J5	Disable PUSH instruction for module
J6 *	Enable PUSH instruction for serial I/O

* Factory setting

Wiring:

The 5 Vdc TTL Output Module uses an external power source. The power must be wired to the swing arm terminal blocks with the positive (+) side of the supply connected to terminals T1, T2, and T3 and the negative (-) side of the supply connected to terminals B1, B2, and B3.

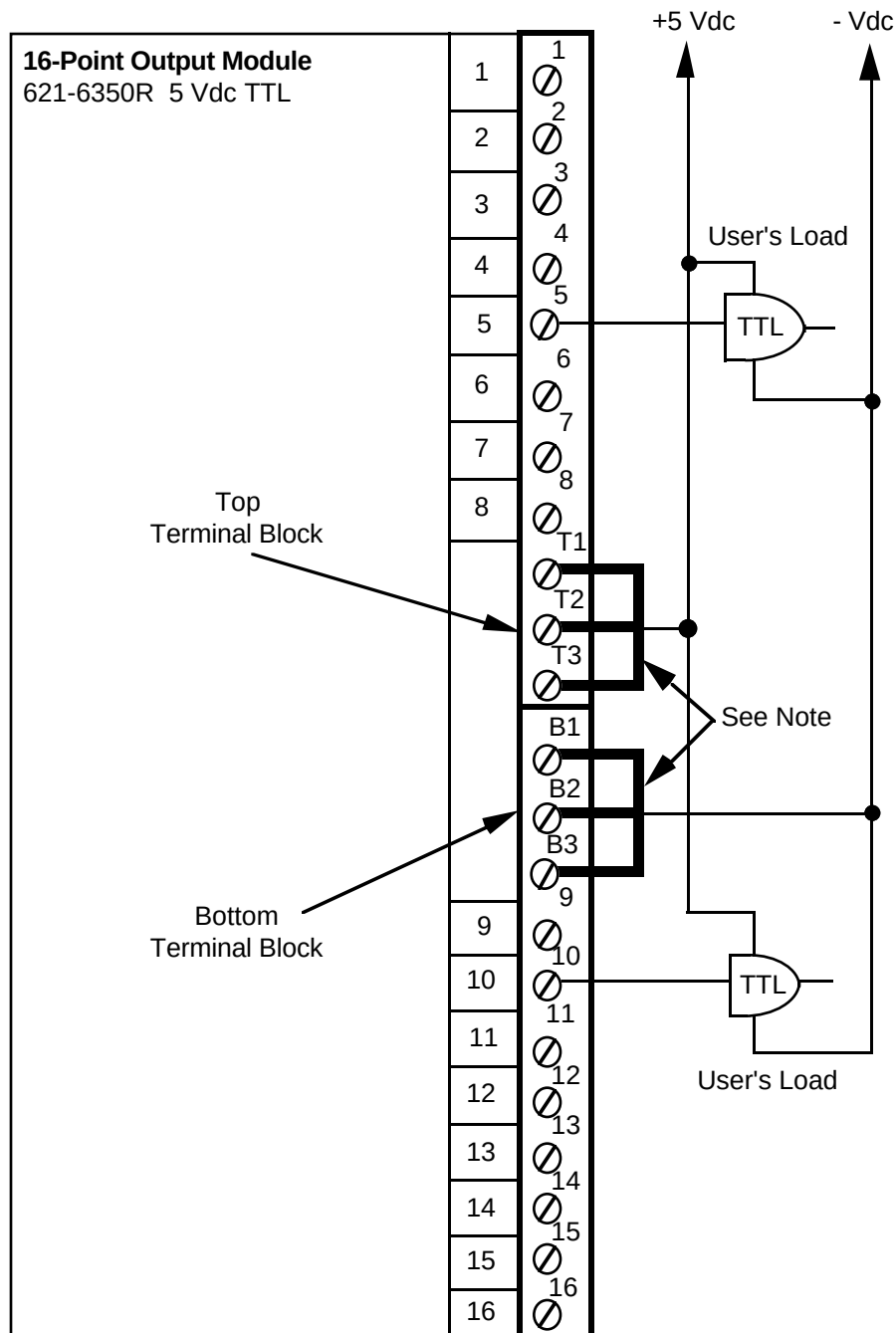
The module may also be used with two power sources, one supplying the top eight outputs, another supplying the bottom eight outputs. The top eight outputs use terminals T1 and T2 as positive connections and terminals B1 and B2 as negative connections. The bottom eight outputs use terminal T3 as a positive connection and terminal B3 as a negative connection. Wiring with two power sources requires the removal of the jumpers between terminals T2 and T3 and terminals B2 and B3 on the swing arm terminal blocks. See Figure 8-24.

CAUTION

The wiring to this module should be isolated or separated from all AC wiring. If the module is used in a noisy environment, or with wire runs of 50 feet or longer, a shielded cable should be used. The suggested cable is Belden 8761, a 2 conductor, 22 gauge, twisted pair shielded cable with a drain wire. When a shielded cable is used, the shield must be tied to earth ground at the external TTL field device. The output module 5 V return (B1, B2, B3), field device return, and field device power supply return must all be tied together.

Specifications:

NUMBER OF OUTPUTS	16
POINTS PER COMMON	8
VOLTAGE	TTL
CURRENT RATING	-20 mA at ≥ 2.5 V, -48 mA at ≥ 2.0 V, 48 mA at ≤ 0.5 V
SURGE CURRENT	-140 mA maximum, rated for only one output at a time
FIELD POWER REQUIREMENT	5 Vdc $\pm 5\%$, 423 mA
OFF-STATE LEAKAGE CURRENT	
Output High	100 μ A
Output Low	-100 μ A
FUSING	None
MODULE OPERATING CURRENT	342 mA plus 13 mA for each energized output, 550 mA maximum
TERMINAL BLOCK REQUIRED	621-9951

**Note:**

To provide separate power sources for the upper and lower sections of the terminal block, remove the terminal block jumpers between terminals T2 and T3 and between terminals B2 and B3. The top eight inputs use terminals T1 and T2 as the positive connections and terminals B1 and B2 as negative connections. The bottom eight inputs use terminal T3 as a positive connection and terminal B3 as a negative connection.

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Figure 8-24 — 16-Point 5 Vdc Discrete Output Module Wiring

8.7.1.9 32-Point Discrete Output Modules

See Table 8-24 and Figure 8-25 for specifications and wiring information for the 32-Point Discrete Output modules.

Table 8-24 — 32-Point Discrete Output Module Specifications

	MODEL NUMBER	
	621-2175R	621-6575R
OUTPUTS	32 Output On at zero crossing	32
POINTS PER COMMON	8 (L1) 16 (L2)	8 (+Vdc) 16 (-Vdc)
SPECIAL FEATURE		Source output
NOMINAL VOLTAGE	115 Vac	24 Vdc
VOLTAGE RANGE	80-140 Vac	18-28 Vdc + 20% ripple max.
MAXIMUM CURRENT RANGE	500 mA per circuit, 3.5 A per group of 8 max., 7 A per group of 16 max.	500 mA per circuit, 3.5 A per group of 8 max., 7 A per group of 16 max.
SURGE CURRENT	10 A, 60 Hz, 1 cycle (nonrepetitive)	2 A per output under 10 ms (nonrepetitive)
FIELD POWER REQUIREMENT		470 mA (typical) *
OFF-STATE LEAKAGE CURRENT	≤5 mA (typical)	≤6 mA (typical)
ON-STATE VOLTAGE DROP	≤2 Vac at 0.5 A	≤1.7 Vac at 0.5 A
FUSING	1 per group of 8 (L1), 5 A fast blow	1 per group of 16, 10 A fast blow
OPERATING CURRENT +5 Vdc	148 mA plus 11 mA for each energized output, 500 mA maximum	50 mA plus 8 mA for each energized output, 306 mA maximum
TERMINAL BLOCK REQUIRED	621-9977	621-9976 or 621-9977

* Each output circuit for the model 621-6575R is protected by overcurrent circuitry instead of fusing. An overcurrent condition is signaled by a card fault. Field power must be cycled to clear the overcurrent fault, and re-enable the output. On power-up of the field power supply, ensure the rise time for the field power is faster than 0.5 ms/volt. Rise times slower than this may cause the overcurrent fault to be latched on the module.

Terminal	621-2175R Vac Output	621-6575R Vdc Source Output
1	Output 1	Output 1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	Output 8	Output 8
9	ac L1	+Vdc
10	ac L1	+Vdc
11	Output 9	Output 9
12	10	10
13	11	11
14	12	12
15	13	13
16	14	14
17	15	15
18	Output 16	Output 16
19	ac L2	-Vdc
20	Output 17	Output 17
21	18	18
22	19	19
23	20	20
24	21	21
25	22	22
26	23	23
27	Output 24	Output 24
28	ac L1	+Vdc
29	ac L1	+Vdc
30	Output 25	Output 25
31	26	26
32	27	27
33	28	28
34	29	29
35	30	30
36	31	31
37	Output 32	Output 32
38	ac L2	-Vdc

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Figure 8-25 — 32-Point Discrete Output Module Wiring

8.7.2 Analog Output Modules

An analog output module accepts numerical data (digital values) from the Logic Manager processor and converts the data into equivalent analog voltage and current, enabling control of field devices (requiring analog signals) connected to the module's output channels. The digital values are translated via 12 bit digital-to-analog converters contained in the module's circuitry.

The converters are addressed by the Logic Manager processor via a PUSH instruction. Once addressed, a converter can translate the digital value to a corresponding analog voltage or current level within microseconds.

Specifications, installation (configuration and wiring) procedures, and operating instructions for each analog output module follow. Note that specific connectors or terminal block sets are required and must be ordered separately for each module unless otherwise noted.

8.7.2.1 4-Point 4-20 mA Analog Output Module

The model 621-0010-AR 4-20 mA Analog Output Module converts digital values presented by a Logic Manager processor to an equivalent analog voltage and current, enabling control of motor speed, valve positions, or other devices requiring an analog signal as a control input. There are four output channels per module and channel is individually selectable. The model 621-0010-AR module is physically and electrically identical to the model 621-0010-VR Analog Output module. The difference is the factory calibration of the module. The model 621-0010-AR module is calibrated for 4 to 20 mA operation, while the model 621-0010-VR module is calibrated for 0-10 Vdc operation.

Table 8-25 illustrates the ranges and the output voltages associated with sample signals. The first two ranges in the table are bipolar outputs. The current output will not function with these bipolar outputs. It functions only with the third range, 0 to +10 Vdc (4 to 20 mA), which is unipolar.

Table 8-25 — Model 621-0010 Discrete to Analog Conversion

RANGE	OUTPUT VOLTAGE (per analog signal)		
	0	2047	4095
-5 to +5 Vdc	-5	0	+5
-10 to +10 Vdc	-10	0	+10
0 to +10 Vdc	0	+5	+10
4 to 20 mA	+4	+12	+20

Specifications:

NUMBER OF OUTPUTS	4 nonisolated
RESOLUTION	12 bits
MODULE POWER REQUIREMENTS	+5 Vdc, 285 mA; -15 Vdc, 120 mA *; +15 Vdc, 85 mA plus 20 mA* for each current output used
OUTPUT RANGES	
Voltage	±10 Vdc, ±5 Vdc, 0–10 Vdc
Current	4–20 mA
LOAD	
Voltage	8k ohm minimum
Current	0–600 ohm
ACCURACY	
Voltage	0.15% of FSR at 25° C after calibration
Current	0.15% of FSR at 25° C after calibration under load
TEMPERATURE COEFFICIENT	
Voltage	±0.006% of FSR per °C plus 3/4 LSB
Current	±0.03% of FSR per °C plus 3/4 LSB
SETTLING TIME	
Voltage	210 μs to within 1 LSB of final value (10 Vdc step change)
SLEW RATE	0.05 Vdc per μs minimum
TERMINAL BLOCK REQUIRED	621-9950

* Voltage outputs are not loaded.

NOTE

For some module fault conditions, the 621-0010-AR/VR module may not be clamped to zero, which is a situation that must be accounted for in the design of the control system. Note that in some applications an interposing relay may be required in series with the output.

Configuration:**CAUTION**

When configuring the I/O system, be certain that the maximum power load for the card file is not exceeded.

Use DIP Switch 5, an 8-position switch, to set each channel for bipolar or unipolar operation. See the Table 8-26 for Switch 5 operation.

CAUTION

Each channel must be configured for bipolar or unipolar operation. Improper setting will cause the module to malfunction.

Table 8-26 — Model 621-0010 Switch 5 Settings

CHANNEL FUNCTION	POSITION STATES
Channel 1 Bipolar	1 Open/Off, 2 Closed/On
Channel 1 Unipolar	1 Closed/On, 2 Open/Off
Channel 2 Bipolar	3 Open/Off, 4 Closed/On
Channel 2 Unipolar	3 Closed/On, 4 Open/Off
Channel 3 Bipolar	5 Open/Off, 6 Closed/On
Channel 3 Unipolar	5 Closed/On, 6 Open/Off
Channel 4 Bipolar	7 Open/Off, 8 Closed/On
Channel 4 Unipolar	7 Closed/On, 8 Open/Off

Each channel can be individually selected for one of three operating ranges. Use DIP switches 1 through 4, one for each channel, to select these ranges. Each is a 2-position switch. Figure 8-26 illustrates the locations of these range selection DIP switches, as well as the offset and span adjustment potentiometers.

Assure that the voltage range selected for each channel, by switches 1 through 4, agrees with the channel bipolar/unipolar settings of switch 5. Table 8-27 defines the operation of the 2-position switches 1 through 4.

Table 8-27 — Model 621-0010 Switches 1-4 Setting Settings

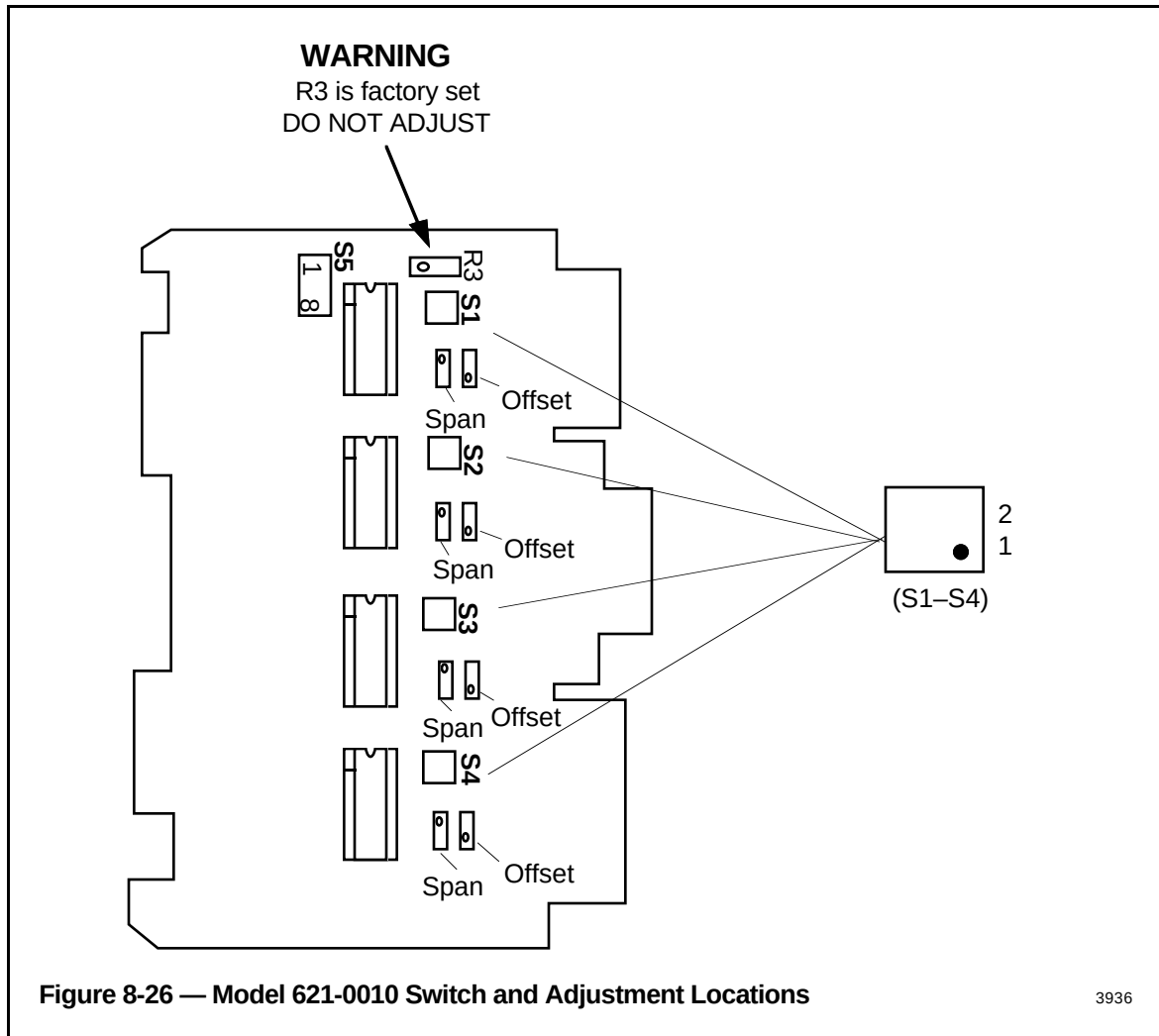
CHANNEL FUNCTION	POSITION STATES
0-10 Vdc (4-20 mA)	1 Closed/On, 2 Open/Off
± 10 Vdc	1 Open/Off, 2 Closed/On
± 5 Vdc	1 Closed/On, 2 Closed/On

NOTE

The 4 to 20 mA output is available only when the 0 to 10 Vdc output is selected. Refer to the Figure 8-27 wiring diagram.

Calibration:

Each model 621-0010-AR has been factory calibrated for 4 mA to 20 mA operation. Operation at other ranges may require that the output be recalibrated. See Appendix A of the *Logic Manager Service* manual in the *Implementation/Logic Manager - 2* binder.



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Wiring:

The I/O card file provides +5 Vdc logic power and a ± 15 Vdc reference voltage for analog output module operation.

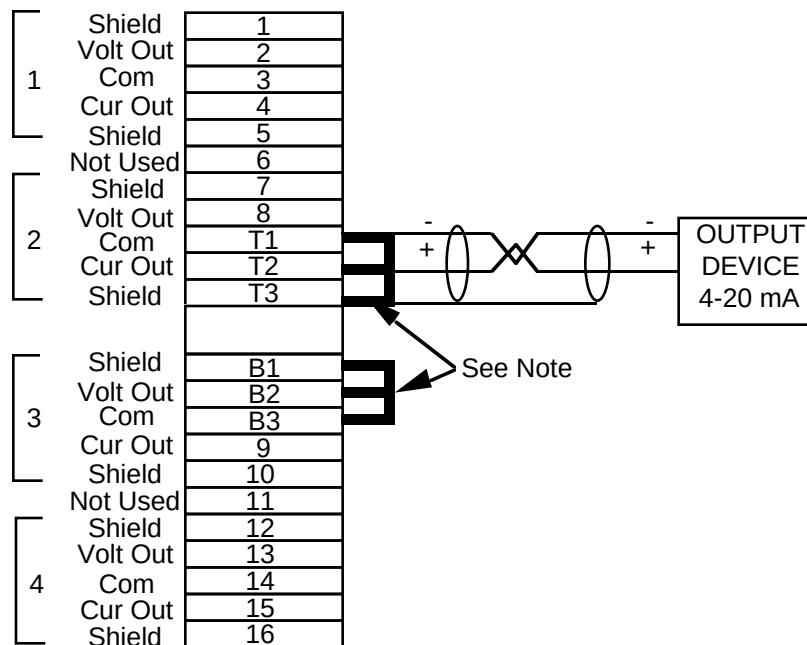
Each field output must be wired to its respective output and common terminals for each analog output channel. Use Belden 8761 or equivalent shielded cable to connect the output devices to the analog output terminals. Ground the cable shield drain wire at the module end only (the terminal labeled SHIELD) as shown in Figure 8-27.

The 4-20 mA outputs use the CUR OUT and COM connections.

More information on wiring can be found in the IEEE Standard 518, subsection 6.3 Wiring Practices.

WARNING

Remove the terminal block jumpers.

**Note:**

Remove the terminal block jumpers.

Figure 8-27 — 4-Point 4-20 mA Analog Output Module Wiring

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8.7.2.2 4-Point 0-10 V Analog Output Module

The model 621-0010-VR 0-10 V Analog Output Module converts digital values presented by the Logic Manager processor to an equivalent analog voltage and current, enabling control of motor speed, valve positions, or other devices requiring an analog signal as a control input. There are four output channels per module and channel is individually selectable. The model 621-0010-VR module is physically and electrically identical to the model 621-0010-AR Analog Output module. The difference is the factory calibration of the module. The model 621-0010-VR module is calibrated for 0 to 10 V operation, while the model 621-0010-AR module is calibrated for 4-20 mA operation.

Table 8-25 illustrates the ranges and the output voltages associated with sample signals. The first two ranges in the table are bipolar outputs. The current output will not function with these bipolar outputs. It functions only with the third range, 0 to +10 Vdc (4 to 20 mA), which is unipolar.

Specifications:

NUMBER OF OUTPUTS	4 nonisolated
RESOLUTION	12 bits
MODULE POWER REQUIREMENTS	+5 Vdc, 285 mA; -15 Vdc, 120 mA *; +15 Vdc, 85 mA plus 20 mA for each current output used
OUTPUT RANGES	
Voltage	±10 Vdc, ±5 Vdc, 0–10 Vdc
Current	4–20 mA
LOAD	
Voltage	8k ohm minimum
Current	0–600 ohm
ACCURACY	
Voltage	0.15% of FSR at 25° C after calibration
Current	0.15% of FSR at 25° C after calibration under load
TEMPERATURE COEFFICIENT	
Voltage	±0.006% of FSR per °C plus 3/4 LSB
Current	±0.03% of FSR per °C plus 3/4 LSB
SETTLING TIME	
Voltage	210 µs to within 1 LSB of final value (10 Vdc step change)
SLEW RATE	0.05 Vdc per µs minimum
TERMINAL BLOCK REQUIRED	621-9950

* Voltage outputs are not loaded.

Configuration:**CAUTION**

When configuring the I/O system, be certain that the maximum power load for the card file is not exceeded.

Use DIP Switch 5, an 8-position switch, to set each channel for bipolar or unipolar operation. See the Table 8-26 for Switch 5 operations.

CAUTION

Each channel must be configured for bipolar or unipolar operation. Improper setting will cause the module to malfunction.

Each channel can be individually selected for one of three operating ranges. Use DIP switches 1 through 4, one for each channel, to select these ranges. Each is a 2-position switch. Figure 8-25 illustrates the locations of these range selection DIP switches, as well as the offset and span adjustment potentiometers.

Assure that the voltage range selected for each channel, by switches 1 through 4, agrees with the channel bipolar/unipolar settings of switch 5. Table 8-27 defines the operation of the 2-position switches 1 through 4.

NOTE

The 4 to 20 mA output is available only when the 0 to 10 Vdc output is selected. Refer to the Figure 8-27 wiring diagram.

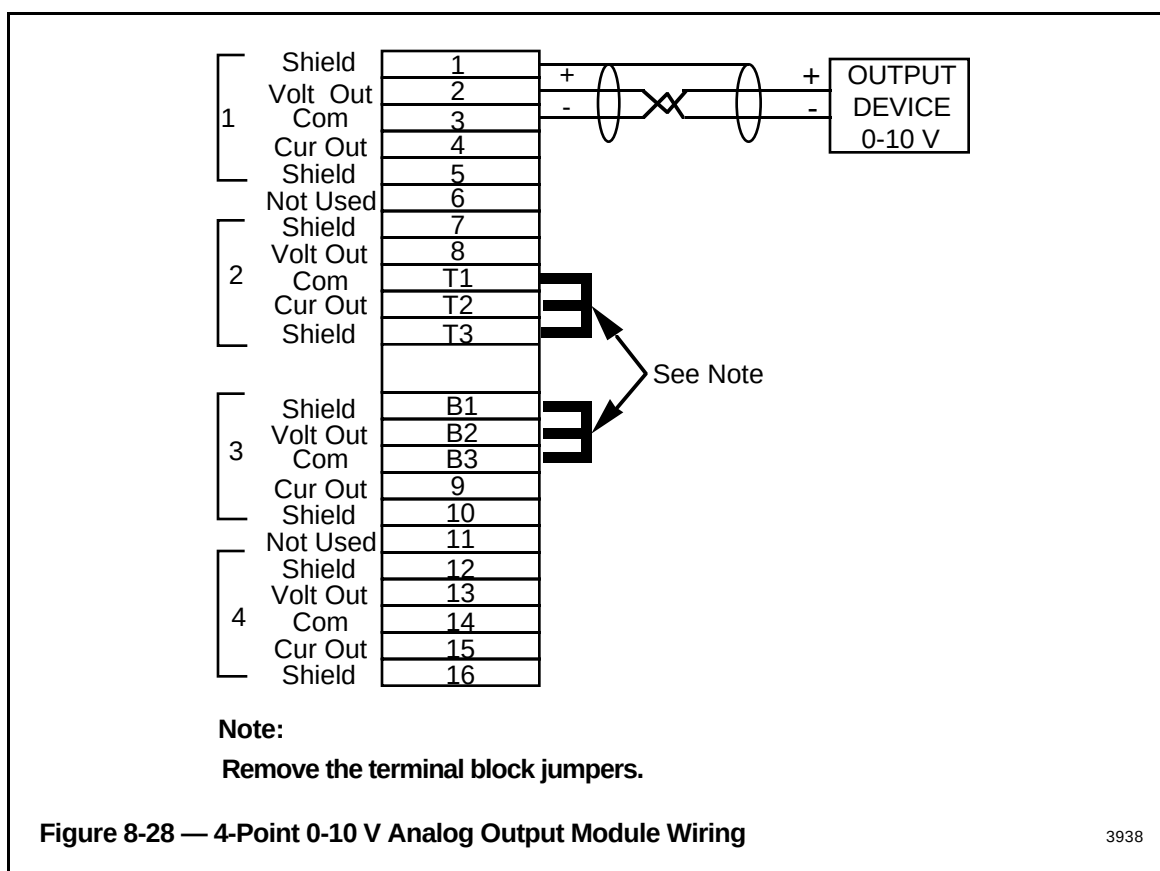
Calibration:

Each model 621-0010-VR has been factory calibrated for 0 to 10 Vdc operation. Operation at other ranges may require that the output be recalibrated. See Appendix A of the *Logic Manager Service* manual in the *Logic Manager Service* binder.

Wiring:

The I/O card file provides +5 Vdc logic power and a ± 15 Vdc reference voltage for analog output module operation.

Each field output must be wired to its respective output and common terminals for each analog output channel. Use Belden 8761 or equivalent shielded cable to connect the output devices to the analog output terminals. Ground the cable shield drain wire at the module end only (the terminal labeled SHIELD) as shown in Figure 8-28. The 0-10 Vdc outputs use the VOLT OUT and COM connections. More information on wiring can be found in the IEEE Standard 518, subsection 6.3 Wiring Practices.



8.8 Special Function Modules

The Logic Manager Special Function I/O Modules provide a variety of control and communications functions for the Logic Manager I/O System. The special function modules use the I/O bus for data transfer with the Logic Manager processor and rely on PUSH/PULL instructions for communications. The PUSH/PULL interface allows these modules to send or receive up to eight 16-bit registers of data at one time.

This section provides individual special function module descriptions, specifications, wiring diagrams, and special instructions for installation and operation. The modules are described in numerical order and are summarized in Table 8-28.

Table 8-28 — Special Function Modules

MODULE DESCRIPTION	CABLE CLASS	MODEL NUMBER
BCD Conversion Module	N/A	621-0006R
Pulse Input Module	DATA 1	621-0008 *
Input Simulator Module	N/A	621-0009R
ASCII Communication Module	DATA 1	621-0012R
Absolute Encoder Module	DATA 1	621-0018R
Pulse/Quadrature Input Module	DATA 1	621-0019R
Pulse Input Module	DATA 1	621-0024R
Resistance Temperature Detector Module	DATA 1	621-0025R
High Speed Counter Module	DATA 1	621-0307R
24 Vdc 24 Sink Input, 8 Source Output Module	DATA 2	621-0576R
Three Port MiniCOP Module	DATA 1	627-1002R
Four Port MiniCOP Module	DATA 1	627-1003R

* Superseded by 621-0024R

8.8.1 BCD Conversion Module

The model 621-0006R BCD Conversion Module (BCM) is an I/O Module that enables the Logic Manager processor to convert BCD numbers to a binary format, and binary numbers to BCD.

This module contains a total of eight 16-bit addressable registers that store data to be converted. The higher four addresses convert BCD data to binary. The lower four addresses convert binary data to BCD data.

A green **ACTIVE** LED on the frontplate illuminates when the module is being accessed.

Specifications:

NUMBER RANGE	
BCD	0-9999
Binary	0-65,535
CONVERSION TIME	less than 0.5 μ s
MODULE OPERATING CURRENT	
+5 Vdc	960 mA (max.)
TERMINAL BLOCK REQUIRED	None required

Installation:

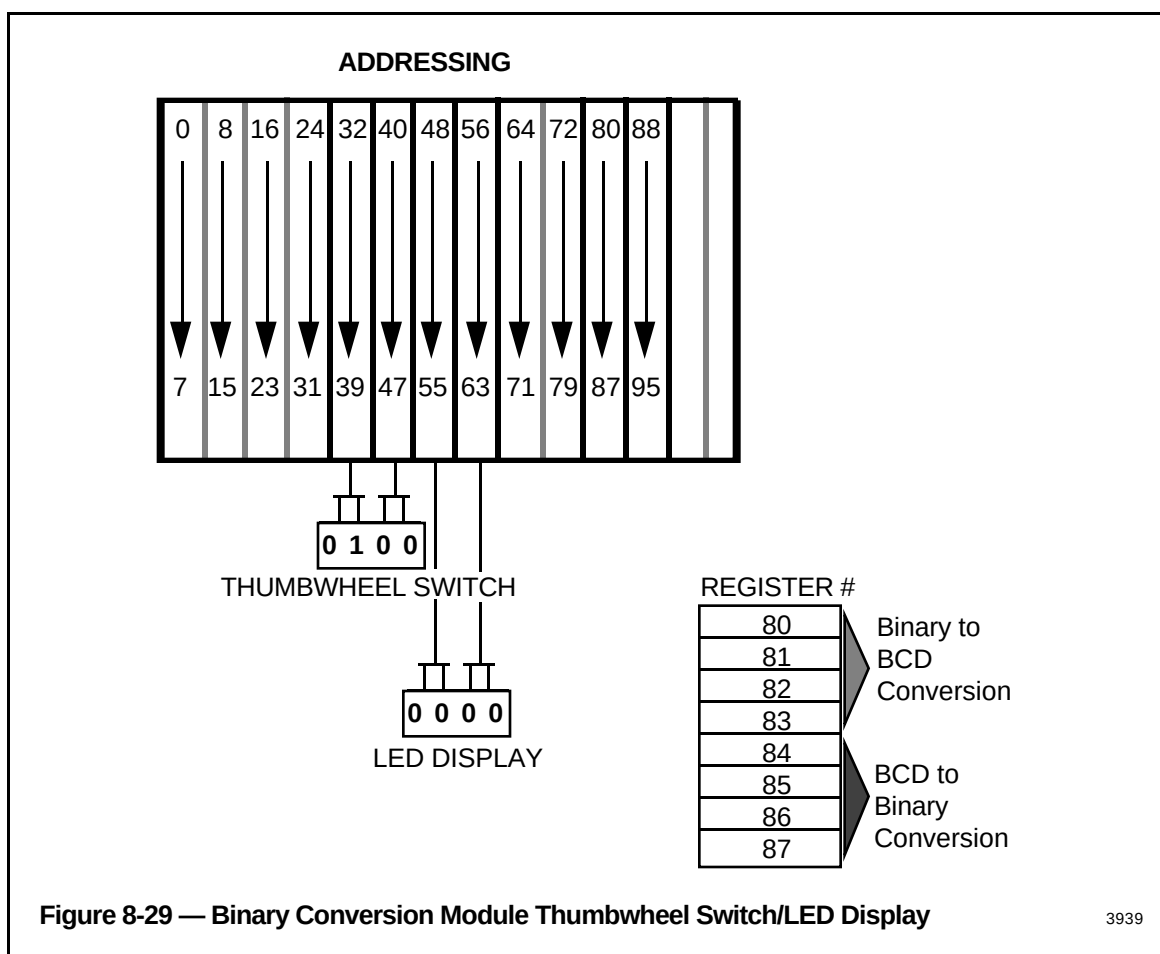
The front coverplate is permanently attached to the printed circuit board. The BCD Conversion Module is installed in any I/O card file module slot. The Card file power load for all I/O modules installed within the card file must not exceed the maximum card file power supply rating.

Application:

Figure 8-29 illustrates this module's ability to read a 4-digit thumbwheel switch for entering a timer preset value. The example also illustrates how the accumulated timer value is displayed on a seven-segment LED display.

The thumbwheel switch is wired to inputs that are addressed from 32 through 47. The LED display is wired to outputs that are addressed from 48 through 63. The BCD Module is located in the card slot with addresses 80 through 87.

When installed in an 8-point slot, the BCM assumes the designed slot addresses. In a 16-point slot, the BCM assumes the lowest eight addresses associated with the slot.



8.8.2 Pulse Input Module (621-0008)

NOTE

The model 621-0008 Pulse Input Module described below has been superseded by the model 621-0024R (see subsection 8.8.7).

The model 621-0008 Pulse Input Module (PIM) is a special function I/O module that counts digital pulses up to a frequency of 100k Hz. Pulses can come from any device with a pulse output such as proximity switch or flow meter. A 16-bit up counter is associated with each input. The counter, which can be read or reset by the processor, has a maximum count capacity of 65,535. There are 3-voltage range selections for each input, 5 V with a 4.5 to 9 volt range, 24 V with a 10 to 32 volt range, and 48 V with a 32 to 60 volt range. The total range of inputs is 5 to 60 Vdc.

Specifications:

INPUTS	4
VOLTAGE RANGES	4.7 to 9 Vdc, 10 to 32 Vdc, 32 to 60 Vdc
CURRENT (typical)	7 mA at 5 V, 8 mA at 12 V, 19 mA at 24 V, 15 mA at 48 V
MAXIMUM INPUT FREQUENCY	
Low (filtered)	200 Hz
High (unfiltered)	100k Hz
MINIMUM PULSE WIDTH	
Filtered	2.5 ms $\pm$ 25%
Unfiltered	5 μ s
INPUT ISOLATION	2500 Vdc optical
MODULE OPERATING CURRENT	850 mA (maximum)
TERMINAL BLOCK REQUIRED	621-9949

Configuration:

Be sure to select the operating frequency and the input voltage before installation. The two operating frequencies are high and low. A high operating frequency allows a pulse from dc to 100 k Hz. The low operating frequency allows a pulse from dc to 200 Hz. The input voltage can be selected for nominal voltages of 5 V, 24 V, or 48 V.

To make any of the above selections, locate the plug headers that reside on the circuit board. They are labeled P1 through P4, one for each counter. See Figure 8-30. Leaving the component side cover on the module, use the shorting bar on each plug header to make the selections. Table 8-30 shows the position of the plug headers for each voltage and frequency selection.

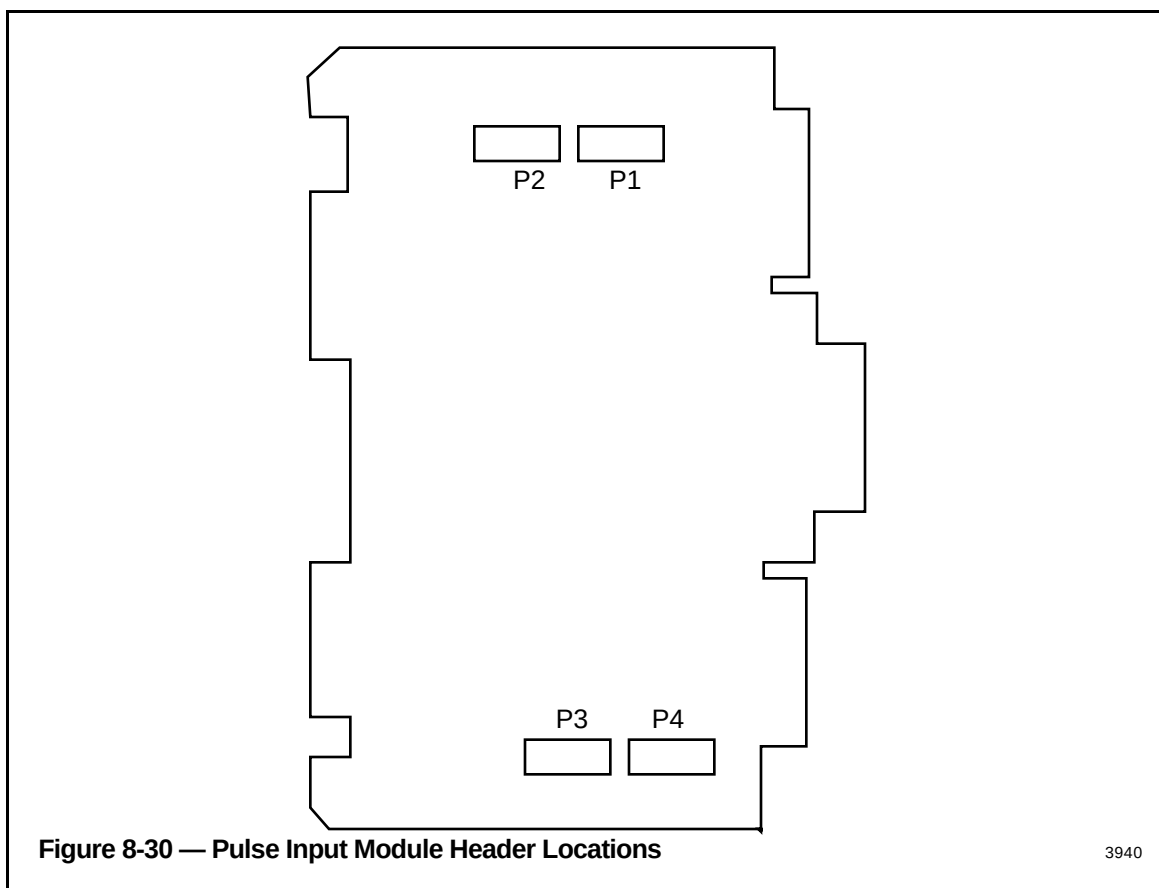


Table 8-29 — Pulse Input Module Voltage and Frequency Selection

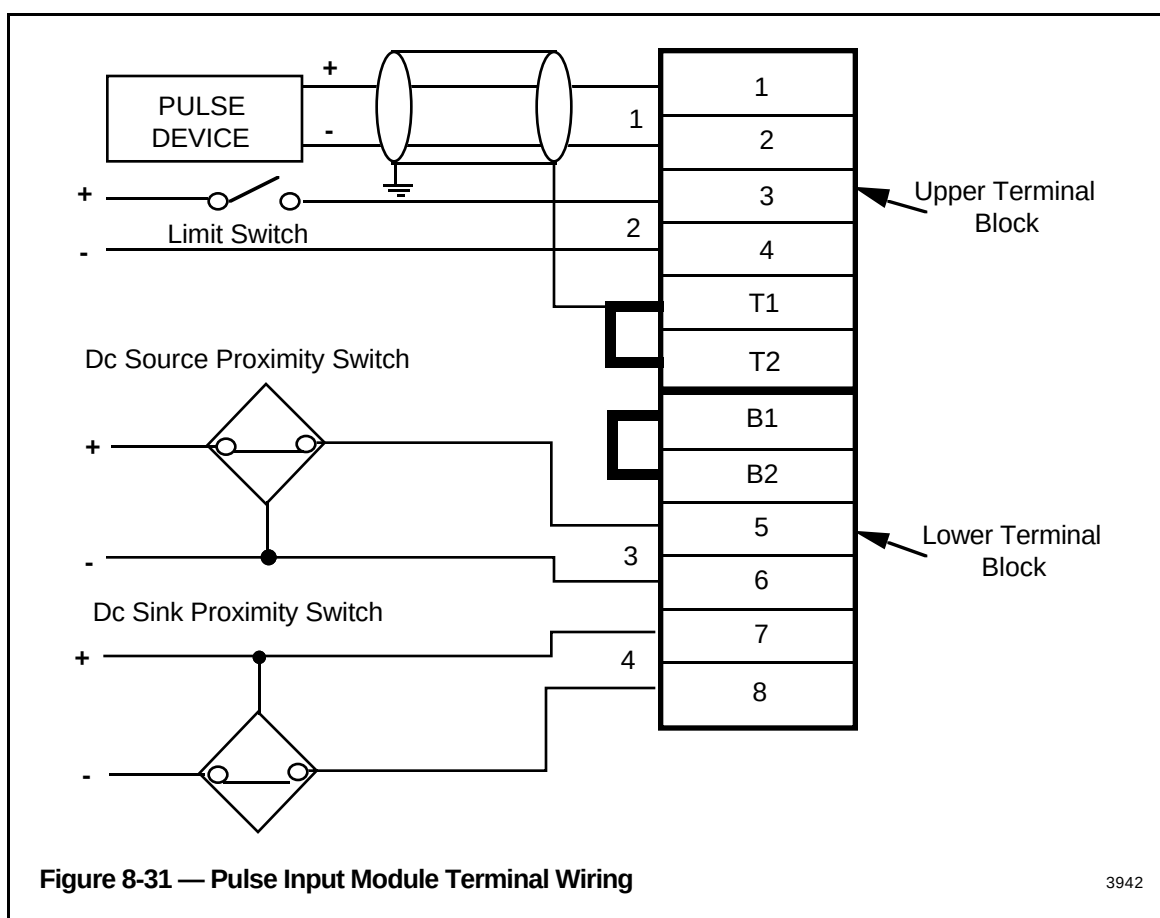
PLUG HEADER	INPUT VOLTAGE			FREQUENCY	
	5 V	24 V	48 V	LOW	HIGH
P1	W3	W2	W1	W4	Unmarked
P2	W7	W6	W5	W8	Unmarked
P3	W11	W10	W9	W12	Unmarked
P4	W15	W14	W13	W16	Unmarked

Wiring:

The Pulse Input Module uses the model 621-9949 8-point terminal block. There are two connections for each signal. The wiring to this module should be isolated from all AC wiring.

If the module is used in an electrically noisy environment or with wire runs of 50 feet or longer, a shielded cable should be used, such as Belden 8761, a 2 conductor, 22 gauge, twisted pair shielded cable with a drain wire. When a shielded cable is used, the shield must be tied to earth ground at both ends of the cable. This protects the cable from radio frequency interference. The earth ground for the shielded cable connections at the module are T1, T2, B1, and B2. See Figure 8-31.

Additional information on wiring can be found in the IEEE Std. 518, subsection 6.3 Wiring Practices.



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8.8.3 Input Simulator Module

The model 621-0009R Input Simulator Module controls the state of inputs and can be used for program development, input module testing, and operator interfacing. Primarily, the Input Simulator is a development tool. It can be used by a programmer to test a program by providing locally controlled inputs. This module can also be used during system troubleshooting. If an input module problem is suspected, the Simulator Module can replace an input module, providing a visual confirmation of the inputs.

The Input Simulator Module is an 8-point module that requires no external wiring or power to operate. Eight switches on the front of the module provide local switch control of inputs that simulate field devices. The status of each input is indicated by a separate LED.

Specifications:

INPUTS	8
OPERATING CURRENT	40 mA plus 3 mA for each energized input (65 mA maximum)

8.8.4 ASCII Communication Module

The ASCII Communication Module (ACM), model 621-0012R, provides an interface between the Logic Manager System and devices that communicate via an asynchronous serial data link. The ACM communicates with peripheral devices using character strings comprised of ASCII and non-ASCII characters. The single board module is installed in the Logic Manager I/O System and is controlled by the Logic Manager processor via the discrete and data I/O buses. Transmission and reception of serial information is provided through a 25-pin D series female connector. This serial port may be selected to operate on RS232, RS422, or 20 mA current loop hardware interface by printed circuit board mounted DIP switches.

The ASCII Communication Module is an I/O module that operates according to the user's ladder logic as interpreted by the module's configuration. The module can be configured for ABC protocol or a user-defined protocol. The ABC protocol is used by Honeywell IACD's Industrial Microcomputers. A user defined protocol allows the user to define all the communication formats.

The module communicates using numeric data and ASCII messages. Data, including information from the module's battery backed real time clock, is transferred between the programmable controller and the module using ladder logic in the programmable controller. Over 12 K of messages may be stored in the module's battery backed memory and can be transferred between the module and a peripheral device using strings of characters.

NOTE

Refer to the Honeywell 621 ASCII Communications Module User Manual, Form 621-8999, for detailed information.

Specifications:

I/O	1 serial port
RS232 Interface	1 peripheral device
RS422 Interface	Up-to-32 peripheral devices
20 mA (active)	Drives up-to-1000 ohm impedance limited to 9600 baud (maximum)
Connector	25-pin D-type female
Baud Rates	150, 300, 600, 1200, 2400, 4800, 9600, 19.2 k baud (19.2 k baud RS232/422 only)
POWER SUPPLY REQUIREMENTS	5 Vdc $\pm$.25 Vdc @ 2.35 A; +15 Vdc $\pm$.45 Vdc @ 100 mA; -15 Vdc $\pm$.45 Vdc @ 100 mA
PROCESSOR	68B09E operating @ 1.84 MHz
MEMORY	16 K EPROM; 16 K CMOS battery backed RAM
OPERATING TEMPERATURE	0° to 60° C
STORAGE TEMPERATURE	-40° to 85° C
HUMIDITY	5% to 95% RH (noncondensing)
TERMINAL BLOCK REQUIRED	None required

8.8.5 Absolute Encoder Module

The Absolute Encoder Module (AEM), model 621-0018R, functions similarly to a mechanical rotary cam switch in that it energizes and de-energizes outputs for a defined region of rotation between 0° and 360°. The AEM reads the rotational input value from the absolute encoder and based on that value controls the ON/OFF status of up-to-eight 24 Vdc, 2 A outputs. The output status for each of the eight outputs, the shaft position, the direction of rotation, and the velocity can be read from the AEM by the Logic Manager processor.

Each output address has two programmable preset registers (16 total) that define the end boundaries for the region of rotation (within 360°), for which the output is either ON or OFF. These preset values are downloaded from the Logic Manager logic program and are unique for each output, which allows independent control for each of the eight outputs.

After the module has been configured, the preset points downloaded from the ladder logic program, and the module outputs enabled, the AEM will directly control its outputs independently of the Logic Manager processor.

Installation:

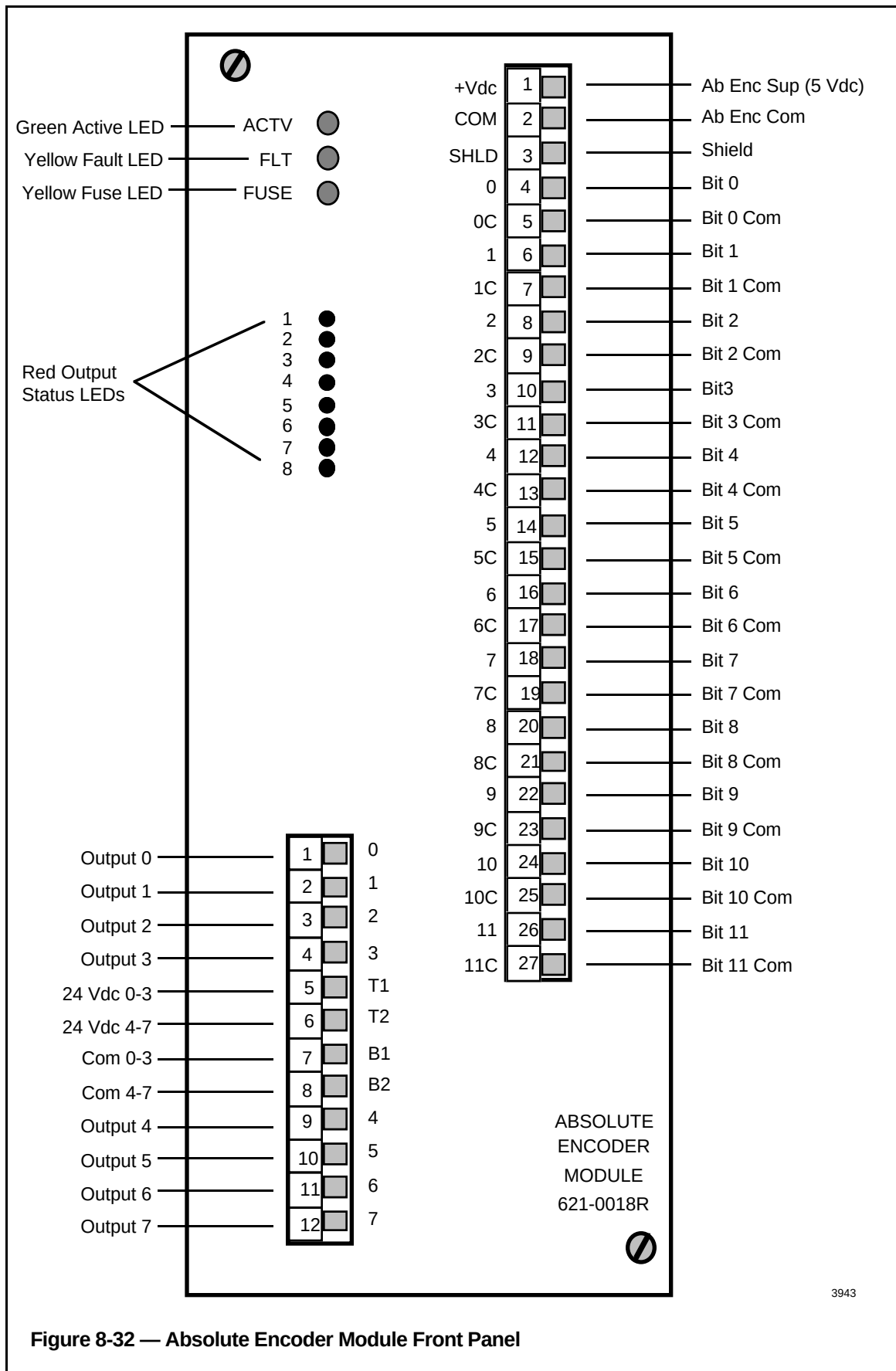
The AEM is a double wide module and may be installed in any two adjacent I/O slots of a I/O card file. The AEM is compatible with 8-, 9-, 10-, 11-, or 12-bit absolute encoders with input format of either BCD, natural binary, or standard gray code. The encoder signal output (AEM input) can be either TTL, open collector, or differential (differential recommended). Up-to-500 mA at 5 Vdc is provided by the AEM for directly powering an absolute encoder. The AEM will function at up to the following maximum velocities, providing full data integrity and maintaining control of each of the eight outputs:

<u>ENCODER RESOLUTION</u>	<u>VELOCITY (max.)</u>
12 bits	830 rpm
11 bits	1660 rpm
10 bits	3320 rpm
9 bits	6640 rpm
8 bits	13280 rpm

Figure 8-32 shows the front panel of the Absolute Encoder module.

NOTE

See the Absolute Encoder User Manual, Form 621-8997, for more detailed information.



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Figure 8-32 — Absolute Encoder Module Front Panel

Specifications:**INPUT****INPUT VOLTAGE RANGE**

Single-ended

Logic 1: 2.3 Vdc to 5.0 Vdc;

Logic 0: 0.0 Vdc to 1.5 Vdc

Differential

0.2 Vdc to 6.0 Vdc

INPUT CURRENT RANGE

Single-ended

-50 μ A (low-0), 300 μ A max.(high 1); each input channel
represents 8 standard TTL loads

Differential

1.3 to 82 mA (typically 13 mA @ 2 V
differential)**ENCODER POWER REQUIREMENTS**5 Vdc $\pm$ 5%, 500 mA (an encoder that
draws more than 500 mA must be
powered externally)**ENCODER CABLING DISTANCE**TTL: 5 ft.; Open Collector: 50 ft.;
Differential: 500 ft. (powered
externally)**MAX. INPUT FREQUENCY**56.7 k Hz (cycle time 17.625 μ s)**ENCODER INPUT FORMAT**BCD, natural binary, or standard
Gray code**ENCODER RESOLUTION**

8, 9, 10, 11, or 12 bits

OUTPUTS

NUMBER OF OUTPUTS	8
VOLTAGE RANGE	18 Vdc to 28 Vdc
MAX. CURRENT RANGE	2 A individual; 6 A per common; 12 A per module
FIELD POWER REQUIREMENTS	<25 mA per energized output 200 mA maximum
OFF-STATE LEAKAGE CURRENT	<5 mA typically
ON-STATE VOLTAGE DROP	<2 V @ 2 A
FUSING	1 per group of 4, 7 A fast blow
MODULE OPERATING CURRENT	1.75 A with all 8 outputs energized (includes both module and encoder requirements).
SURGE CURRENT	8 A for 10 ms (nonrepetitive)
OUTPUT DELAY	OFF to ON: <30 μ s; ON to OFF: <500 μ s
RESOLUTION	Binary and Gray code: 1 part in 4096 (0-4095); BCD: 1 part in 1000 (0-999)The Absolute Encoder Module (AEM) interfaces directly to absolute encoders, providing high-speed control independently of the Logic Manager processor.
TERMINAL BLOCK REQUIRED	Supplied with the module (621-9953)

8.8.6 Pulse/Quadrature Input Module

The 621-0019R Pulse/Quadrature Input Module (PQIM) is a special function module that counts high-speed pulse streams from devices that produce pulsed outputs. These devices fall into two categories: encoders with differential outputs, such as quadrature type incremental shaft encoders or separate pulse devices, such as limit switches with single-ended outputs. A maximum of three devices may be connected to the Pulse/Quadrature Input Module; all three may be encoders, all may be separate pulse devices, or they may be a mixture of both types.

Module Features:

- Compatible with Parallel and 32-point Serial I/O Systems
- Installation in any single I/O slot configured for 16 points
- Pulse counts may be read by the Logic Manager processor with a PULL instruction
- Three counters, each of which may be configured as a 32-bit or a 16-bit counter
- Four count modes: quadrature input x1, x2, x4, and separate count
- LEDs that show up/down counts for each channel
- Module fault reporting
- Input pulse frequencies of up-to-200 Hz, 100 k Hz, or 200 k Hz

Specifications:

NUMBER OF INPUTS	3 pulse/quadrature encoders
SYSTEM POWER UNIT LOAD	
Encoders Powered Externally	0.75 A, +5 Vdc
Encoders Powered by Module	2.25 A, +5 Vdc (500 mA/channel)
COUNTERS	
Number of Counters	1 per channel
Maximum Count Range	32 bits per channel
Maximum Offset Value	32 bits per channel
VOLTAGE RANGE	
Differential	Logic 1: 200 mV to 6.0 Vdc; Logic 0: -200 mV to -6.0 Vdc
Single-Ended	Logic 1: 2.0 Vdc to 12.0 Vdc; Logic 0: -10.0 Vdc to 1.0 Vdc
COMMON MODE VOLTAGE	±12 Vdc
INPUT SENSITIVITY	±200 mV
HYSTERESIS	±50 mV
MAXIMUM INPUT FREQUENCY	200 k Hz, unfiltered; 100 k Hz, filtered; 200 Hz, filtered
MINIMUM PULSE WIDTH	3 μs, unfiltered
SIGNAL INPUT ISOLATION	None

SIGNAL THROUGHPUT RATE

Normal Mode

0.25 ms per channel

Enhanced Throughput Mode

25.0 μ s per channel

TERMINAL BLOCK REQUIRED

None required

NOTE

Refer to the 621-0019R Pulse/Quadrature Input Module User Manual (Form No. 621-8990) for more information about this module.

8.8.7 Pulse Input Module (621-0024R)

CAUTION

You may remove the 621-0024R from, or insert it into, a **serial** I/O card file *while the system is powered*, provided you use a 621-9938R Serial I/O Module (SIOM). **DO NOT DO THIS** with a parallel I/O card file, or a serial I/O card file containing a 621-9938 or 621-9940 SIOM.

The model 621-0024R Pulse Input Module (PIM) supersedes the 621-0008 PIM. It is a special function I/O module that counts digital pulses up to a frequency of 100 kHz. Pulses can come from any device with a pulse output such as a proximity switch or flow meter. A 16-bit up counter is associated with each input. The counter, which can be read or reset by the processor, has a maximum count capacity of 65,535. There are three voltage range selections for each input: 4.7 to 7.5 Vdc, 10 to 25 Vdc, and 32 to 60 Vdc.

Specifications:

INPUTS	4
VOLTAGE RANGES	4.7 to 7.5 Vdc, 10 to 25 Vdc, 32 to 60 Vdc
CURRENT (typical)	≥12 mA at 5 Vdc, ≥9 mA at 12 Vdc, ≥20 mA at 24 Vdc, ≥16 mA at 48 Vdc
MAXIMUM INPUT FREQUENCY	
Low (filtered)	200 Hz
Medium (filtered)	5 kHz
High (unfiltered)	100 kHz
MINIMUM PULSE WIDTH	
ON	
Low	2.5 ms ±25%
Medium	40 μs
High	5 μs
OFF	
Low	2.5 ms ±25%
Medium	85 μs
High	4 μs
INPUT ISOLATION	2500 Vdc optical
MODULE OPERATING CURRENT	850 mA (maximum)
TERMINAL BLOCK REQUIRED	621-9949

Configuration:

Be sure to select the operating frequency and the input voltage before installation. The three operating frequency ranges are high, medium, and low. A high operating frequency range allows a pulse rate up to 100 kHz. The medium range allows a rate up to 5 kHz. The low range allows a rate up to 200 Hz. The input voltage can be selected for nominal voltages of 5 Vdc, 24 Vdc, or 48 Vdc.

To make any of the above selections, locate the plug headers that reside on the circuit board. They are labeled P4 through P7, one for each counter. See Figure 8-33. Leaving the component side cover on the module, use the shorting bar on each plug header to make the selections. Table 8-30 shows the position of the plug headers for each voltage and frequency selection.

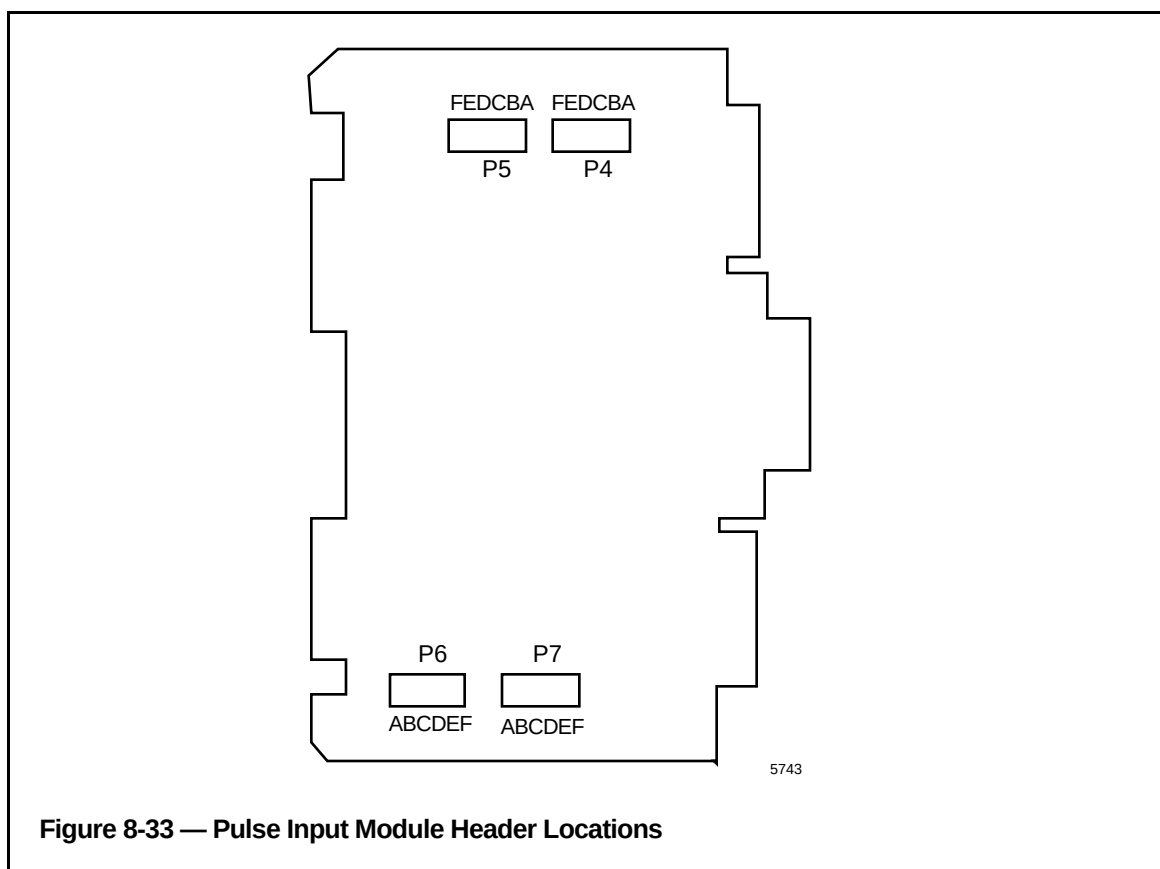


Table 8-30 — Pulse Input Module Voltage and Frequency Selection

CHANNEL	PLUG HEADER	INPUT VLOTAGE			FREQUENCY		
		5 Vdc	24 Vdc	48 Vdc	200 Hz	5 kHz	100 kHz
1	P4	C	B	A *	D	E	F *
2	P5	C	B	A *	D	E	F *
3	P6	C	B	A *	D	E	F *
4	P7	C	B	A *	D	E	F *

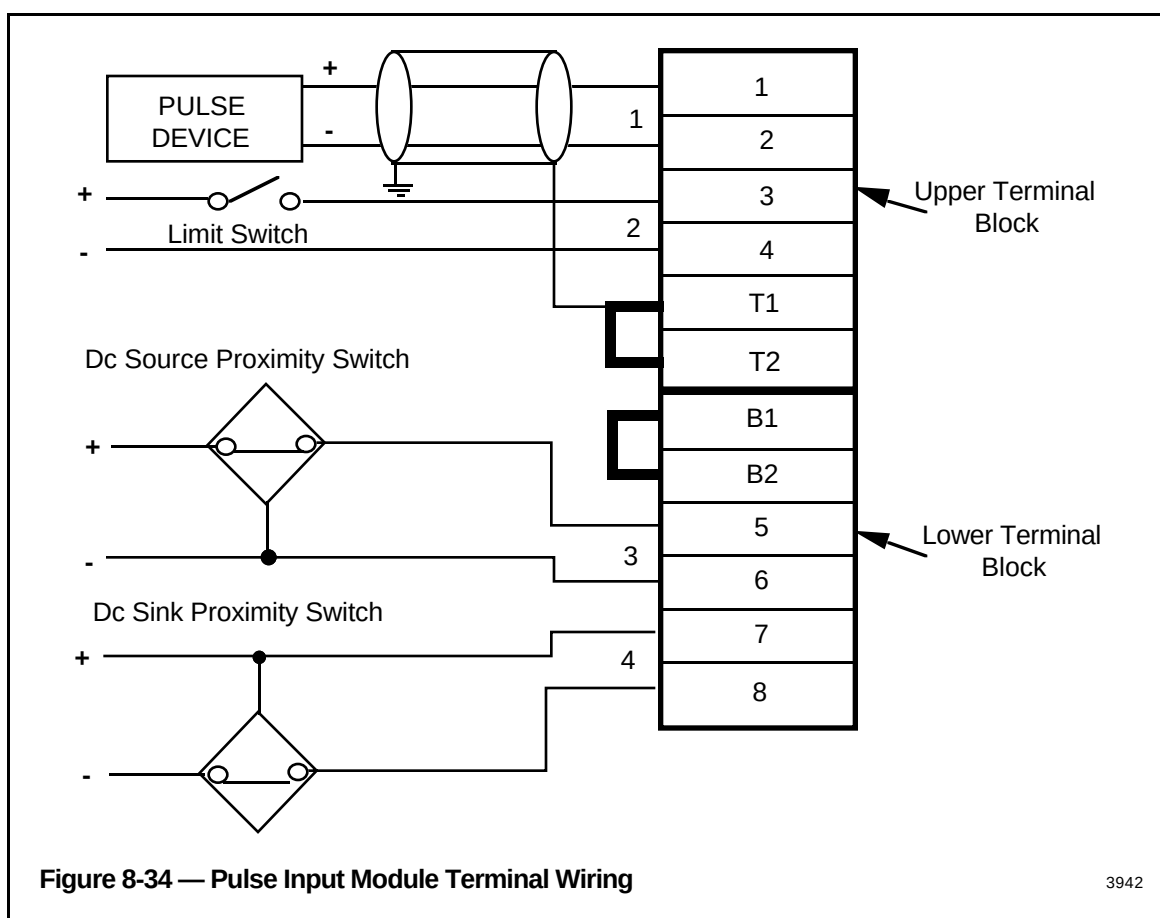
* Factory setting

Wiring:

The Pulse Input Module uses the model 621-9949 8-point terminal block. There are two connections for each signal. The wiring to this module should be isolated from all ac wiring.

If the module is used in an electrically noisy environment or with wire runs of 50 feet or longer, a shielded cable should be used, such as Belden 8761, a two-conductor, 22 gauge, twisted pair shielded cable with a drain wire. When a shielded cable is used, the shield must be tied to earth ground at both ends of the cable. This protects the cable from radio frequency interference. The earth ground for the shielded cable connections at the module are T1, T2, B1, and B2. See Figure 8-34.

Additional information on wiring can be found in the IEEE Std. 518, subsection 6.3 Wiring Practices.



8.8.8 Resistance Temperature Detector Module

CAUTION

You may remove the 621-0025R from, or insert it into, a **serial** I/O card file *while the system is powered*, provided you use a 621-9938R Serial I/O Module (SIOM). DO NOT DO THIS with a parallel I/O card file, or a serial I/O card file containing a 621-9938 or 621-9940 SIOM.

The model 621-0025R Resistance Temperature Detector Module (RTDM) is an input device that can read resistance sent from RTD probes. As the module reads the signals, an analog-to-digital converter within the RTDM changes the signals into binary counts. A Control Processor Module can then read the counts and determine how to control the process that the probe is monitoring.

The RTDM is typically used in process and batch process applications that require precise temperature measurements for long periods of time. Reactors and ovens are examples of devices that the RTDM can monitor. The range of temperatures that the RTDM can measure is -300 to 900°F (-184 to 482°C).

For detailed application information, refer to *621-0025R Resistance Temperature Detector Module and 623-0025 Configuration Utility User Manual*, Form No. 621-8985.

RTD Probes:

Types of probes:	100 ohm platinum 200 ohm platinum 500 ohm platinum 10 ohm copper
Lead wires:	2, 3, or 4 (does not support 4-wire probes that include separate lead resistance loops)
Alphas:	European (IEC) American (SAMA) Japanese (JIS) 10 ohm copper (GE)

Specifications:

NUMBER OF INPUTS	8
OUTPUT WORD SIZE	12 to 15 bits, configurable
RTD PROBE LEAD WIRES	2, 3, or 4 (does not support 4-wire probes that include separate lead resistance loops)
RTD SIGNAL ACCURACY	
10 ohm copper	2.0°F (1.11°C)
All others	0.8°F (0.48°C)
RTD TEMPERATURE STABILITY	
10 ohm copper	0.12°/°C
All others	0.05°/°C
RELATIVE HUMIDITY:	
SIGNAL ACCURACY	0.3°F (0.2°C)
TEMPERATURE STABILITY	0.02°/°C
ELECTRICAL ISOLATION	300 Vac peak, channel to channel; 300 Vac peak, input to logic
INPUT FILTER	1 ms RC nominal
CONVERSIONS PER SECOND	20 at 60 Hz, 16.7 at 50 Hz
MODULE OPERATING CURRENT	125 mA at 5 Vdc, 70 mA at +15 Vdc, 70 mA at -15 Vdc

Configuration:

The RTDM is designed to accommodate a wide variety of input applications. To support this flexibility, Honeywell provides the *623-0025 Configuration Utility*. This software is used to individually configure up to 8 input channels. There are no hardware DIP switches to be configured. However, two-wire and three-wire RTD probes require jumper wires to be installed on the input terminal block, as illustrated in Figure 8-35.

For information on the 623-0025 Configuration Utility, refer to *621-0025R Resistance Temperature Detector Module and 623-0025 Configuration Utility User Manual*, Form No. 621-8985.

Wiring:

The RTDM input cables fall into the category of DATA 1, as defined in IEEE Standard 518. These cables carry low-level voltages. Be sure to separate noise-carrying cables (those with time-varying voltage or time-varying current) from the RTDM input data cables. Table 8-31 lists minimum recommended distances between the RTDM's DATA 1 cables and other classes of cables. Measure these distances from the top of one tray/conduit to the bottom of the tray/conduit above, or between sides of adjacent trays/conduits. Trays containing DATA 1 or DATA 2 cables should have solid bottoms and covers to complete shielding. Only trays containing DATA 3 cables may have ventilation slots or louvers.

Table 8-31 — Cable Spacing

CABLE CLASS	MINIMUM DISTANCE FROM DATA 1 CABLES		
	TRAYS	TRAY-CONDUIT	CONDUIT
DATA 2	<i>Note 1</i>	1 inch	1 inch
DATA 3	6 inches	4 inches	3 inches
DATA 3A	6 inches	4 inches	3 inches
DATA 4	26 inches	18 inches	12 inches
DATA 4A	26 inches	18 inches	12 inches

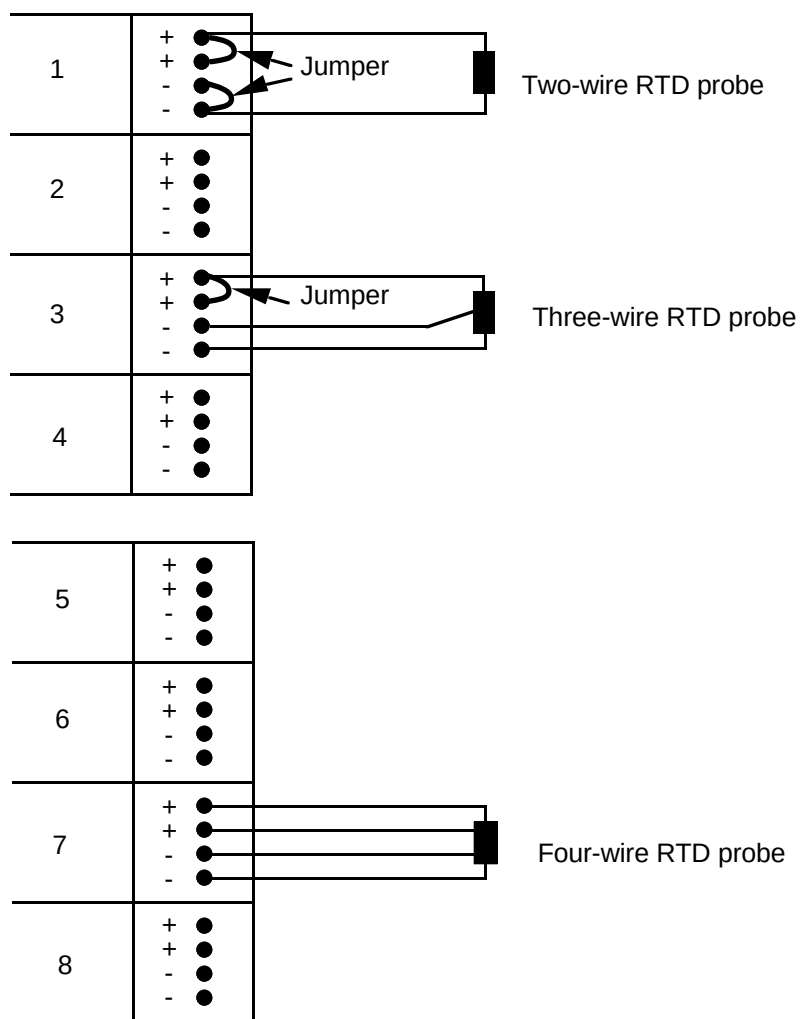
Note 1: You may run DATA 1 and DATA 2 cables in the same tray if you separate them with a grounded steel barrier.

Each module includes two 16-position terminal blocks that can be removed to facilitate wiring and module removal and installation. Each input has four connections on the terminal block. Table 8-32 and Figure 8-35 detail how to wire these connections. The terminal blocks are compatible with 14 through 22 AWG stranded wire. They also accommodate two 18 AWG stranded wires or a single 12 AWG solid wire. Strip each wire 3/8 inch $\pm$ 1/8 inch before inserting it into the terminal block and tightening the clamp screw.

Two- and three-wire RTD inputs require external jumpers (supplied with the RTDM). Refer to Table 8-32 and Figure 8-35 for information on wiring the RTD probes and placement of the jumpers.

Table 8-32 — RTDM Wiring and Jumper Placement

NUMBER OF WIRES	CONNECTIONS	JUMPERS
2	<i>Top Wire:</i> either of the two positive connections <i>Bottom Wire:</i> either of the two negative connections	One across the two positive connections and one across the two negative connections
3	<i>Top Wire:</i> either of the two positive connections <i>Middle Wire:</i> first negative connection <i>Bottom Wire:</i> second negative connection	One across the two positive connections
4	One wire to each of the four connections	None (remove jumpers)



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Figure 8-35 — Sample RTDM Wiring Diagram

8.8.9 High-Speed Counter Module

The model 621-0307R High-Speed Counter Module (HSC) allows the Logic Manager System to count high-speed pulse streams, primarily from quadrature type incremental shaft encoders. Other devices providing a pulsed output may also be used.

The HSC module occupies two card slots in the I/O card file. It accepts pulses from one encoder with either a 5 Vdc TTL or a differential output. The HSC contains two user program presettable 16-bit counters that respond to a single pulse input device. Each counter has an associated 5-15 Vdc output that either turns ON or OFF when the preset is reached. Each 16-bit counter is controlled separately by the control program. External control inputs provide external count reset and count latch functions for each counter. Counter accumulated values may be read by the user program. The count range is 0 to 65,535 counts, however, overflow and underflow status bits can be used to extend this count range.

Input Operation:

High-Speed Counter inputs are divided into two groups: pulse source inputs and external counter control inputs.

The pulse source inputs are located on the top field wiring connector. The HSC may be selected via a DIP switch for pulse inputs from a device providing either a TTL or differential output. There are three pulse inputs; the A Channel input, the B Channel input, and the C Channel input (marker input). A and B Channel inputs cause the two 16-bit counters to operate. The marker input is used in conjunction with the external counter reset control inputs. The marker signal, provided by most shaft encoders once every revolution, must be in the high state when the external counter reset inputs are energized, before the counters are set to zero.

The pulse source inputs can accurately follow a square wave input at a 50% duty cycle up to a 100 k Hz rate. These inputs may be filtered via a DIP switch to reduce the input frequency to no greater than 200 Hz at 50% duty cycle.

The pulse source inputs when operated in the differential mode utilize SN74175 differential line receivers. This device has the following specifications:

- Common Mode Voltage ± 12 Vdc
- Max Differential Input Voltage ± 25 Vdc
- Input Sensitivity ± 200 mV
- Input Hysteresis ± 50 mV

Counter control inputs provide external control of the HSC 16-bit counters. There are two reset counter inputs and two latch counter inputs, for control of both 16-bit counters.

When both the reset input and the marker input for a given counter are in the high state, the respective counter is reset to zero. The count will resume when the reset input is turned off. The latch input will cause its respective counter to stop counting when the latch input transitions from a low to a high state. Once the counter receives an external latch command, the latch may be reset through the user program by means of one of the following:

1. A clear latch control output command (counter not reset)
2. A reset counter control output command
3. The external reset counter input

The counter will resume counting when the clear latch control output is turned on, or when the reset control output or external counter reset input is turned off.

The reset counter and latch counter inputs have a 5-15 Vdc operating range. The turn on voltage level is 4.5 Vdc. The turn off voltage level is 1.5 Vdc. Electrically isolated from each other, these inputs are filtered and have the following input delay times:

- OFF to ON: 2.4 ms $\pm$ 25%
- ON to OFF: 17.5 ms $\pm$ 25%

The reset counter and latch counter inputs each have a **STATUS** LED that illuminates when the respective input is in the high state. The latch counter **STATUS** LED indicates that input state, not the latch condition (i.e., the latch input may be off, but the counter may still be latched).

Output Operation

The HSC contains two solid-state outputs, each responding to its respective 16-bit counter. The outputs are DIP switch selectable to turn on or off when the preset value of each counter is reached. These outputs may also be forced on or off by the user control program.

Each output can be configured for either sink or source operation. See the INSTALLATION section of the High-Speed Counter User Manual (Form No. 621-8992) for proper jumper settings with sink/source operation. Each output has an associated status LED that illuminates when the output is energized.

Wiring

Two connectors are mounted on the HSC. These connectors are two piece construction for input and output terminations. The male half may be removed to facilitate wiring and for easy module installation and removal.

All pulse source inputs are wired to the 10-position top connector. See Figure 8-36 for typical differential or TTL wiring.

Pulse Source Input Cabling

For pulse source inputs the voltage is provided by the pulse source. TTL pulse source inputs, however, may also be activated by switching the inputs to ground through a contact closure, as these inputs are pulled high by internal pull-up resistors.

Pulse source devices should be connected to the HSC using shielded cable. Cable shields must be grounded at the pulse source.

Belden number 9730 and 9729 are three and two twisted-pair cables, respectively, suitable for long distance differential operation (less than 2000 feet). TTL operation should be limited to distances less than 100 feet, due to electrical noise present in most applications, which may affect pulse signal quality.

Specifications:

SIGNAL INPUTS —CHANNELS A, B, C—TTL OPERATION

Inputs A, B, and C are TTL compatible, incorporating a 1.3 k pull-up resistor to 5 Vdc.

DIFFERENTIAL OPERATION

Input Range (max.)	0.5–12 Vdc
Common Mode Voltage	±12 Vdc
Max. Differential Input Voltage	25 Vdc
Input Sensitivity	±200 mV
Hysteresis	±50 mV

MAXIMUM INPUT FREQUENCY 100k Hz, unfiltered; 200 Hz, filtered

MINIMUM PULSE WIDTH 3 μ s, unfiltered

SIGNAL INPUT ISOLATION None

CONTROL INPUTS—RESET AND LATCH COUNTER

INPUT VOLTAGE	5–15 Vdc $\pm$ 10%
Turn-On Voltage	4.5 Vdc
Turn-Off Voltage	1.5 Vdc

INPUT CURRENT 5 mA minimum,
 34 mA maximum,
 7.5 mA (typical) at 5 Vdc,
 24 mA (typical) at 12 Vdc

INPUT DELAY

OFF to ON	2.4 ms $\pm$ 25%
ON to OFF	17.5 ms $\pm$ 25%

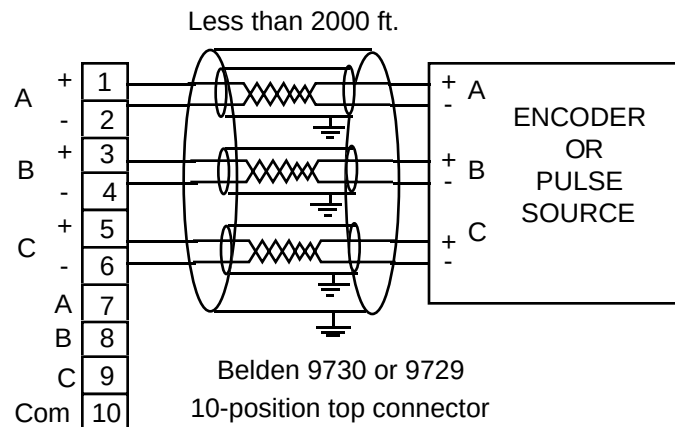
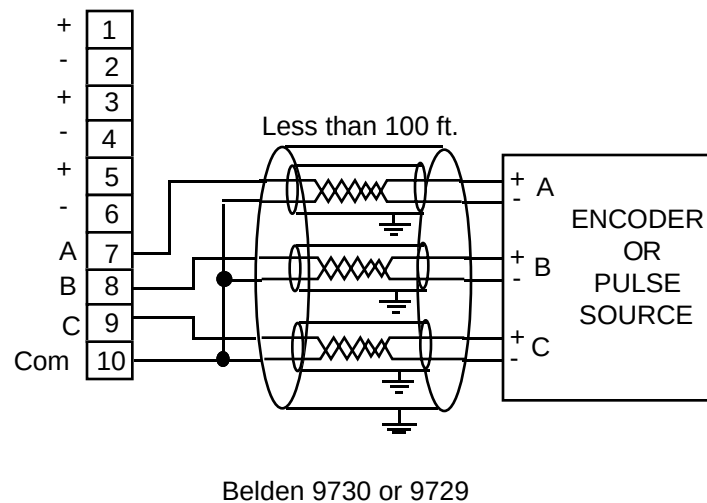
INPUT ISOLATION 2500 Vdc optical

COUNTER OUTPUTS

NUMBER OF OUTPUTS	2 per module
OUTPUT VOLTAGE	5–15 Vdc
OUTPUT CURRENT	150 mA source or sink (External Suppression Diode required for Inductive Loads)
OUTPUT DELAY	
OFF to ON	15 μ s maximum
ON to OFF	350 μ s maximum (Measured with max. resistive load from Vcc to 10% of Vcc)
OUTPUT ISOLATION	2500 Vdc optical

GENERAL SPECIFICATIONS

NUMBER OF INPUTS	1 quadrature encoder
SYSTEM POWER UNIT LOAD	1 A at +5 Vdc
EXT. POWER SUPPLY REQ.	Inputs and outputs will operate over a 5–15 Vdc range from external power source
NUMBER OF COUNTERS	2 per module
MAXIMUM PRESET VALUE	65,535
TEMPERATURE RANGE	0–60°C ambient
HUMIDITY RANGE	5–95% noncondensing
TERMINAL BLOCK REQUIRED	Supplied with module

DIFFERENTIAL SIGNAL INPUT WIRING**TTL SIGNAL INPUT WIRING****NOTE**

Encoder and HSC chassis grounds
must be at the same potential.

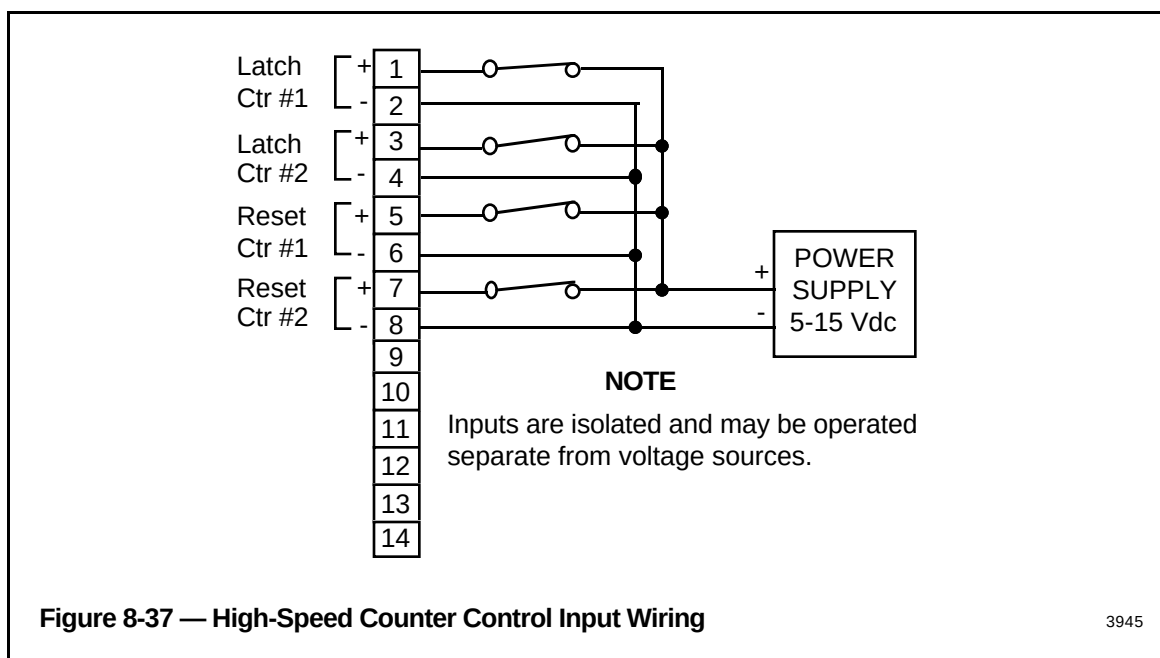
Figure 8-36 — High-Speed Counter Signal Input Wiring

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External Counter Control Input Cabling

The external counter control inputs (reset and latch for each counter) are wired to terminals 1 through 8 on the lower 14-position connector. Each input is isolated from the other. See Figure 8-37.

External counter control devices should be connected to the HSC using shielded twisted pair cable (Belden model number 8761 or equivalent) when operating the inputs at 5 Vdc. When operating the inputs at 12 Vdc, shielded cable is not normally required. However, avoid running input wiring with high voltage wiring (115 Vac or above) in the proximity.



Output Cabling

HCS outputs are wired to terminals 9 through 14 on the lower connector. Each output is selected for either sink or source operation by jumper placement on the module. Figure 8-38 shows wiring for HSC outputs in either current sinking or sourcing configurations.

WARNING

HSC outputs are not fuse protected. Make certain that connections are correct before applying power. Diode suppression is required for inductive loads. Use Littlefuse type 3AG catalog number 312.250 or equivalent. Use diode 1N4001.

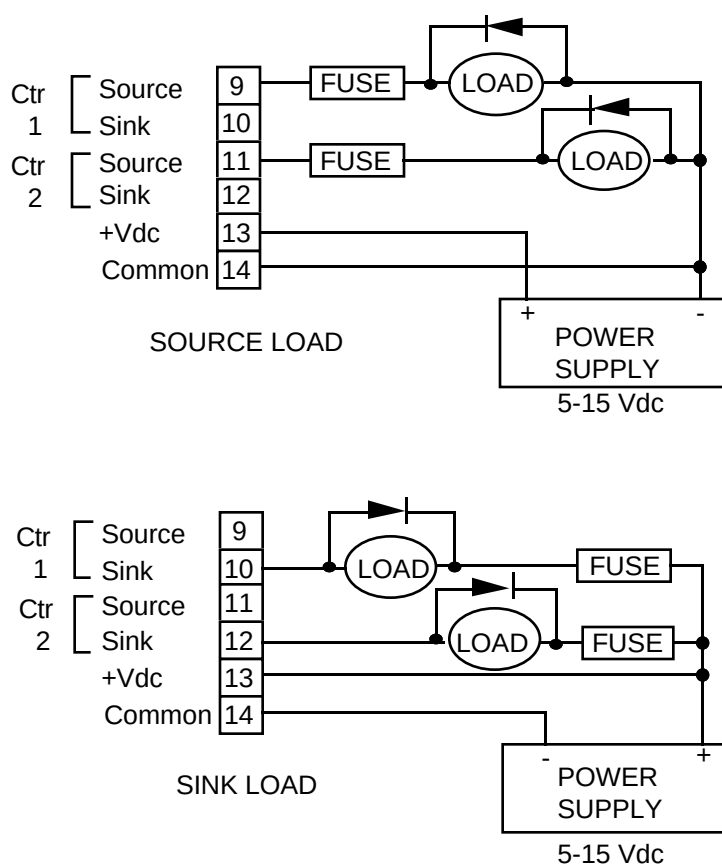


Figure 8-38 — High-Speed Counter Output Wiring

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NOTE

Refer to the High-Speed Counter Module User Manual, Form 621-8992, for more information.

8.8.10 24 Vdc Sink Input, Source Output Module

The model 621-0576R 24 Vdc Sink Input, Source Output Module provides twenty-four (24) 24 Vdc inputs and eight 24 Vdc, 2 A outputs in one module.

The eight 24 Vdc outputs are individually protected by overcurrent circuitry. When an output goes into overcurrent, the output is automatically shutdown, and a card fault is reported back to the Logic Manager processor. The field power to the module must be cycled to clear the overcurrent condition and re-enable the module output.

CAUTION

When turning on the field power supply, ensure the rise time for the field power is faster than 0.5 ms/V. Rise times slower than this may cause the overcurrent fault to be latched on the module.

Terminal 28 provides the +V for outputs 1-4, and terminal 29 provides +V for outputs 5-8. Terminal 19 provides the common for inputs 1-16, and terminal 38 provides the common for inputs 17-24 and outputs. See Figure 8-39.

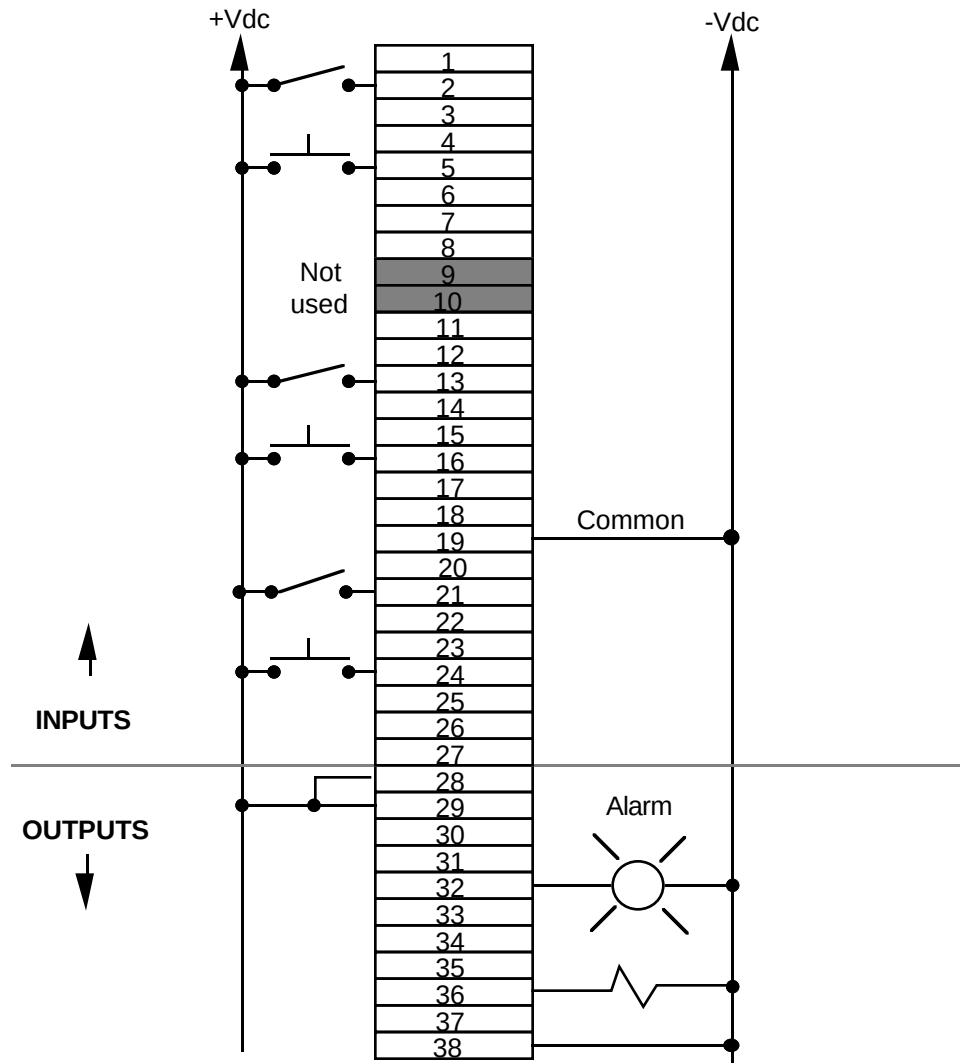


Figure 8-39 — 24 Vdc Sink Input, Source Output Module Terminal Block

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Specifications:

INPUTS	24 per module; terminals 1-8, 11-18, and 20-27
VOLTAGE RANGE	18-28 Vdc
CURRENT RANGE	6-20 mA (10 mA at 24 V typical)
SWITCHING LEVEL	
Logic 1	18 Vdc
Logic 0	11 Vdc
ALLOWABLE LEAKAGE CURRENT	2.5 mA at 18 Vdc
INPUT DELAY	
Off to On	2.4 ms $\pm$ 20%
On to Off	18 ms $\pm$ 28%
OUTPUTS	8 per module; terminals 30-37
VOLTAGE RANGE	18-28 Vdc +20% ripple (max.)
MAXIMUM CURRENT RANGE	2 A per circuit, 8 A per group of 4 maximum
SURGE CURRENT	8 A per output under 10 ms (nonrepetitive)
FIELD POWER REQUIREMENT	28 Vdc +20% ripple maximum, 125 mA (typical), 0.5 us/V
OFF-STATE LEAKAGE CURRENT	$\leq$ 6 mA (typical)
ON-STATE LEAKAGE CURRENT	$\leq$ 1.7 Vdc at 2 A
FUSING	1 per group of 4, 10 A fast blow
POINTS PER COMMON	
Commons (+Vdc)	4
Commons (-Vdc)	16
MODULE OPERATING CURRENT	
+5 Vdc	300 mA maximum
TERMINAL BLOCK REQUIRED	621-9976 or 621-9977

8.8.11 3-Port MiniCOP Module

The MiniCOP module, model 627-1002R, is a self-contained, full-function industrial microcomputer, designed for ASCII communications. The model 627-1002R is a double-wide module that may be installed in any two contiguous I/O slots in the I/O card file.

The MiniCOP uses the OS-9 operating system and supports BASIC09 or Assembly Language. The OS-9 operating system utilities are functionally identical to 627 COP Microcomputer utilities.

The module contains 16 K ROM with 48 K of user work space and 256 k COPRAM. The MiniCOP memory configuration has an 8192 bit data table (512 16-bit registers). OS-9 utilities allow high-level language access to the MiniCOP's data table locations, which may be referenced as single bits, 16-bit registers, or both.

The module communicates with peripheral devices through asynchronous serial data ports. The hardware selectable serial port parameters include signal type (RS232, RS422, or 20 mA), full/half duplex communications, and multidrop/point-to-point configuration. Baud rate, data bits, stop bits, and parity are software selectable. The model 627-1002R module has three 25-pin D-series female connectors for serial communications.

NOTE

Refer to the Honeywell 627 MiniCOP/COP System User Manual (Form Number 627-8991) for detailed operating instructions.

Specifications:

MICROPROCESSOR	68B09E operating at 1.84 MHz
BATTERY BACKUP	9 month continuous backup at full memory; AA size lithium (model 629-3002)
MEMORY	16 K EPROM (operating system); 48 K user memory; 256 K CMOS RAM
FUSE	100 mA, 250 V
POWER	+5 Vdc $\pm$.25 Vdc @ 3.8 A; +15 Vdc $\pm$.48 Vdc @ .285 A; -15 Vdc $\pm$.48 Vdc @ .285 A
INTERFACE	RS232, RS422/485, 20 mA (DIP selectable)
BAUD RATES	150-19.2 k baud, 9600 maximum for 20 mA
DATA WORD	7 or 8 data bits; 1 or 2 stop bits; even, odd, or no parity (all software selectable)
OPERATING TEMPERATURE	0° to 60°C
STORAGE TEMPERATURE	-40°C to 85°C
HUMIDITY	5% to 95% noncondensing
NOISE TEST	NEMA 2-230, IEEE 472-1974, VDE 0871 Class A
ELECTRICAL CODES	UL, CSA, VDE

8.8.12 4-Port MiniCOP Module

The 4-Port MiniCOP Module, model 627-1003R, is a self-contained, full-function industrial microcomputer, designed for ASCII communications. The model 627-1003R is a triple-wide module and may be installed in any three contiguous I/O slots in an I/O card file.

The MiniCOP uses the OS-9 operating system and supports BASIC09 or Assembly Language. The OS-9 operating system utilities are functionally identical to 627 COP Microcomputer utilities.

The module contains 16 K ROM with 48 K of user work space and 256 K COPRAM. The MiniCOP memory configuration has an 8192-bit data table (512 16-bit registers). OS-9 utilities allow high-level language access to the MiniCOP's data table locations, which may be referenced as single bits, 16-bit registers, or both.

The module communicates with peripheral devices through asynchronous serial data ports. The hardware selectable serial port parameters include signal type (RS232, RS422, or 20 mA), full/half duplex communications, and multidrop/point-to-point configuration. Baud rate, data bits, stop bits, and parity are software-selectable. The model 627-1003R has three 25-pin D-series female ports for serial communications, an additional 25-pin D-connector port for video output/keyboard input, a modular keyboard port, three BNC type video ports for RGB color monitor signals and one BNC port for monochrome signals. All video ports are adjustable from the front panel.

NOTE

Refer to the Honeywell 627 MiniCOP/COP System User Manual (Form Number 627-8991) for detailed operating instructions.

Specifications:

MICROPROCESSOR	68B09E operating at 1.84 MHz
VIDEO CONTROLLER	MC6845
BATTERY BACKUP	9 month continuous backup at full memory; AA size lithium (model 629-3002)
MEMORY	16 K EPROM (operating system); 48 K user memory; 256 K CMOS RAM
FUSE	100 mA, 250 V
POWER	+5 Vdc $\pm$.25 Vdc @ 5.5 A (typical); +15 Vdc $\pm$.48 Vdc @ .379 A (typical); -15 Vdc $\pm$.48 Vdc @ .379 A (typical)
INTERFACE	RS232, RS422/485, 20 mA (DIP selectable)
BAUD RATES	150-9.2 k baud, 9600 k maximum for 20 mA (software-selectable)
DATA WORD	7 or 8 data bits; 1 or 2 stop bits; even, odd, or no parity (all software selectable)
SERIAL CONNECTORS	(4) 25-pin subminiature D female; (1) modular keyboard jack*
VIDEO CONNECTORS	(3) BNC connectors to drive a RGB monitor; (1) BNC connector to drive a monochrome monitor
VIDEO SIGNAL STANDARD	RS-170 (analog)
OPERATING TEMPERATURE	0° to 60°C
STORAGE TEMPERATURE	-40°C to 85°C
HUMIDITY	5% to 95% (noncondensing)
NOISE TEST	NEMA 2-230, IEEE 472-1974, VDE 0871 Class A
ELECTRICAL CODES	UL, CSA, VDE

* When the keyboard jack is used, only three serial ports are available.

UCN CONNECTION Section 9

This section describes the installation procedure for connection of the Logic Manager System, both single and redundant, to the Universal Control Network.

9.1 UCN NODE ADDRESS SELECTION

A DIP switch on the daughter board of the the two card Logic Manger Module (LMM) located in slots A and B of the Logic Manager Processor card file is used to select the desired UCN node address. See subsection 7.3.1 for the location and selection procedure for the DIP switch.

9.2 UCN INSTALLATION

The installation procedure and cabling information can be found in the *UCN Site Planning and Installation* manual in the *UCN Site Planning and Installation* binder. Section 4 details the installation of the cabling, while Section 5 covers cable fabrication.

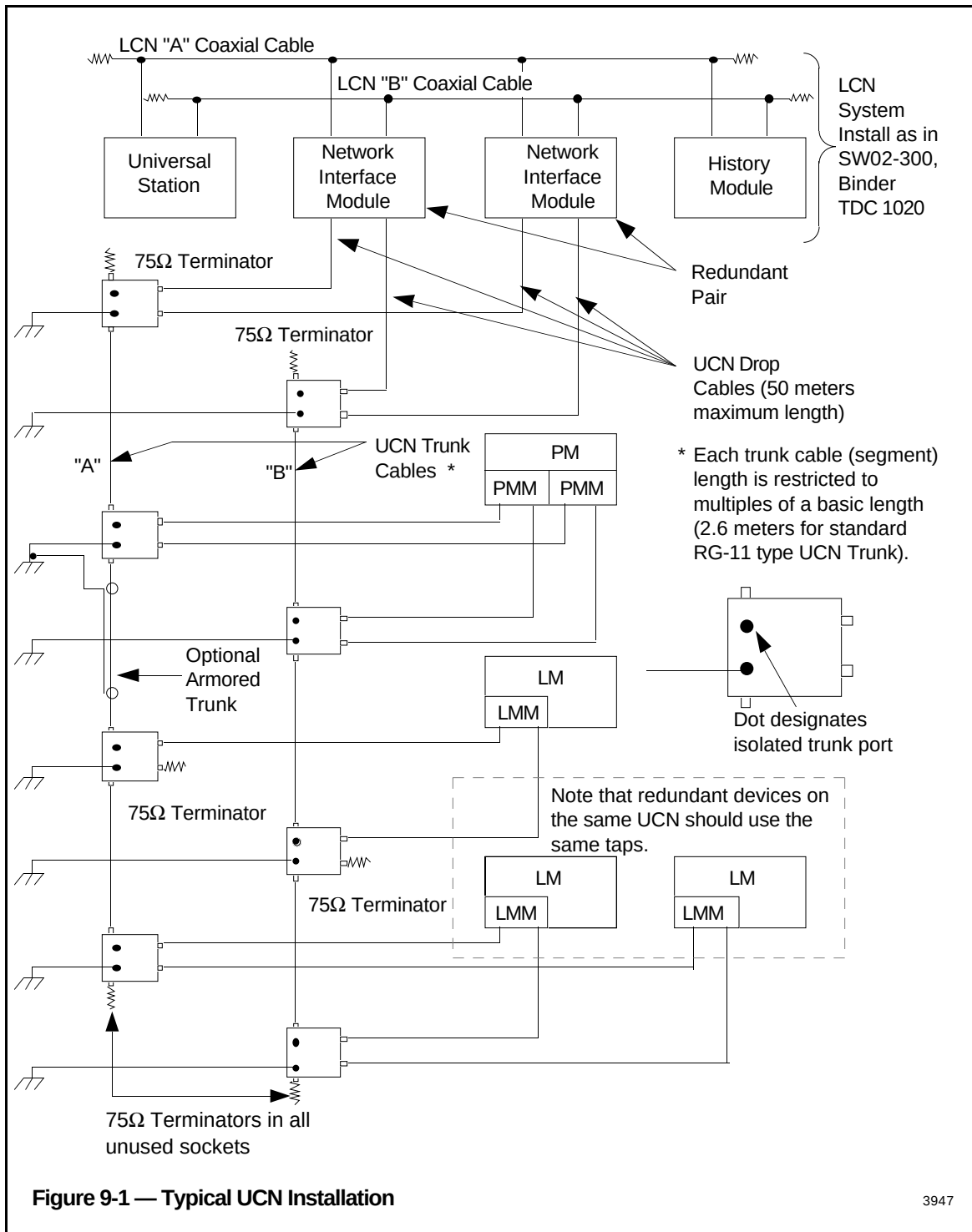
Figure 9-1 shows a typical UCN installation. Connection of the UCN system is made at the front panel of the Logic Manager Module.

9.3 UCN TRUNK CABLE CONNECTION

Connection of the UCN trunk lines are made at the the front panel of the Logic Manager Module. See Figure 7-2 for more detail. Use the procedure that is found in the *UCN Site Planning and Installation* manual as the installation guide.

9.4 REDUNDANT LMM CONSIDERATIONS

When redundant Logic Manager Modules are installed in a Logic Manager System, both LMMs separately connect to the UCN trunk and care must be taken to select the same UCN network taps for both modules. Observe also that the same nodal address is selected with the DIP switch on the daughter board of the Logic Manager Module.



SERIAL I/O SCAN RATE CALCULATION Appendix A

The serial I/O system scans asynchronously to the logic controller scan. This appendix provides formulas to calculate the serial I/O scan rate for various system configurations.

A.1 GENERAL FORMULAS

- System configured to update 1 PUSH/PULL card per card file per I/O scan:
 1. $T_t = 3.14N_r + 0.016N_a + 2.02N_{rp} + 7$ (nonredundant system)
 2. $T_t = 3.14N_r + 0.016N_a + 2.02N_{rp} + 12$ (redundant system)
- System configured to update ALL PUSH/PULL cards per card file per I/O scan:
 3. $T_t = 3.14N_r + 0.016N_a + 1.92N_p + 23$ (nonredundant system)
 4. $T_t = 3.14N_r + 0.016N_a + 1.92N_p + 35$ (redundant system)

where T_t = total serial I/O scan rate in milliseconds
 N_r = number of I/O card files on the link
 N_a = number of addresses on the link
 N_{rp} = number of card files on the link with PUSH/PULL cards
 N_p = number of PUSH/PULL cards on the link

NOTE

Each 621-9939 Serial Link Module (SLM) in the processor card file can have two serial channels. If the calculated serial scan rate is too long, it can be reduced by splitting the one link into two links and recalculating each link's scan rate. The calculated scan rates may vary from the actual time by ± 10 percent due to retries on the link and other link disturbances. Note that when the system is configured to update 1 PUSH/PULL card per card file per I/O scan, and a card file contains more than 1 PUSH/PULL card, the total update time for the PUSH/PULL cards in that card file is the total serial scan rate times the number of PUSH/PULL cards in that card file.

A.2 EXAMPLES

A.2.1 Example 1

For a nonredundant system with four I/O card files, each containing twelve 16-point I/O modules and no PUSH/PULL cards, use Equation 1:

$$N_r = 4$$

$$N_a = 4 \text{ card files} * (12 \text{ modules/card file} * 16 \text{ addresses/module}) = 768 \text{ addresses}$$

$$N_{rp} = 0$$

$$T_t = (3.14 * 4) + (0.016 * 768) + (2.02 * 0) + 7$$

$$T_t = 32\text{ms}$$

A.2.2 Example 2

For a redundant system with six I/O card files, each containing twelve 16-point I/O modules, and two of those card files each containing two analog (PUSH/PULL) modules, use Equation 2 (system is configured to update 1 PUSH/PULL card per card file per scan):

$$N_r = 6$$

$$\begin{aligned} N_a &= [4 \text{ card files} * (12 \text{ modules} * 16 \text{ points})] + [2 \text{ card files} * (10 \text{ modules} * 16 \text{ points})] \\ &\quad + [2 \text{ card files} * (2 \text{ modules} * 8 \text{ points})] \\ &= 1120 \text{ addresses} \end{aligned}$$

$$N_{rp} = 2$$

$$T_t = (3.14 * 6) + (0.016 * 1120) + (2.02 * 2) + 12$$

$$T_t = 53\text{ms}$$

A.2.3 Example 3

For a redundant system with two I/O card files, each containing eight 16-point I/O modules and eight analog (PUSH/PULL) modules, use Equation 4 (system is configured to update ALL PUSH/PULL cards per card file per scan):

$$N_r = 2$$

$$\begin{aligned} N_a &= [2 \text{ card files} * (8 \text{ modules} * 16 \text{ points})] + [2 \text{ card files} * (8 \text{ modules} * 8 \text{ points})] \\ &= 384 \text{ addresses} \end{aligned}$$

$$N_p = 16$$

$$T_t = (3.14 * 2) + (0.016 * 384) + (1.92 * 16) + 35$$

$$T_t = 78\text{ms}$$

**FIBER OPTIC SERIAL I/O
Appendix B****NOTE**

For Fiber Optic Communications, please contact:

Weed Fibroptics
800 Paloma
Suite 240
Round Rock, Texas 78664
512-388-2522

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